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AIUB, MATH 2 mid term

Math-2

Indefinite Integrals!

ch-05

Topic Name :

Day :

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Formula:

1. $\int dx = x + c$

2. $\int 1 dx = 1x + c$

3. $\int a dx = ax + c$

4. $\int x^n \cdot dx = \frac{x^{n+1}}{n+1} + c$

5. $\int e^x \cdot dx = e^x + c$

6. $\int \frac{1}{x} \cdot dx = \ln|x| + c$

7. $\int \frac{1}{\sqrt{x}} \cdot dx = 2\sqrt{x} + c$

8. $\int a^x \cdot dx = \frac{a^x}{\ln a} + c$

9. $\int \sin x \cdot dx = -\cos x + c$

10. $\int \cos x \cdot dx = \sin x + c$

11. $\int \sec^2 x \cdot dx = \tan x + c$

12. $\int \operatorname{cosec}^2 x \cdot dx = -\cot x + c$

13. $\int \sec x \cdot (\tan x) \cdot dx = \sec x + c$

14. $\int \operatorname{cosec} x \cdot (\cot x) \cdot dx = -\operatorname{cosec} x + c$

15. $\int \frac{1}{1+x^2} \cdot dx = \tan^{-1} x + c$

16. $\int \frac{1}{\sqrt{1-x^2}} \cdot dx = \sin^{-1} x + c$

17. $\int \frac{1}{|x|\sqrt{x^2-1}} \cdot dx = \sec^{-1} x + c$

18. $\int \frac{1}{a^2+x^2} \cdot dx = \frac{1}{a} + \tan^{-1} \frac{x}{a} + c$

19. $\int \frac{1}{\sqrt{a^2-x^2}} \cdot dx = \sin^{-1} \frac{x}{a} + c$

20. $\int \cosh x \cdot dx = \sinh x + c$

21. $\int \sinh x \cdot dx = \cosh x + c$

22. $\int e^{mx} \cdot dx = \frac{1}{m} \times e^{mx} + c$

23. $\int \sin mx \cdot dx = -\frac{1}{m} \times \cos mx + c$

24. $\int e^{\frac{mx}{n}} \cdot dx = \frac{n}{m} e^{\frac{mx}{n}} + c$

Evaluate the following integral:-

* $\int (x^3+e^{5x}+\sin 2x)\cdot dx$

$= \int x^3\cdot dx + \int e^{5x}\cdot dx + \int \sin 2x \cdot dx$

$= \frac{1}{4}x^4 + \frac{1}{5}e^{5x} + (-\frac{1}{2}\cos 2x) + c$

$= \frac{x^4}{4} + \frac{e^{5x}}{5} - \frac{1}{2}\cos 2x + c$

* $\int e^{\frac{2x}{3}}\cdot dx$

$= \frac{3}{2}e^{\frac{2x}{3}} + c$

* $\int \frac{\cos\theta}{\sin^2\theta}\cdot d\theta$

$= \int \frac{\cos\theta}{\sin\theta}\cdot\frac{1}{\sin\theta}\cdot d\theta$

$= \int \cot\theta\cdot \operatorname{cosec}\theta\cdot d\theta$

$= -\operatorname{cosec}\theta + c$

* $\int \frac{2t+t^2\sqrt{t}+2}{t^2}\cdot dt$

$= \int 2\times\frac{1}{t}\cdot dt + \int \sqrt{t}\cdot dt + \int 2\cdot\frac{1}{t^2}dt$

$= 2\cdot\ln|t| + \frac{t^{\frac{1}{2}+1}}{\frac{1}{2}+1} + 2\times\frac{t^{-2+1}}{-2+1} + c$

$= 2\cdot\ln|t| + \frac{2}{3}\times t^{\frac{3}{2}} - 2\times\frac{1}{t} + c$

$$*\int \frac{2t^2+t^2\sqrt{t}+2}{t^2}\,dt$$

$$= 2t + \frac{t^{\frac{1}{2}+1}}{\frac{1}{2}+1} + 2 \times \frac{t^{-2+1}}{-2+1} + c$$

$$= 2t + \frac{2}{3}t^{\frac{3}{2}} - 2/t + c$$

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(9) $\int (u+4)(2u+1) \cdot du$

$= \int (2u^2 + u + 8u + 4) \cdot du$

$= 2 \times \frac{1}{3}u^3 + \frac{1}{2}u^2 + 8 \times \frac{1}{2}u^2 + 4u + c$

$= \frac{2}{3}u^3 + \frac{1}{2}u^2 + 4u^2 + 4u + c$

(12) $\int \left(x^2 + 1 + \frac{1}{x^2+1}\right) \cdot dx$

$= \frac{1}{3}x^3 + x + \tan^{-1}x + c$

(13) $\int (\sin 2x + \sinh x) \cdot dx$

$= -\frac{1}{2}\cos 2x + \cosh x + c$

(15) $\int (2 + \tan^2\theta) \cdot d\theta$

$= 2\theta + \int \tan^2\theta \cdot d\theta$

$= 2\theta + \int (\sec^2\theta - 1) \cdot d\theta$

$= 2\theta + \tan\theta - \theta + c$

$= \theta + \tan\theta + c$

(16) $\int 2^t(1+5^t)\cdot dt$

$= \int 2^t \cdot dt + \int 10^t \cdot dt$

$= \frac{2^t}{\ln 2} + \frac{10^t}{\ln 10} + c$

$\Rightarrow 2^t \times 5^t = 10^t$

$\Rightarrow 2^{3t} \times 2^{6t} = 2^{3t+6t}$

(18) $\int \frac{\sin 2x}{\sin x} \cdot dx$

$= \int \frac{2\sin x \cdot \cos x}{\sin x} \cdot dx$

$= 2\sin x + c$

Exercise:

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(5) $\int (x^{1\cdot 3} + 7x^{2\cdot 5}) \cdot dx$

$= \frac{x^{1\cdot 3+1}}{1\cdot 3+1} + 7 \cdot \frac{x^{2\cdot 5+1}}{2\cdot 5+1} + c$

$= \frac{1}{2\cdot 3} \cdot x^{2\cdot 3} + 7 \times \frac{x^{3\cdot 5}}{3\cdot 5} + c$

(6) $\int \sqrt[4]{x^5} \cdot dx$

$= \int (x^5)^{\frac{1}{4}} \cdot dx$

$= \frac{x^{\frac{5}{4}+1}}{\frac{5}{4}+1} + c$

$= \frac{4}{9} x^{\frac{9}{4}} + c$

(7) $\int \left(5+\frac{2}{3}x^2+\frac{3}{4}x^3\right)\cdot dx$

$= 5x+\frac{2}{3}\times\frac{1}{3}x^3+\frac{3}{4}\times\frac{1}{4}\times x^4+C$

$= 5x+\frac{2}{9}x^3+\frac{3}{16}x^4+C$

(8) $\int \left(u^6-2u^5-u^3+\frac{2}{7}\right)\cdot du$

$= \frac{1}{7}u^7-2\times\frac{1}{6}u^6-\frac{1}{4}u^4+\frac{2}{7}u+C$

$= \frac{1}{7}u^7-\frac{1}{3}u^6-\frac{1}{4}u^4+\frac{2}{7}u+C$

(10) $\int t(t^2-3t+2)\cdot dt$

$= \int (t^3-3t^2+2t)\cdot dt$

$= \frac{1}{4}t^4-3\times\frac{1}{3}t^3+2\times\frac{1}{2}t^2+C$

$= \frac{1}{4}t^4-t^3+t^2+C$

(11) $\int \frac{1+\sqrt{x}+x}{x}\cdot dx$

$= \ln|x|+x^{-\frac{1}{2}+1}/\left(-\frac{1}{2}+1\right)+x+C$

$= \ln|x|+2\times x^{\frac{1}{2}}+x+C$

(14) $\int \left(\frac{1+x}{x}\right)^2 \cdot dx$

$= \int \frac{(1+x)^2}{x^2} \cdot dx$

$= \int \frac{1+2x+x^2}{x^2} \cdot dx$

$= \frac{x^{-2+1}}{-2+1} + 2\ln|x| + x + c$

$= -\frac{1}{x} + 2\ln|x| + x + c$

(17) $\int 2^t(1+5^t) \cdot dt$

$= \int (2^t + 10^t)\, dt$

$= \frac{2^t}{\ln 2} + \frac{10^t}{\ln 10} + c$

(16) $\int \sec t \cdot (\sec t + \tan t) \cdot dt$

$= \int (\sec^2 t + \sec t \cdot \tan t) \cdot dt$

$= \tan t + \sec t + c$

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Integration by Substitution:—

For Indefinite integral:—

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① $\int x\sqrt{1-x^2}\,dx$

$= \int \sqrt{z}\cdot -\frac{1}{2}\cdot dz$

$= -\frac{1}{2} \times \frac{z^{\frac{1}{2}+1}}{\frac{1}{2}+1} + c$

$= -\frac{1}{2} \times \frac{2}{3} \times z^{\frac{3}{2}} + c$

$= -\frac{1}{3} z^{\frac{3}{2}} + c$

$= -\frac{1}{3}(1-x^2)^{\frac{3}{2}} + c$

Let:—

$z = 1-x^2$

$\frac{dz}{dx} = -2x$

$-\frac{1}{2}\cdot dz = x\cdot dx$

② $\int (1-2x)^9\cdot dx$

$= -\frac{1}{2}\int z^9\cdot dx$

$= -\frac{1}{2} \times \frac{1}{10} \times z^{10} + c$

$= -\frac{1}{20} \times (1-2x)^{10} + c$

Let

$z = 1-2x$

$\frac{dz}{dx} = -2$

$-\frac{1}{2} \times dz = dx$

(10) $\int \sin t \cdot \sqrt{1+\cos t} \cdot dt$

$= -\int \sqrt{z} \cdot dz$

$= -\dfrac{z^{\frac{1}{2}+1}}{\frac{1}{2}+1} + c$

$= -\dfrac{2}{3} \times z^{\frac{3}{2}} + c$

$= -\dfrac{2}{3} \times (1+\cos t)^{\frac{3}{2}} + c$

Let

$z = 1 + \cos t$

$\dfrac{dz}{dt} = -(\sin t)$

$-dz = \sin t \cdot dt$

(14) $\int y^2 (4-y^3)^{\frac{2}{3}} \cdot dy$

$= \dfrac{1}{3} \int z^{\frac{2}{3}} \cdot dz$

$= \dfrac{1}{3} \times \dfrac{z^{\frac{2}{3}+1}}{\frac{2}{3}+1} + c$

$= \dfrac{1}{3} \times \dfrac{3}{5} \times (4+y^3)^{\frac{5}{3}} + c$

$= \dfrac{1}{5} (4+y^3)^{\frac{5}{3}} + c$

Let.

$z = (4+y^3)$

$\dfrac{dz}{dy} = 3y^2$

$\dfrac{1}{3} dz = y^2 \cdot dy$

(15) $\int \cos^3\theta \cdot \sin\theta \cdot d\theta$

$= -\int z^3 \cdot dz$

$= -\dfrac{1}{4} \times (\cos\theta)^4 + c$

Let:-

$z = \cos\theta$

$\dfrac{dz}{d\theta} = -\sin\theta$

$-dz = \sin\theta \cdot d\theta$

(17) $\int \frac{e^x}{(1-e^x)^2} \cdot dx$

$= -\int \frac{1}{z^2} \cdot dz$

$= -\frac{z^{-2+1}}{-2+1} + c$

$= \frac{1}{z} + c$

$= \frac{1}{1-e^x} + c$

Let:-

$z = (1-e^x)$

$\frac{dz}{dx} = -e^x$

$-\frac{dz}{1} = e^x \cdot dx$

(28) $\int e^{\cos t} \cdot \sin t \cdot dt$

$= -\int e^z \cdot dz$

$= -e^{\cos t} + c$

Let

$\cos t = z$

$\frac{dz}{dt} = -\sin t$

$-dz = \sin t \cdot dt$

(43) $\int \frac{dx}{\sqrt{1-x^2}} \times \frac{1}{\sin^{-1} x}$

$= \int \frac{1}{z} \cdot dz$

$= \ln|z| + c$

$= \ln|\sin^{-1} x| + c$

Let:-

$z = \sin^{-1} x$

$\frac{dz}{dx} = \frac{1}{\sqrt{1-x^2}}$

$dz = \frac{1}{\sqrt{1-x^2}} \times dx$

(44) $\int \frac{x}{1+x^4} \cdot x$

$= \int \frac{x}{1+(x^2)^2} \cdot dx$

$= \frac{1}{2}\int \frac{1}{1+z^2} \cdot dz$

$= \frac{1}{2} \times \tan^{-1}(z) \cdot + c$

Let

$z = x^2$

$\frac{dz}{dx} = 2x$

$\frac{1}{2} dz = x \cdot dx$

$= \frac{1}{2} \times \tan^{-1}(x^2) + c$

Ans:

Exercise:

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(7) $\int x\sqrt{1-x^2}\,dx$

Let:
$z = 1 - x^2$
$\frac{dz}{dx} = -2x$
$-\frac{1}{2}dz = x \cdot dx$

$= -\frac{1}{2}\int \sqrt{z} \cdot dz$

$= -\frac{1}{2} \times \frac{z^{\frac{1}{2}+1}}{\frac{1}{2}+1} + c$

$= -\frac{1}{2} \times \frac{2}{3} \times z^{\frac{3}{2}} + c$

$= -\frac{1}{3} \times z^{\frac{3}{2}} + c$

(8) $\int x^2 e^{x^3} \cdot dx$

Let:
$z = x^3$
$\frac{dz}{dx} = 3x^2$
$\frac{1}{3}dz = x^2 \cdot dx$

$= \frac{1}{3}\int e^z \cdot dz$

$= \frac{1}{3} \times e^z + c$

$= \frac{1}{3} \times e^{x^3} + c$

(9) $\int \cos\left(\frac{\pi t}{2}\right) \cdot dt$

Let:
$z = \left(\frac{\pi t}{2}\right)$
$\frac{dz}{dt} = \frac{\pi}{2}$
$\frac{dz}{1} \times \frac{2}{\pi} = dt$

$= \frac{2}{\pi}\int \cos z \cdot dz$

$= \frac{2}{\pi}\sin\left(\frac{\pi t}{2}\right) + c$

(12) $\int \sec^2 2\theta \cdot d\theta$

$= \frac{1}{2}\int \sec^2 z \cdot dz$

$= \frac{1}{2}\tan z + c$

$= \frac{1}{2}\tan(2\theta) + c$

Let:-

$z = 2\theta$

$\frac{dz}{d\theta} = 2$

$\frac{1}{2}dz = d\theta$

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(13) $\int \frac{dx}{5-3x}$

$= -\frac{1}{3}\int \frac{1}{z} \cdot dz$

$= \frac{1}{3}\ln|z| + c$

$= -\frac{1}{3}\ln|5-3x| + c$

Let

$z = 5-3x$

$\frac{dz}{dx} = -3$

$-\frac{1}{3}dz = dx$

(15) $\int \frac{\sin\sqrt{x}}{\sqrt{x}} \cdot dx$

$= 2\int \sin z \cdot dz$

$= -2\cos z + c$

$= -2\cos\sqrt{x} + c$

Let:-

$z = \sqrt{x}$

$\frac{dz}{dx} = \frac{1}{2\sqrt{x}}$

$2dz = \frac{1}{\sqrt{x}} \times dx$

(20) $\int \frac{z^2}{z^3+1} \cdot dz$

$= \frac{1}{3}\int \frac{1}{x} \cdot dx$

$= \frac{1}{3}\ln|x| + c$

$= \frac{1}{3}\ln|z^3+1| + c$

Let:-

$x = z^3+1$

$\frac{dx}{dz} = 3z^2$

$\frac{1}{3}dx = z^2 \cdot dz$

(21) $\int \frac{(\ln x)^2}{x} \cdot dx$

$= \int \frac{z^2}{1} \cdot dz$

$= \frac{1}{3} \times z^3 + c$

$= \frac{1}{3} (\ln x)^3 + c$

Let:-

$z = \ln x$

$\frac{dz}{dx} = \frac{1}{x}$

$dz = \frac{1}{x} \cdot dx$

(27) $\int (x^2+1)(x^3+3x)^4 \cdot dx$

$= \frac{1}{3} \int z^4 \cdot dz$

$= \frac{1}{3} \times \frac{1}{5} \times z^5 + c$

$= \frac{1}{15} (x^3+3x)^5 + c$

Let:-

$z = x^3 + 3x$

$\frac{dz}{dx} = 3x^2 + 3$

$= 3(x^2+1)$

$\frac{1}{3} dz = (x^2+1) \cdot dx$

(31) $\int \frac{(\arctan x)^2}{x^2+1} \cdot dx$

$= \int \frac{(\tan^{-1} x)^2}{x^2+1} \cdot dx$

$= \int z^2 \cdot dz$

$= \frac{1}{3} (\tan^{-1} x)^3 + c$

$= \frac{1}{3} (\arctan x)^3 + c$

Let:-

$z = \tan^{-1} x$

$\frac{dz}{dx} = \frac{1}{1+x^2}$

$dz = \frac{1}{1+x^2} \cdot dx$

(32) $\int \frac{x}{x^2+4} \cdot dx$

$= \frac{1}{2} \int \frac{1}{z} \cdot dz$

$= \frac{1}{2} \ln|x^2+4| + c$

Let:-

$z = x^2 + 4$

$\frac{dz}{dx} = 2x$

$\frac{1}{2} \cdot dz = x \cdot dx$

Definite Integrals

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Fundamental Theorem of Definite integrals:-

* $\int_0^1 x^3 \cdot dx$

$= \frac{1}{3+1} \times [x^{3+1}]_0^1$

$= \frac{1}{4} [x^4]_0^1$

$= \frac{1}{4} [1^4 - 0^4]$

$= \frac{1}{4}$

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(27) $\int_0^{\pi} (5e^x + 3\sin x) \cdot dx$

$= 5[e^x]_0^{\pi} + 3 \times [-\cos x]_0^{\pi}$

$= 5(e^{\pi} - e^0) - 3[\cos\pi - \cos 0]$

$= 5e^{\pi} - 5 - 3\cos\pi + 3\cos 0$

$= 5e^{\pi} - 5 + 3 + 3$

$= 5e^{\pi} + 1$

(30) $\int_0^1 \frac{4}{1+p^2} \cdot dp$

$\Rightarrow 4[\tan^{-1} p]_0^1$

$= 4 \times (\tan^{-1} 1 - \tan^{-1} 0)$

$= 4 \times \frac{\pi}{4}$

$= \pi$

(৩৭) $\int_0^1 (5x - 5^x)\cdot dx$

$= \left[5\times\frac{1}{2}\times x^2 - \frac{5^x}{\ln 5}\right]_0^1$

$= \left(\frac{5}{2}\times 1^2\right) - \left(\frac{5^1}{\ln 5} - \frac{5^0}{\ln 5}\right)$

$= \frac{5}{2} - \left(\frac{5-1}{\ln 5}\right)$

$= \frac{5}{2} - \frac{4}{\ln 5}$

(৩৯) $\int_1^2 \left(\frac{x}{2} - \frac{2}{x}\right)\cdot dx$

$= \left[\frac{1}{2}\times\frac{1}{2}\times x^2 - 2\times\ln|x|\right]_1^2$

$= \frac{1}{4}(2^2 - 1^2) - 2\{\ln|2| - \ln|1|\}$

$= \frac{3}{4} - 2\ln 2$

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Exercise:

(11) $\int_1^3 (x^2 + 2x - 4)\cdot dx$

$= \left[\frac{1}{3}x^3 + 2\times\frac{1}{2}x^2 - 4x\right]_1^3$

$= \frac{1}{3}(3^3 - 1^3) + (3^2 - 1^2) - 4\times(3-1)$

$= \frac{26}{3} + 8 - 8$

$= \frac{26}{3}$

(23) $\int_1^9 \sqrt{x}\cdot dx$

$= \left[\dfrac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1}\right]_1^9$

$= \dfrac{2}{3}\left[x^{\frac{3}{2}}\right]_1^9$

$= \dfrac{2}{3}\left[9^{\frac{3}{2}} - 1^{\frac{3}{2}}\right]$

$= \dfrac{2}{3} \times 26$

$= \dfrac{52}{3}$

(24) $\int_1^8 x^{-\frac{2}{3}} \cdot dx$

$= \left[\dfrac{x^{-\frac{2}{3}+1}}{-\frac{2}{3}+1}\right]_1^8$

$= 3 \times \left[x^{\frac{1}{3}}\right]_1^8$

$= 3 \times \left[8^{\frac{1}{3}} - 1^{\frac{1}{3}}\right]$

$= 3$

(25) $\int_{\pi/6}^{\pi} \sin\theta \cdot d\theta$.

$= -\left[\cos\theta\right]_{\pi/6}^{\pi}$

$= -\left[\cos\pi - \cos\dfrac{\pi}{6}\right]$

$= 1 + \dfrac{\sqrt{3}}{2}$

$= \dfrac{2+\sqrt{3}}{2}$

(27) $\int_0^1 (u+2)(u-3)\cdot dx$

$= \int_0^1 (u^2 - 3u + 2u - 6)\cdot dx$

$= \int_0^1 (u^2 - u - 6)\cdot dx$

$= \left[\dfrac{1}{3}u^3 - \dfrac{1}{2}u^2 - 6x\right]_0^1$

$= \dfrac{1}{3}\times 1^3 - \dfrac{1}{2}1^2 - 6\times 1$

$= \dfrac{1}{3} - \dfrac{1}{2} - 6$

$= -\dfrac{37}{6}$

(28) $\int_1^4 \dfrac{2+x^2}{\sqrt{x}} \cdot dx$

$= \int_1^4 \dfrac{2dx}{\sqrt{x}} + \int_1^4 \sqrt{x}\cdot x \cdot dx$

$= \int_1^4 2\times\dfrac{dx}{\sqrt{x}} + \int_1^4 x^{\frac{1}{2}+1}\cdot dx$

$= \int_1^4 2\times\dfrac{1}{x^{\frac{1}{2}}}\cdot dx + \int_1^4 x^{\frac{3}{2}}\cdot dx$

$= 2\left[\dfrac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1}\right]_1^4 + \left[\dfrac{x^{\frac{3}{2}+1}}{\frac{3}{2}+1}\right]_1^4$

$= 4\left(4^{\frac{1}{2}} - 1^{\frac{1}{2}}\right) + \dfrac{2}{5}\left(4^{\frac{5}{2}} - 1^{\frac{5}{2}}\right)$

$= \dfrac{82}{5}$

(35) $\int_1^2 \frac{u^3+3u^6}{u^4} \cdot du$

$= \left[\ln|u| + 3 \times \frac{1}{3}u^3\right]_1^2$

$= \ln|2| - \ln|1| + 2^3 - 1^3$

$= 7 + \ln|2|$

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(21) $\int_{-2}^{3} (x^2 - 3) \cdot dx$

$= \left[\frac{1}{3}x^3 - 3x\right]_{-2}^{3}$

$= \frac{1}{3}(3^3 - (-2)^3) - 3 \times (3+2)$

$= -\frac{10}{3}$

(22) $\int_1^2 (4x^3 - 3x^2 + 2x) \cdot dx$

$= \left[4 \times \frac{1}{4}x^4 - 3 \times \frac{1}{3}x^3 + 2 \times \frac{1}{2}x^2\right]_1^2$

$= (2^4 - 1^4) - (2^3 - 1^3) + (2^2 - 1^2)$

$= 11$

(23) $\int_{-2}^{0} \left(\frac{1}{2}t^4 + \frac{1}{4}t^3 - t\right) \cdot dt$

$= \left[\frac{1}{2} \times \frac{1}{5}t^5 + \frac{1}{4} \times \frac{1}{4} \times t^4 - \frac{1}{2} \times t^2\right]_{-2}^{0}$

$= \frac{1}{10}(0^5 - (-2)^5) + \frac{1}{16} \times (0^4 - (-2)^4) - \frac{1}{2}(0^2 - (-2)^2)$

$= \frac{1}{10} \times 32 + \frac{1}{16} \times -16 - \frac{1}{2} \times -4$

$= \frac{21}{5}$

(24) $\int_0^3 (1 + 6w^2 - 10w^4) \cdot dw$

$= \left[w + 6 \times \frac{1}{3}w^3 - 10 \times \frac{1}{5} \cdot w^5\right]_0^3$

$= 3 + 2 \times (3^3) - 2 \times 3^5$

$= -429$

(25) $\int_0^2 (2x-3)(4x^2+1) \cdot dx$

$= \int_0^2 (8x^3 + 2x - 12x^2 - 3) \cdot dx$

$= \left[8 \times \frac{1}{4} \times x^4 + 2 \times \frac{1}{2}x^2 - 12 \times \frac{1}{3}x^3 - 3x\right]_0^2$

$= \frac{8}{4}(2^4) + (2^2) - 12 \times \frac{1}{3} \times 2^3 - 3 \times 2$

$= -10$

(26) $\int_{-1}^{1} t(1-t)^2 \, dt$

$= \int_{-1}^{1} t(1^2 - 2t + t^2) \cdot dt$

$= \int_{-1}^{1} (t - 2t^2 + t^3) \cdot dt$

$= \left[\frac{1}{2} \times t^2 - 2 \times \frac{1}{3}t^3 + \frac{1}{4} \times t^4\right]_{-1}^{1}$

$= \frac{1}{2} \times 0 - \frac{2}{3} \times 2 + 0$

$= -\frac{4}{3}$

Ans:-

(28) $\int_1^2 \left(\frac{1}{x^2} - \frac{4}{x^3}\right) \cdot dx$

$= \left[-\frac{1}{x} - 4\times -\frac{1}{2}\times\frac{1}{x^2}\right]_1^2$

$= -\left[\frac{1}{2} - \frac{1}{1}\right] + 2\times\left(\frac{1}{2^2} - \frac{1}{1^2}\right)$

$= -1$

(31) $\int_0^1 x\left(\sqrt[3]{x} + \sqrt[4]{x}\right) dx$

$= \int_0^1 \left(x^{\frac{1}{3}+1} + x^{\frac{1}{4}+1}\right) \cdot dx$

$= \int_0^1 \left(x^{\frac{4}{3}} + x^{\frac{5}{4}}\right) \cdot dx$

$= \left[\frac{3}{7} x^{\frac{7}{3}} + \frac{4}{9} x^{\frac{9}{4}}\right]_0^1$

$= \frac{3}{7} + \frac{4}{9}$

$= \frac{55}{63}$

(32) $\int_1^4 \frac{\sqrt{y} - y}{y^2} \cdot dy$

$= \int_1^4 \frac{1}{y^{\frac{1}{2}+1}} \cdot dy - \int_1^4 \frac{1}{y} \cdot dy$

$= \left[\frac{y^{-\frac{3}{2}+1}}{-\frac{3}{2}+1}\right]_1^4 - \left[\ln|y|\right]_1^4$

$= -2\left(4^{-\frac{1}{2}} - 1^{-\frac{1}{2}}\right) - \ln|4|$

$= 1 - \ln|4|$

(35) $\int_0^1 (x^{10}+10^x)\cdot dx$

$= \left[\frac{1}{11}x^{11} + \frac{10^x}{\ln 10}\right]_0^1$

$= \frac{1}{11}1^{11} + \frac{10}{\ln 10} - \frac{1}{\ln 10}$

$= \frac{1}{11} + \frac{9}{\ln 10}$

(36) $\int_0^{\frac{\pi}{4}} \sec\theta\cdot\tan\theta\cdot d\theta$

$= \left[\sec\theta\right]_0^{\frac{\pi}{4}}$

$= \sec\frac{\pi}{4} - \sec 0$

$= \sqrt{2}-1$

(37) $\int_0^{\frac{\pi}{4}} \frac{1+\cos^2\theta}{\cos^2\theta}\cdot d\theta$

$\Rightarrow \int_0^{\frac{\pi}{4}} \left(\frac{1}{\cos^2\theta}+1\right)\cdot d\theta$

$= \int_0^{\pi/4} (\sec^2\theta+1)\cdot d\theta$

$= \left[\tan\theta\right]_0^{\frac{\pi}{4}} + \left[\theta\right]_0^{\pi/4}$

$= \tan\frac{\pi}{4} - \tan 0 + \frac{\pi}{4}$

$= 1 - 0 + \frac{\pi}{4}$

$= \frac{5\pi}{4}$

(38) $\int_0^{\frac{\pi}{3}} \frac{\sin\theta+\sin\theta\cdot\tan^2\theta}{\sec^2\theta}\cdot d\theta$

$= \int_0^{\pi/3} \frac{\sin\theta(1+\tan^2\theta)}{\sec^2\theta}\cdot d\theta$

$= \int_0^{\pi/3} \frac{\sin\theta\cdot\sec^2\theta}{\sec^2\theta}\cdot d\theta$

$= \int_0^{\pi/3} \sin\theta\cdot d\theta$

$= -\left[\cos\theta\right]_0^{\frac{\pi}{3}}$

$= -\left[\cos\frac{\pi}{3} - \cos 0\right]$

$= \frac{1}{2}$

(39) $\int_1^8 \frac{2+t}{\sqrt[3]{t^2}}\cdot dt$

$= \left[2\times\frac{t^{-\frac{2}{3}+1}}{-\frac{2}{3}+1} + \frac{t^{\frac{1}{3}+1}}{\frac{1}{3}+1}\right]_1^8$

$= \left[6t^{\frac{1}{3}} + \frac{3}{4}\times t^{\frac{4}{3}}\right]_1^8$

$= 6\left(8^{\frac{1}{3}} - 1^{\frac{1}{3}}\right) + \frac{3}{4}\left(8^{\frac{4}{3}} - 1^{\frac{4}{3}}\right)$

$= \frac{69}{4}$

(40) $\int_{-10}^{10} \frac{2e^x}{\sinh x+\cosh x} \cdot dx$

here
$\sinh x = \frac{1}{2}(e^x - e^{-x})$
$\cosh x = \frac{1}{2}(e^x + e^{-x})$

$= \int_{-10}^{10} \frac{2e^x}{\frac{1}{2}(e^x - e^{-x}) + \frac{1}{2}(e^x + e^{-x})} \cdot dx$

$= \int_{-10}^{10} \frac{4e^x}{2e^x} \cdot dx$

$= \left[\frac{4x}{2}\right]_{-10}^{10}$

$= \frac{4}{2} \times (10 - (-10))$

$= 2 \times 20$

$= 40$

(41) $\int_0^{\frac{\sqrt{3}}{2}} \frac{dx}{\sqrt{1-x^2}}$

$= [\sin^{-1} x]_0^{\frac{\sqrt{3}}{2}}$

$= \sin^{-1}(\sqrt{3}/2) - \sin^{-1}(0)$

$= \frac{\pi}{3} - 0$

$= \frac{\pi}{3}$

(42) $\int_1^2 \frac{(x-1)^3}{x^2} \cdot dx$

$= \int_1^2 \frac{x^3 - 3x^2 + 3x - 1}{x^2} \cdot dx$

$= \left[\frac{1}{2}x^2 - 3x + 3 \times \ln|x| + \frac{1}{x}\right]_1^2$

$= \frac{1}{2}(4-1) - 3 \times (1) + 3 \times (\ln|2| - \ln|1|) + \left(\frac{1}{2} - \frac{1}{1}\right)$

$= \frac{3}{2} - 3 + 3\ln|2| - \frac{1}{2}$

$= 3\ln|2| - 2$

(43) $\int_0^{1/\sqrt{3}} \frac{t^2-1}{t^4-1} \cdot dt$

$= \int_0^{1/\sqrt{3}} \frac{(t^2-1)}{(t^4-1)} \cdot dt$

$= \int_0^{1/\sqrt{3}} \frac{(t^2-1)}{(t^2)^2-1^2} \cdot dt$

$= \int_0^{1/\sqrt{3}} \frac{(t^2-1)}{(t^2+1)\cdot(t^2-1)} \cdot dt$

$= \int_0^{1/\sqrt{3}} \frac{1}{t^2+1} \cdot dt$

$= \left[\tan^{-1} t\right]_0^{\frac{1}{\sqrt{3}}}$

$= \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) - \tan^{-1}(0)$

$= \frac{\pi}{6}$

(44) $\int_0^2 |2x-1| \cdot dx$

$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

$|2x-1| = \begin{cases} 2x-1 & \text{if } 2x-1 \geq 0, x \geq \frac{1}{2} \\ -(2x-1) & \text{if } 2x-1 < 0, x < \frac{1}{2} \end{cases}$

$= \int_0^{\frac{1}{2}} -(2x-1) \cdot dx + \int_{\frac{1}{2}}^2 (2x-1) \cdot dx$

$= -2\left[\frac{1}{2}x^2\right]_0^{\frac{1}{2}} - [x]_0^{\frac{1}{2}} + 2 \times \frac{1}{2}\left[x^2\right]_{\frac{1}{2}}^2 - [x]_{\frac{1}{2}}^2$

$= -\frac{1}{4} - \frac{1}{2} + \left[4 - \frac{1}{4}\right] - \left[2 - \frac{1}{2}\right]$

$= \frac{3}{2}$

(45) $\int_{-1}^{2}(x-2|x|)\,dx$

$= \int_{-1}^{0}(x+2x)\cdot dx + \int_{0}^{2}(x-2x)\cdot dx$

Let

$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

$x - 2|x| = \begin{cases} x-2x & \text{if } x-2x \geq 0,\ x=0 \\ x-2(-x) & \text{if } x-2x<0,\ x\neq 0 \end{cases}$

$= \left[\frac{1}{2}x^2 + 2\times\frac{1}{2}x^2\right]_{-1}^{0} + \left[\frac{1}{2}x^2 - \frac{1}{2}\times 2x^2\right]_{0}^{2}$

$= -\frac{1}{2} + \quad 0^2 - (-1)^2 + \frac{1}{2}(2^2 - (0)^2) - \quad 4 - (0)^2$

$= -\frac{1}{2} - 1 + 4/2 + (-4)$

$= -\frac{7}{2}$

(46) $\int_{0}^{3\pi/2} |\sin x|\,dx$

$= \int_{0}^{\pi} \sin x \cdot dx + \int_{\pi}^{3\pi/2} -\sin x \cdot dx$

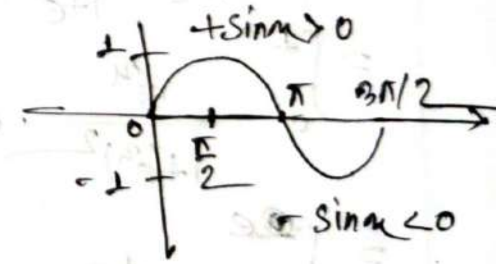
Let:—

$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

$|\sin x| = \begin{cases} \sin x & \text{if } \sin x \geq 0 \\ \sin(-x) & \text{if } \sin x < 0 \end{cases}$

$= -[\cos x]_{0}^{\pi} \quad + [\cos x]_{\pi}^{3\pi/2}$

$= -[\cos\pi - \cos 0] + \cos\frac{3\pi}{2} - \cos\pi$



$= -\cos\pi + \cos 0 + \cos\frac{3\pi}{2} - \cos\pi$

$= 3$

Integral by Substitution

For Definite integral:—

Page: 422:—

(18) $\int_0^2 y^2\sqrt{1+y^3}\cdot dy$

$= \int_1^9 \frac{1}{3}\cdot\sqrt{z}\cdot dz$

$= \frac{1}{3}\left[\frac{z^{\frac{1}{2}+1}}{\frac{1}{2}+1}\right]_1^9$

$= \frac{1}{3}\times\frac{2}{3}\left[9^{\frac{3}{2}} - 1^{\frac{3}{2}}\right]$

$= \frac{52}{9}$

Let:—

$z = 1+y^3$

$\frac{dz}{dy} = 3y^2$

$\frac{1}{3}\cdot dz = y^2\cdot dy$

y	2	0
z	9	1

(24) $\int_0^1 \frac{e^x}{1+e^{2x}}\cdot dx$

$= \int_0^1 \frac{e^x}{1+(e^x)^2}\cdot dx$

$= \int_1^e \frac{1}{1+z^2}\cdot dz$

$= \left[\tan^{-1} z\right]_1^e$

$= \tan^{-1} e - \tan^{-1} 1$

$= \tan^{-1} e + \frac{\pi}{4}$

Let:—

$z = e^x$

$\frac{dz}{dx} = e^x$

$dz = e^x\cdot dx$

x	1	0
z	e	1

(69) $\int_{e}^{e^4} \frac{dx}{x\sqrt{\ln x}}$

$= \int_{1}^{4} \frac{1}{z^{\frac{1}{2}}} \cdot dz$

$= \left[\frac{z^{-\frac{1}{2}+1}}{-\frac{1}{2}+1}\right]_{1}^{4}$

$= 2\left[4^{\frac{1}{2}} - 1\right]$

$= 2$

Let:-

$z = \ln x$

$\frac{dz}{dx} = \frac{1}{x}$

$dz = \frac{1}{x} \cdot dx$

x	e^4	e
z	4	1

(70.) $\int_{0}^{2} (x-1)\, e^{(x-1)^2} \cdot dx$

$= \int_{1}^{1} \frac{1}{2} e^{z} \cdot dz$

$= 0$

Let:-

$z = (x-1)^2$

$\frac{dz}{dx} = 2(x-1)$

$dz \frac{1}{2} = (x-1) \cdot dx$

x	2	0
y	1	1

Exercise:-

Page: 418: + 1 = 419:-

(71) (h.m) $\int_{0}^{1} \cos\left(\frac{\pi t}{2}\right) \cdot dt$

$= \int_{0}^{\frac{\pi}{2}} \cos z \;\; 2/\pi \cdot dz$

$= \frac{2}{\pi} \left[\sin z\right]_{0}^{\frac{\pi}{2}}$

$= \frac{2}{\pi} \left(\sin \pi/2 - \sin 0\right)$

$= \frac{2}{\pi}$

Let:-

$z = \frac{\pi t}{2}$

$\frac{dz}{dt} = \frac{\pi}{2}$

$dz \times \frac{2}{\pi} = dt$

t	1	0
z	$\frac{\pi}{2}$	0

(54) $\int_0^1 (3t-1)^{50}\cdot dt$

$= \int_{-1}^{2} z^{50} \times \frac{1}{3}\cdot dz$

$= \frac{1}{3}\left[\frac{1}{51}\cdot z^{51}\right]_{-1}^{2}$

$= \frac{1}{153}\left[2^{51}+1\right]$

Let:-

$z = 3t - 1$

$\frac{dz}{dt} = 3$

$\frac{1}{3}\cdot dz = dt$

t	1	0
z	2	-1

(55) $\int_0^1 \sqrt[3]{1+7x}\cdot dx$

$= \int_1^8 z^{\frac{1}{3}} \cdot \frac{1}{7}\cdot dz$

$= \frac{1}{7}\left[\frac{z^{\frac{1}{3}+1}}{\frac{1}{3}+1}\right]_1^8$

$= \frac{1}{7}\times\frac{3}{4}\left[z^{\frac{4}{3}}\right]_1^8$

$= \frac{3}{28}\left[8^{\frac{4}{3}}-1\right]$

$= \frac{45}{28}$

Let:-

$z = 1+7x$

$\frac{dz}{dx} = 7$

$\frac{1}{7}\times dz = dx$

x	1	0
z	8	1

(56) $\int_1^2 \frac{e^{\frac{1}{x}}}{x^2}\cdot dx$

$= -\int_1^{\frac{1}{2}} e^z\cdot dz$

$= -\left[e^{\frac{1}{2}} - e^1\right]$

$= e - e^{\frac{1}{2}}$

Let:-

$z = \frac{1}{x}$

$\frac{dz}{dx} = -\frac{1}{x^2}$

$-dz = \frac{1}{x^2}\times dx$

x	2	1
z	$\frac{1}{2}$	1

(60) $\int_0^1 xe^{-x^2} \cdot dx$

$= -\frac{1}{2}\int_0^{-1} e^{z} \cdot dz$

$= -\frac{1}{2}\left[e^{-1} - e^{0}\right]$

$= -\frac{1}{2}\left[\frac{1}{e} - 1\right]$

$= -\frac{1}{2}\left[\frac{1-e}{e}\right]$

$= \frac{e-1}{2e}$

Ans:-

Let

$z = -x^2$

$\frac{dz}{dx} = -2x$

$-\frac{1}{2}dz = x \cdot dx$

x	1	0
z	-1	0

(71) $\int_0^1 \frac{e^z+1}{e^z+z} \cdot dz$

$= \int_0^{e+1} \frac{1}{x} \cdot dx$

$= \left[\ln|x|\right]_0^{e+1}$

$= \ln|e+1|$

Let:-

$x = e^z + z$

$\frac{dx}{dz} = e^z + 1$

$dx = (e^z+1) \cdot dx$

z	1	0
x	$e+1$	0

Properties of Integrals

Some properties of Definite integrals:—

(1) if $f(x)$ is any integrable function then

$$\int_a^a f(x)\cdot dx = 0$$

(2) if $f(x)$ is integrable any interval containing three points a, b, c and $a<c<b$ then.

$$\int_a^b f(x)\cdot dx = \int_a^c f(x)\cdot dx + \int_c^b f(x)\cdot dx$$

(3) if $f(x)$ is an even function, that is $f(-x) = f(x)$ for all x then

$$\int_{-a}^a f(x)\cdot dx = 2\int_0^a f(x)\cdot dx$$

Ex:- $\int_{-2}^{2} x^2\cdot dx$

$= 2\int_0^2 x^2\cdot dx$

$= \frac{2}{3}[x^3]_0^2$

$= \frac{2}{3}[2^3 - 0]$

$= \frac{16}{3}$

Let

$f(x) = x^2$

$f(-x) = (-x)^2$

$= x^2$

$f(-x) = f(x)$

So, it's a even function

(4) if $f(x)$ is a odd function that is if $f(-x) = -f(x)$ for all x then :-

$$\int_{-a}^{a} f(x)\cdot dx = 0$$

Ex:- $\int_{-3}^{3} x^3 \cdot dx$

$= 0$

Det:-
$f(x) = x^3$
$f(-x) = (-x)^3$
$f(-x) = -x^3$
$f(-x) = -f(x)$
So, it's a odd function

Application of Properties:-

Page:- 400:-

(43) $\int_0^{\pi} f(x)\cdot dx$ where $f(x) = \begin{cases} \sin x & \text{if } 0 \le x \le \frac{\pi}{2} \\ \cos x & \text{if } \frac{\pi}{2} \le x \le \pi \end{cases}$

$= \int_0^{\frac{\pi}{2}} \sin x \cdot dx + \int_{\frac{\pi}{2}}^{\pi} \cos x \cdot dx$

$= -[\cos x]_0^{\frac{\pi}{2}} + [\sin x]_{\pi/2}^{\pi}$

$= -[\cos\frac{\pi}{2} - \cos 0] + [\sin\pi - \sin\pi/2]$

$= -\cos\frac{\pi}{2} + \cos 0 + \sin\pi - \sin\pi/2$

$= 0$

Page NO: 419:—

(61) $\int_{-\pi/4}^{\pi/4} (x^3+x^4\tan x)\cdot dx$

$= 0$

Let

$f(x) = x^3+x^4\tan x$.

$f(-x) = -x^3-x^4\tan x$

$= -(x^3+x^4\tan x)$

$= -f(x)$

So, It's a odd function

(66) $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} x^4\cdot\sin x\cdot dx$

$= 0$

Let:—

$f(x) = x^4\cdot\sin x$

$f(-x) = -(x^4\sin x)$

$= -f(x)$

So, It's a odd function.

Page NO: 422:—

(22) $\int_{-1}^{1} \frac{\sin x}{1+x^2}\cdot dx$

$= 0$

Let:—

$f(x) = \frac{\sin x}{1+x^2}$

$f(-x) = \frac{\sin(-x)}{1+(-x)^2}$

$= -\frac{\sin x}{1+x^2}$

$f(-x) = -f(x)$

So, It's a odd function

(23) $\int_{-\pi/4}^{\pi/4} \frac{t^4\tan t}{2+\cos t}\cdot dt$

$= 0$

Let,—

$f(x) = \frac{t^4\cdot\tan t}{2+\cos t}$

$f(-x) = \frac{(-t)^4\tan(-t)}{2+\cos(-t)}$

$= -\left(\frac{t^4\cdot\tan t}{2+\cos t}\right)$

$f(-x) = -f(x)$

Practice now:

$*\int_{-2}^{2}(2x^2+3)\cdot dx$

$= 2\int_{0}^{2}(2x^2+3)\cdot dx$

$= 2\left[2\times\frac{1}{3}x^3+3x\right]_0^2$

$= \left[4/3\,x^3+6x\right]_0^2$

$= \frac{4}{3}(2^3-0^3)+6\times 2$

$= \frac{68}{3}$

Let:-

$f(x) = 2x^2+3$

$f(-x) = 2(-x)^2+3$

$= 2x^2+3$

$f(-x) = f(x)$

So, it's a even function

$*\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\cos x\cdot dx$

$= 2\int_{0}^{\frac{\pi}{4}}\cos x\cdot dx$

$= 2\left[\sin x\right]_0^{\frac{\pi}{4}}$

$= 2\left[\sin\frac{\pi}{4}-\sin 0\right]$

$= \sqrt{2}$

Let:-

$f(x) = \cos x$

$f(-x) = \cos(-x)$

$= \cos x$

$f(-x) = f(x)$

So, it's a even function.

$*\int_{-2}^{2}x(1+x+x^2)\cdot dx$

$= \int_{-2}^{2}(x+x^2+x^3)\cdot dx$

$= \left[\frac{1}{2}x^2+\frac{1}{3}x^3+\frac{1}{4}x^4\right]_{-2}^{2}$

$= \frac{1}{2}(2^2-2^2)+\frac{1}{3}(2^3+2^3)+\frac{1}{4}(2^4-2^4)$

$= \frac{16}{3}$

Let:-

$f(x) = x+x^2+x^3$

$f(-x) = -x+x^2-x^3$

$= -(x-x^2+x^3)$

So it's a neither function

Riemann Sum And Trapezoidal rule

Riemann Sum:—

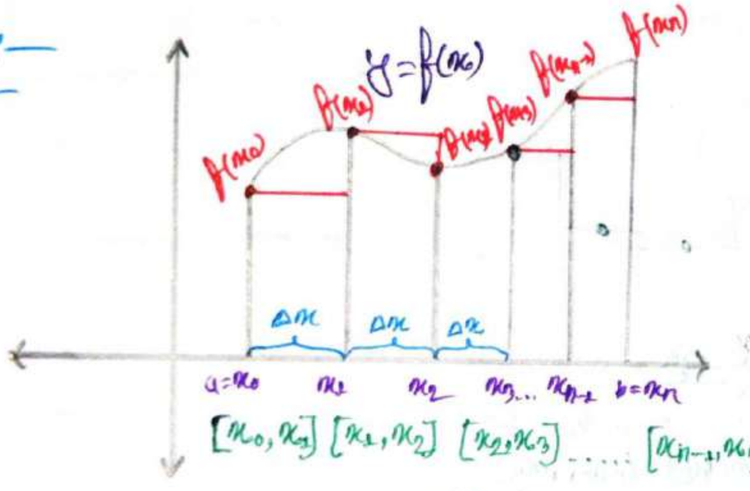
⇒ A Riemann Sum is a method for approximating the total area underneath a curve on a graph.

⇒ This method is also known as taking an integral

⇒ There are 3 forms of Riemann Sums:—

1. Left
2. Right
3. Middle

Left:—



$$\Delta x = \frac{b-a}{n}$$

b = the upper limit

a = the lower limit

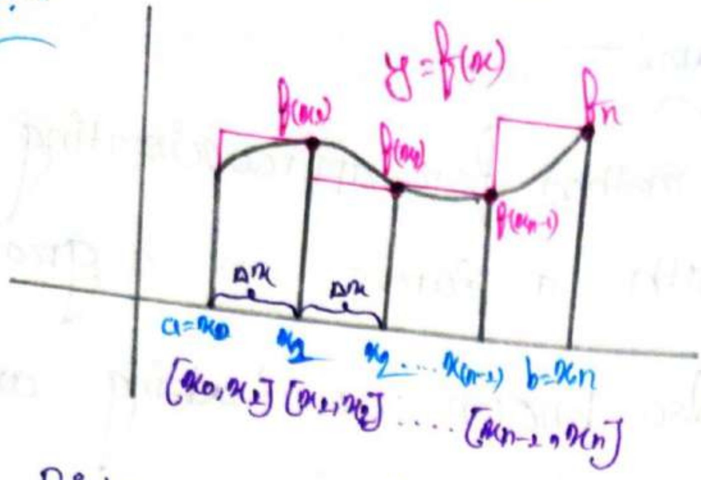
n = the number of subintervals.

$\therefore$ Left $R_S = \Delta x * f(x_0) + \Delta x * f(x_1) + \Delta x * f(x_2) + \Delta x f(x_3) + \cdots\cdots + \Delta x * f(x_{n-1})$

$$\boxed{R_S = \Delta x\left(f(x_0) + f(x_1) + f(x_2) + f(x_3) + \cdots\cdots + f(x_{n-1})\right)}$$

$$R_S = \Delta x \sum_{r=0}^{n-1} f(x_r)$$

Right:-

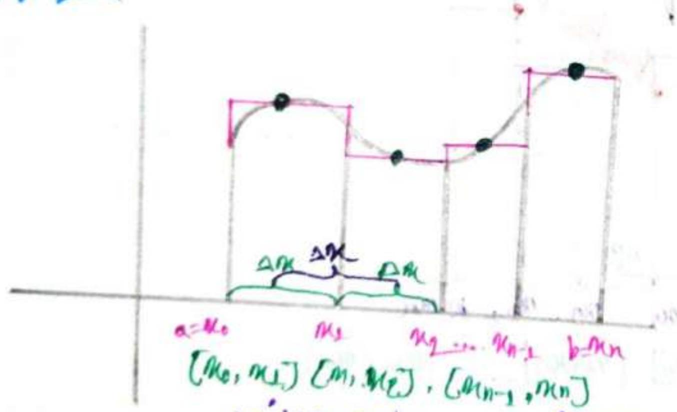


Right $R_S = \Delta x * f(x_1) + \Delta x * f(x_2) + \cdots\cdots + \Delta x f(x_n)$

$$\boxed{R_S = \Delta x \left(f(x_1) + f(x_2) + \cdots\cdots + f(x_n) \right)}$$

$$R_S = \Delta x \sum_{r=1}^{n} f(x_r)$$

Middle:-



middle point $= \dfrac{x_0 + x_1}{2}$

*Let a function $f(x)$ is defined in the closed interval $[a,b]$. In evaluation of Réemann Sum we commonly use equal Subintervals, Dividing $[a,b]$ into n equal subintervals of the length

$$\Delta x = \frac{b-a}{n}$$

Réemann Sum of $f(x)$ Over the interval $[a,b]$ is R_s or S_n

$$R_s/S_n = \sum_{k=1}^{n} f(x_k) * \Delta x = \Delta x \sum_{k=1}^{n} f(x_k)$$

For all k	The Sum R_s will be called
$x_k = c_{k-1}$	Left Réemann Sum
$x_k = c_k$	Right Réemann Sum
$x_k = \frac{c_k + c_{k-1}}{2}$	Middle Réemann Sum.

Q:- Find the Area Under the Curve $f(x) = x^4 - 3x^2 + 3$ by Using diffrent Riemann Sum Over the interval $[0, 1.6]$ using 8 subinterval.

$\Rightarrow$ is given:—

$$f(x) = x^4 - 3x^2 + 3$$

$$a = 0$$

$$b = 1.6$$

$$n = 8$$

$$\therefore \Delta x = \frac{b-a}{n} = \frac{1.6-0}{8} = 0.2$$

Sub Interval	Left Riemann Sum		Right Riemann Sum		Middle Riemann Sum	
	x_r	$f(x_r)$	x_r	$f(x_r)$	x_r	$f(x_r)$
$[0, 0.2]$	0	3	0.2	2.8816	0.1	2.9701
$[0.2, 0.4]$	0.2	2.8816	0.4	2.5456	0.3	2.7381
$[0.4, 0.6]$	0.4	2.5456	0.6	2.0496	0.5	2.3125
$[0.6, 0.8]$	0.6	2.0496	0.8	1.4896	0.7	1.7701
$[0.8, 1.0]$	0.8	1.4896	1.0	1.000	0.9	1.2261
$[1.0, 1.2]$	1.0	1.0000	1.2	0.7536	1.1	0.8341
$[1.2, 1.4]$	1.2	0.7536	1.4	0.9616	1.3	0.7861
$[1.4, 1.6]$	1.4	0.9616	1.6	1.8736	1.5	1.3125
$\sum f(x_r)$	14.6816		13.5552		13.8805	

∴ Area for left Riemann Sum :—

$$R_s = \Delta x \sum_{r=1}^{n} f(x_{r-1})$$

$$= 0.2 \times 14.6816$$

$$= 2.93632$$

∴ Area for Right Riemann Sum :—

$$R_s = \Delta x \cdot \sum_{r=1}^{n} f(x_r)$$

$$= 0.2 \times 13.5552$$

$$= 2.71104$$

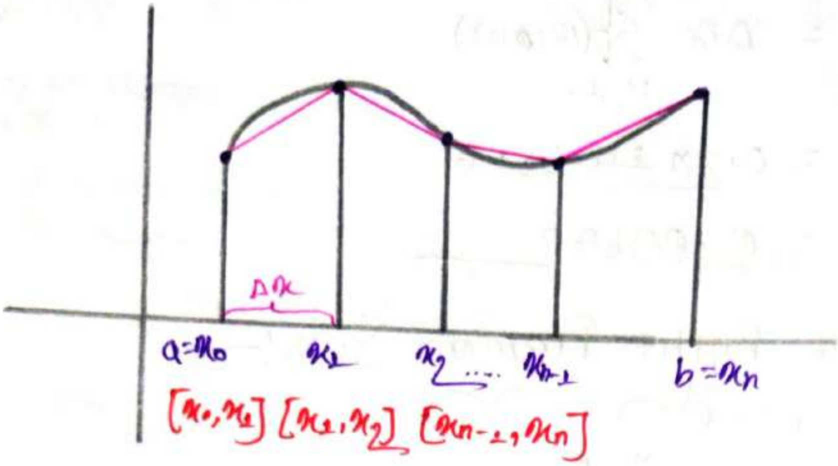
∴ Area for Middle Riemann Sum :—

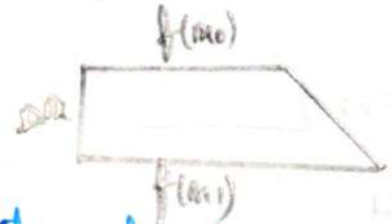
$$R_s = \Delta x \sum_{r=1}^{n} f(x_r)$$

$$= 0.2 \times 13.8805$$

$$= 2.7761$$

Trapezoidal rule:





$S_1 = \frac{1}{2} \times \Delta x \times [f(x_0) + f(x_1)]$

$S_2 = \frac{1}{2} \times \Delta x \times [f(x_1) + f(x_2)]$

$S_3 = \frac{1}{2} \times \Delta x \times [f(x_2) + f(x_3)]$

$S_4 = \frac{1}{2} \times \Delta x \times [f(x_3) + f(x_4)]$

$S_{n-1} = \frac{1}{2} \times \Delta x \times [f(x_{n-2}) + f(x_{n-1})]$

$S_n = \frac{1}{2} \times \Delta x \times [f(x_{n-1}) + f(x_n)]$

$\therefore S = S_1 + S_2 + S_3 + S_4 + \cdots\cdots + S_{n-1} + S_n$

$= \frac{1}{2} \times \Delta x [f(x_0) + f(x_1) + f(x_1) + f(x_2) + f(x_2) + f(x_3) + \cdots\cdots + f(x_{n-2}) + f(x_{n-1}) + f(x_{n-1}) + f(x_n)]$

$$\boxed{S = \frac{1}{2} \times \Delta x [f(x_0) + f(x_n) + 2 \times \{f(x_1) + f(x_2) + f(x_3) + \cdots + f(x_{n-1})\}]}$$

Let $f(x)$ be a Continuas on $[a,b]$ The trapezoidal Rule for Approximation $\int_a^b f(x)\cdot dx$ is given by

$$T_s/S_n = \frac{b-a}{2n}\left[f(a)+2f(x_1)+2f(x_2)+\cdots\cdots+f(x_n)\right]$$

$$\boxed{T_s/S_n = \frac{1}{2}\Delta x\left[y_0+2(y_1+y_2+y_3+\cdots\cdots+y_{n-1})+y_n\right]}$$

Q:- Use the trapezoidal rule with $n=5$ to approximate the integral $\int_{0.5}^{1}\sqrt{1+e^{x^2}}\cdot dx$ to 3 decimal places.

$\Rightarrow$ is, given:—

$b=1$
$a=0.5$
$n=5$
$f(x)=\sqrt{1+e^{x^2}}$

0.5 0.6 0.7 0.8 0.9 1
[0.5,0.6] [0.6,0.7] [0.7,0.8] [0.8,0.9] [0.9,1]

$\therefore \Delta x = \frac{b-a}{n} = \frac{1-0.5}{5} = 0.1$

x	0.5	0.6	0.7	0.8	0.9	1.0
$f(x)$	1.5113	1.5560	1.6225	1.7020	1.8022	1.9283

using trapezoidal rule:—

$$T_s/S_n = \frac{1}{2}\times\Delta x\times\left[f(0.5)+f(1)+2\left(f(0.6)+f(0.7)+f(0.8)+f(0.9)\right)\right]$$
$$= \frac{1}{2}\times 0.1\times\left[1.5113+1.9283+2\{1.5560+1.6225+1.7020+1.8022\}\right]$$
$$= \frac{1}{2}\times 0.1\times(1.5113+1.9283+13.405)$$
$$= 0.84223 = 0.842$$

Also Find the left, Right and middle Reemann Sum of the Question?

$\Rightarrow$ is given:—

$$b = 1,\quad a = 0,\quad n = 5 \;;\; f(x) = \sqrt{1+e^{x^2}} \;;\; \Delta x = \frac{b-a}{5} = \cdot 1$$

Left Reemann Sum:—

$$R_S = \sum_{n=1}^{n} x_n \times \Delta x$$
$$= (1\cdot 5113 + 1\cdot 5560 + 1\cdot 6225 + 1\cdot 7020 + 1\cdot 8022) \times \Delta x$$
$$= 8\cdot 194 \times 0\cdot 1$$
$$= 0\cdot 819$$

Right Reemann Sum:—

$$R_S = \Delta x \sum_{n=1}^{n} f(x_n)$$
$$= \Delta x \times (1\cdot 5560 + 1\cdot 6225 + 1\cdot 7020 + 1\cdot 8022 + 1\cdot 9283)$$
$$= 0\cdot 1 \times 8\cdot 611$$
$$= 0\cdot 861$$

middle Reemann Sum:—

x_n	$0\cdot 55$	$\cdot 65$	$0\cdot 75$	$0\cdot 85$	$\cdot 95$
$f(x_n)$	$1\cdot 5341$	$1\cdot 5890$	$1\cdot 6599$	$1\cdot 7492$	$1\cdot 8617$

$$\therefore R_S = \Delta x \times (1\cdot 5341 + 1\cdot 5890 + 1\cdot 6599 + 1\cdot 7492 + 1\cdot 8617)$$
$$= 0\cdot 1 \times 8\cdot 3939$$
$$= 0\cdot 839$$

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① Evaluate the Riemann sum for $f(x) = x-1$, $-6 \le x \le 4$, with five subintervals, taking the sample to be right end points.

⇒ is given,

$$a = -6 \qquad n = 5$$
$$b = 4$$
$$f(x) = x - 1$$

$$\therefore \Delta x = \frac{b-a}{n}$$
$$= \frac{4-(-6)}{5}$$
$$= 2$$

[-6, -4] [-4, -2] [-2, 0] [0, 2] [2, 4]

x_r	-4	-2	0	2	4		
$f(x_r)$	-5	-3	-1	1	3		

$$\therefore RRS = \Delta x \times (-5-3-1+1+3)$$
$$= 2(-5)$$
$$= -10$$

(3) If $f(x) = x^2 - 4$; $0 \le x \le 3$ Find the Riemann Sum with $n=6$. Taking the sample point to be midpoints.

$\Rightarrow$ is given:—

$$a = 0 \qquad f(x) = x^2 - 4$$
$$b = 3$$
$$n = 6$$

$$\Delta x = \frac{b-a}{n} = \frac{3-0}{6} = 0.5$$

0 0.5 1.0 1.5 2.0 2.5 3.0

[0, 0.5] [0.5, 1.0] [1.0, 1.5] [1.5, 2.0] [2.0, 2.5] [2.5, 3.0]

x_r	0.25	0.75	1.25	1.75	2.25	2.75
$f(x_r)$	-3.9375	-3.4375	-2.4375	-0.9375	1.0625	3.5625

$\therefore MRS = \Delta x \times (-3.9375 - 3.4375 - 2.4375 - 0.9375 + 1.0625 + 3.5625)$

$= (-6.155) \times 0.5$

$= -3.0775$

(7.) A table of Values of an increasing function f is shown. Use the table to Find the lower and upper estimates for $\int_{10}^{30} f(x) \cdot dx$.

x	10	14	18	22	26	30
$f(x)$	-12	-6	-2	1	3	8

$\Rightarrow$ in given:-

$a = 10$
$b = 30$
$\Delta x = \frac{30-10}{5}$
$= 4$

$\therefore LRs = \Delta x \times (-12 - 6 - 2 + 1 + 3)$
$= 4 \times -16$
$= -64$

$RRs = \Delta x (-6 - 2 + 1 + 3 + 8)$
$= 4 \times 4$
$= 16$

(*) Use the midpoint Rule with the given value of n to approximate the integral. Round the answer to 4 decimal places.

(i) $\int_0^8 \sin\sqrt{x} \cdot dx \; ; \; n=4$

$\Rightarrow$ is given:—

$a = 0$
$b = 8$
$n = 4$
$f(x) = \sin\sqrt{x}$

$\Delta x = \frac{b-a}{n}$
$= \frac{8-0}{4}$
$= 2$

0 2 4 6 8

[0,2] [2,4] [4,6] [6,8]

x_n	1	3	5	7	
$f(x)$	0.0175	0.0303	0.0340	0.0462	

$\therefore MP_s = \Delta x \times [f(1) + f(3) + f(5) + f(7)]$
$= 2 \times (0.0175 + 0.0303 + 0.0340 + 0.0462]$
$= 2 \times (0.128)$
$= 0.256$

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(7.) $\int_1^2 \sqrt{x^3-1}\cdot dx$ $n=4$ Find the LRS, RRS, MRS, TR

⇒ is given:—

$b=2$

$a=1$

$f(x)=\sqrt{x^3-1}$

$n=4$

$\therefore \Delta x = \frac{b-a}{n} = \frac{2-1}{4} = 0.25$

1 1.25 1.5 1.75 2

[1, 1.25] [1.25, 1.5], [1.5, 1.75], [1.75, 2]

Sub Intervals	LRS		RRS		MRS		TR	
	x_n	$f(x_n)$	x_n	$f(x_n)$	x_n	$f(x_n)$	x	$f(x)$
[1, 1.25]	1	0	1.25	0.076	1.125	0.651	1	0
[1.25, 1.5]	1.25	0.076	1.5	1.541	1.375	1.264	1.25	0.076
[1.5, 1.75]	1.5	1.541	1.75	2.087	1.625	1.814	1.5	1.541
[1.75, 2]	1.75	2.087	2	2.645	1.875	2.364	1.75	2.087
							2	2.645
$\sum f(x_n)$	4.604		7.249		6.093		7.249	
$R_S=\Delta x \sum f(x_n)$	1.151		1.812		1.523			

$TR = \Delta x \times \frac{1}{2} \times \{f(1)+f(2)+2[f(1.25)+f(1.5)+f(1.75)]\}$

$= .25 \times \frac{1}{2} \times 11.853$

$= 1.481$

(8) $\int_0^2 \frac{1}{1+x^6} \cdot dx$; $n=4$ Find LRS, RRS, MRS, TR

$\Rightarrow$ in given:—

$b = 2$

$a = 0$

$n = 4$

$f(x) = \frac{1}{1+x^6}$

$\therefore \Delta x = \frac{b-a}{n}$

$= \frac{2-0}{4}$

$= 0.5$

0   0.5   1.0   1.5   2

[0, 0.5] [0.5, 1.0] [1.0, 1.5] [1.5, 2]

Sub Interval	LRS		MRS		RRS	
	x_R	$f(x_R)$	x_R	$f(x_R)$	x_R	$f(x_R)$
[0, 0.5]	0	1	0.25	0.999	0.5	0.984
[0.5, 1.0]	0.5	0.984	0.75	0.848	1.0	0.5
[1.0, 1.5]	1.0	0.5	1.25	0.2076	1.5	0.080
[1.5, 2]	1.5	0.080	1.75	0.033	2	0.015
$\sum f(x_R)$	2.564		2.0876		1.579	
$RS = \Delta x \sum f(x_R)$	1.282		1.0438		0.7895	

For: TR:—

x	0	0.5	1.0	1.5	2
$f(x)$	1	0.984	0.6	0.080	0.015

$\therefore TR = \frac{1}{2} \times \Delta x \times [f(0) + f(2) + \{2 \times (f(0.5) + f(1.0) + f(1.5))\}]$

$= \frac{1}{2} \times .5 \times [1 + 0.015 + 2 \times (0.984 + 0.5 + 0.080)]$

$= 1.03575$

(1) $\int_0^2 \frac{e^x}{1+x^2} \cdot dx$ $n=4$, find LRS, RRS, MRS, TR

⇒ is given:—

$a = 0$

$b = 2$

$n = 4$

$f(x) = \frac{e^x}{1+x^2}$

$\therefore \Delta x = \frac{b-a}{n} = \frac{2}{4} = 0.5$

$0 \quad 0.5 \quad 1.0 \quad 1.5 \quad 2$

For TR:—

x	0	0.5	1.0	1.5	2
$f(x)$	1	1.31	1.3591	1.37890	1.4778

$TR = \frac{1}{2} \Delta x \times (f(0) + f(2) + 2 \times \{f(0.5) + f(1) + f(1.5)\})$

$= \frac{1}{2} \times 0.5 \times (10.5738) = 2.64345$

For LRS:—

$$LRS = \Delta x \sum f(x_n)$$
$$= 0.5 \times (1 + 1.31 + 1.3591 + 1.3787)$$
$$= \underline{2.524}$$

For RRS:—

$$RRS = \Delta x \sum f(x_n)$$
$$= 0.5 \times (1.31 + 1.3591 + 1.3787 + 1.4778)$$
$$= \underline{2.7628}$$

For MRS:—

x_n	0.25	0.75	1.25	1.75
$f(x_n)$	1.2084	1.3548	1.3620	1.4165

$$MRS = \Delta x \times (f(.25) + f(0.75) + f(1.25) + f(1.75))$$
$$= 0.5 \times (1.2084 + 1.3548 + 1.3620 + 1.4165)$$
$$= 0.5 \times 5.3417$$
$$= \underline{2.67085}$$

lecture : 4

\# Estimate $\int_3^5 \frac{1}{1-\ln x}$ using

① Left Riemann Sum

② Right Riemann Sum

③ Middle - Riemann sum

④ Trapezoidal Rule with $n=4$ Subintervals.

⇒ is given :—

$$b=5 \qquad n=4$$
$$a=3 \qquad f(x)=\frac{1}{1-\ln x}$$

$$\therefore \Delta x=\frac{b-a}{n}$$
$$=\frac{5-3}{4}$$
$$=0.5$$

④

[3, 3.5] [3.5, 4] [4, 4.5] [4.5, 5]

3 3.5 4 4.5 5

x	3	3.5	4	4.5	5
$f(x)$	-10.14	-3.75	-2.58	-1.78	-1.64

$$\therefore TR=\frac{1}{2}\times\Delta x\times[f(3)+f(5)+\{2\times(f(3.5)+f(4)+f(4.5))\}]$$
$$=\frac{1}{2}\times0.5\times[-10.14+(-1.64)+2\{-3.75-2.58-1.78\}]$$
$$=\frac{1}{2}\times0.5\times[-11.78-17.02]$$
$$=-7.2$$

1

$$LRs = \Delta x \times (f(3) + f(3.5) + f(4) + f(4.5))$$
$$= 0.5 \times (-10.14 - 3.75 - 2.58 - 1.78)$$
$$= -9.325$$

2

$$RRs = \Delta x \times (f(3.5) + f(4) + f(4.5) + f(5))$$
$$= 0.5 \times (-3.75 - 2.58 - 1.78 - 1.64)$$
$$= -5.075$$

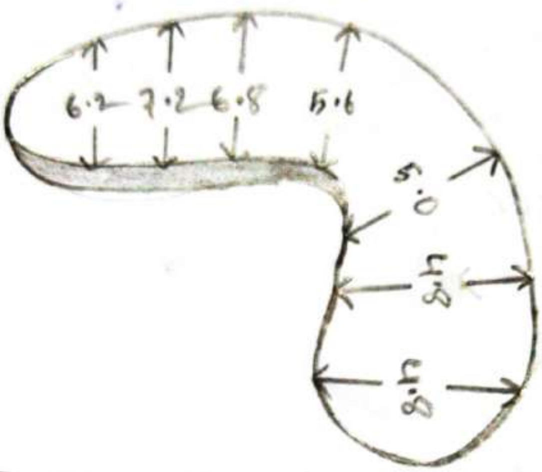
3

m_k	3.25	3.75	4.25	4.75	
$f(m_k)$	-5.59	-3.10	-2.23	-1.79	

$$MRs = \Delta x \times \{f(3.25) + f(3.75) + f(4.25) + f(4.75)\}$$
$$= 0.5 \times \{-5.59 - 3.10 - 2.23 - 1.79\}$$
$$= -6.355$$

Page: [illegible]:435:—

(48) The width (in meters) of a kidney-shaped swimming pool were measured at 2-meter intervals as indicated in the figure. Use the midpoint rule to estimate the area of the pool.



⇒ is given:—

$\Delta x = 2$

we know:—

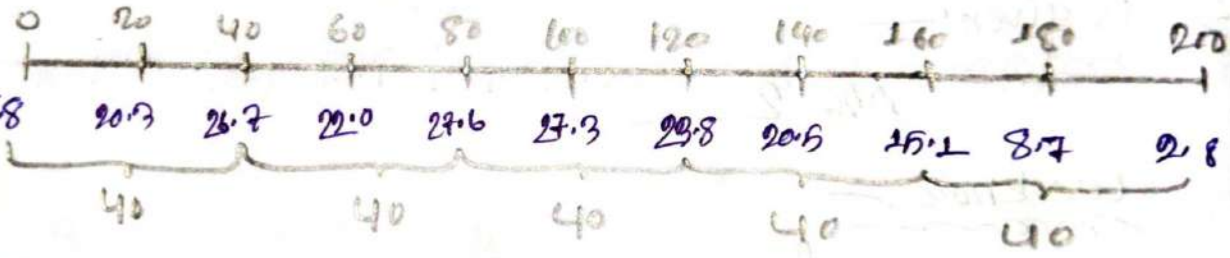
$MR_8 = \Delta x \sum f(x_k)$

$= 2 \times (6.2 + 7.2 + 6.8 + 5.6 + 5.0 + 4.8 + 4.8)$

$= 80.8 m^2$

(40.) A cross-Section of an airplane wing is shown. Measurements of the thickness of the wing, in centimeters, at 20 Centimeters intervals are 5.8, 20.3, 26.7, 29.0, 27.6, 27.3, 23.8, 20.5, 15.1, 8.7, and 2.8. Use midpoint Rule to estimate the area of the wing's cross-section.

⇒ is given:-



here:-

$a = 0$

$b = 200$

$n = 5$

$\therefore \Delta x = \dfrac{200-0}{5} = 40$

We know:-

$M_{RS} = \Delta x \sum f(x_R)$

$= 40 \times (20.3 + 29.0 + 27.3 + 20.5 + 8.7)$.

$= 4224 \text{ cm}^2$

APPLICATIONS of Integration

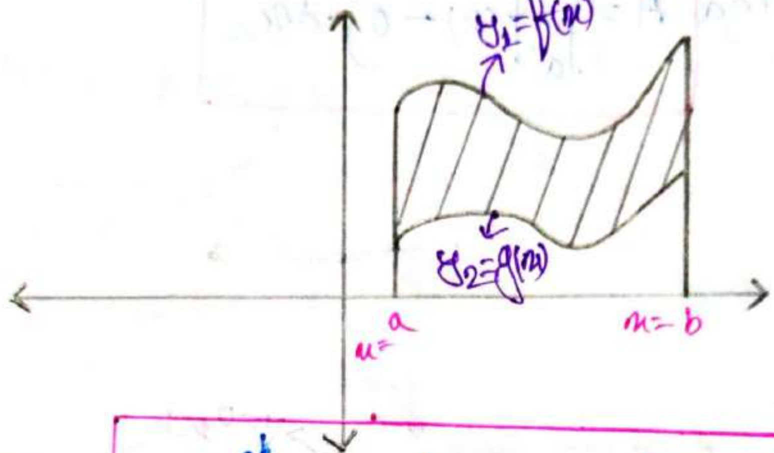
Ch no. 6

Applications of Integration:—

For Definite Integral:—

Area Between two Curves:—

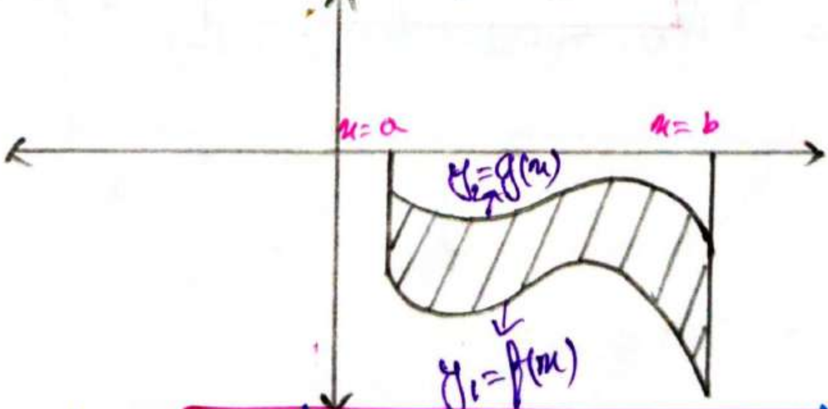
(1)



$\therefore$ Area $\boxed{A = \int_a^b (\text{upper function} - \text{lower function}) \cdot dx}$

$$\therefore A = \int_a^b \left(f(x) - g(x)\right) \cdot dx$$

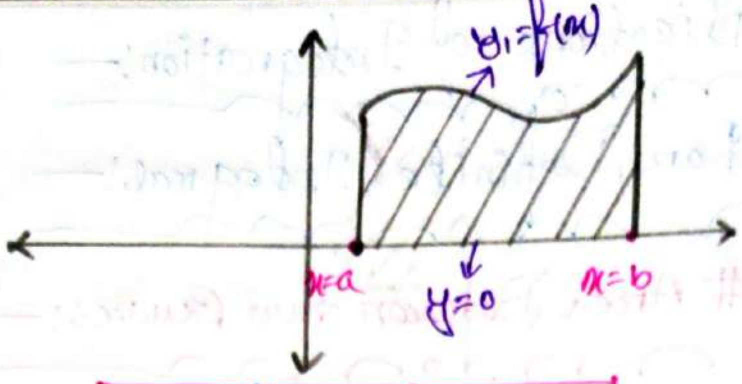
(2)



$\therefore$ Area $\boxed{A = \int_a^b \left|(\text{upper functio}) - (\text{lower function})\right| \cdot dx}$

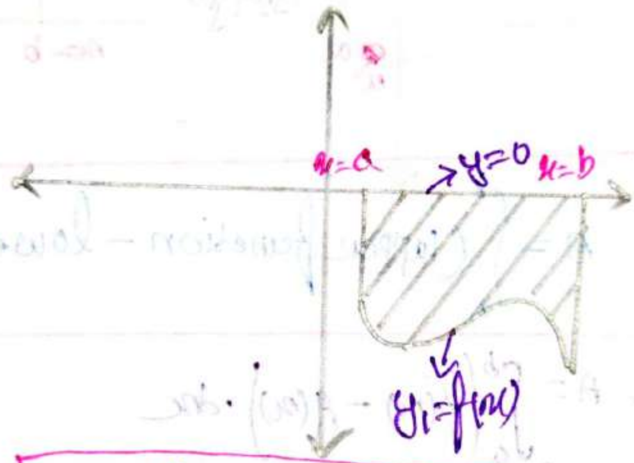
$$\therefore A = \int_a^b \left|g(x) - f(x)\right| \cdot dx$$

(3)



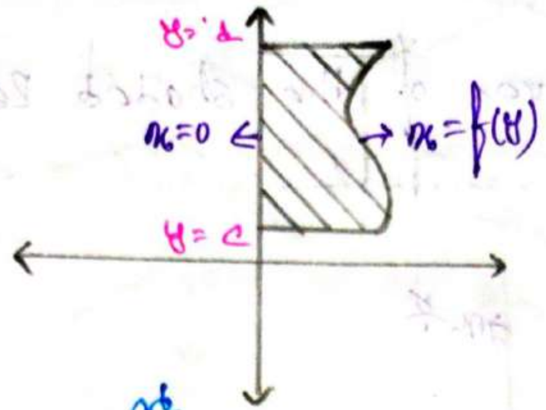
$\therefore$ Area $\boxed{A = \int_a^b [f(x) - 0] \cdot dx}$

(4)



$\therefore$ Area $\boxed{A = \int_a^b |-f(x)| \cdot dx}$

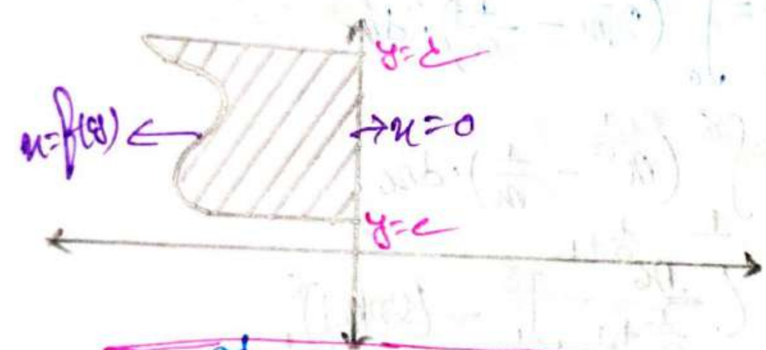
(5)



$\therefore$ Area $\cdot A = \int_c^d (f(y) - 0) \cdot dy$

$$\boxed{A = \int_c^d [(\text{Right function}) - (\text{Left function})] \; dy}$$

(6)



$\therefore$ Area $\boxed{A = \int_c^d |(\text{Right function}) - (\text{left function})| \cdot dy}$

$A = \int_c^d |-f(y)| \cdot dy$

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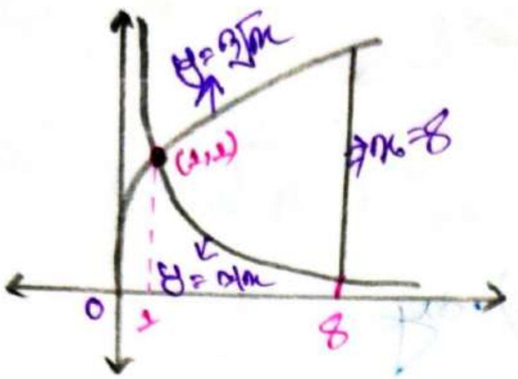
Q. Find the the area of the shaded Region:—

(1.)



⇒ is given:—

$a=1$
$b=8$

we know:—

Area $A=\int_a^b \left(\sqrt[3]{x}-\frac{1}{x}\right)\cdot dx$

$=\int_1^8 \left(x^{\frac{1}{3}}-\frac{1}{x}\right)\cdot dx$

$=\left[\frac{x^{\frac{1}{3}+1}}{\frac{1}{3}+1}\right]_1^8-\left[\ln|x|\right]_1^8$

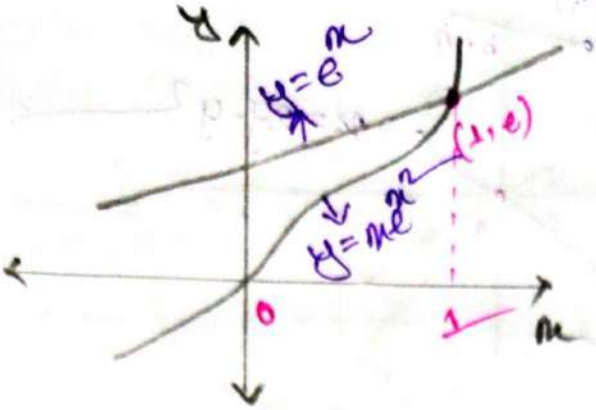
$=\frac{3}{4}\left[x^{\frac{4}{3}}\right]_1^8-(\ln 8-\ln 1)$

$=\frac{3}{4}\left[8^{\frac{4}{3}}-1^{\frac{4}{3}}\right]-\ln 8$

$=\frac{45}{4}-\ln 8$

$=9.17$

2



⇒ is given:—

$b = 1$

$a = 0$

we know:—

$$\text{Area } A = \int_a^b (e^x - xe^{x^2}) \cdot dx$$

$$\therefore A = \int_0^1 (e^x - xe^{x^2}) \cdot dx$$

$$= \int_0^1 e^x \cdot dx - \int_0^1 xe^{x^2} \cdot dx$$

$$= [e^x]_0^1 - \frac{1}{2}[e^z]_0^1$$

$$= e^1 - e^0 - [\tfrac{1}{2}e^1 - e^0]$$

$$= e^1 - e^0 - \frac{e^1}{2} + e^0$$

$$= 1.3591$$

Let:—

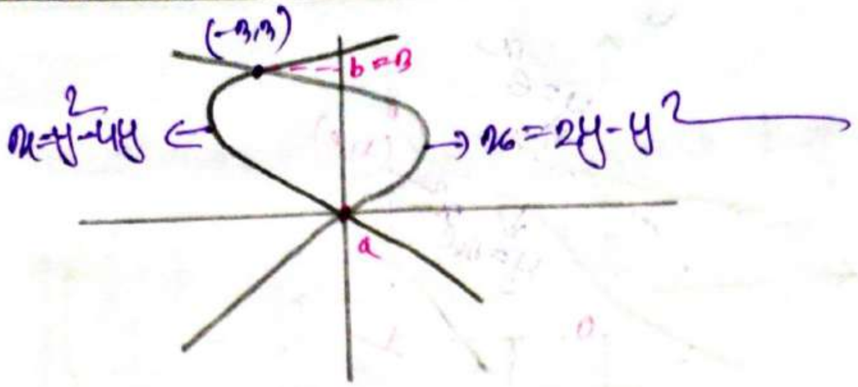
$z = x^2$

$\frac{dz}{dx} = 2x$

$dz \times \frac{1}{2} = x \cdot dx$

x	1	0
z	1	0

(4.)



$\Rightarrow$ is given:-

$b=3$
$a=0$

We know:-

Area $A=\int_a^b [(2y-y^2)-(y^2-4y)]\cdot dy$

$A=\int_0^3 (2y-y^2-y^2+4y)\cdot dy$

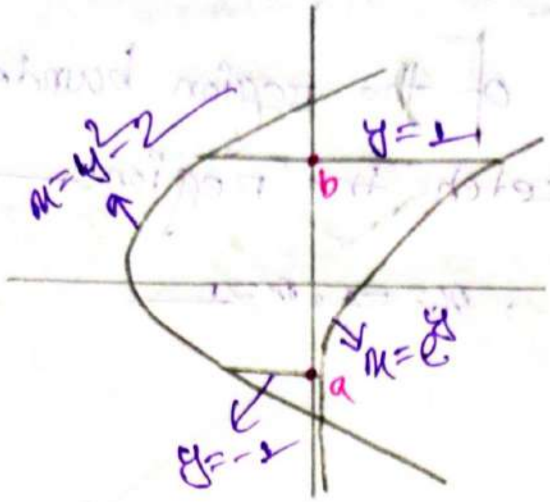
$=\int_0^3 [-2y^2+6y]\cdot dy$

$=-2\times\frac{1}{2+1}[y^3]_0^3+6\times\frac{1}{2}[y^2]_0^3$

$=-\frac{2}{3}[3^3-0^3]+3[3^2-0^3]$

$=9$

③



is given:-

$b=1$

$a=-1$

we know

$$\text{Area } A=\int_a^b (e^y-y^2+2)\cdot dy$$

$$\therefore A=\int_{-1}^{1}(e^y-y^2+2)\cdot dy$$

$$=[e^y]_{-1}^{1}-\frac{1}{3}[y^3]_{-1}^{1}+2[y]_{-1}^{1}$$

$$=e^1-e^{-1}-\frac{1}{3}(1+1)+2\times(1+1)$$

$$=e^1-\frac{1}{e^1}-\frac{2}{3}+4$$

$$=5.6837$$

Function:

$\Rightarrow$ Area, $A = \pi r^2$

$A \propto r^2$

$r = 1$; $A = 1$

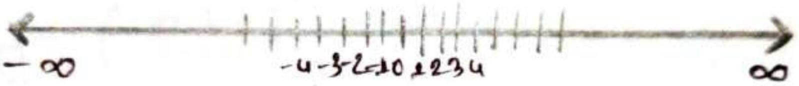
$r = 2$; $A = 4$

$f(x) = x^2$

$f(t) = t^2$

$f(z) = z^2$

Number line:



$[1, 4] \rightarrow 1 \leq x \leq 4$

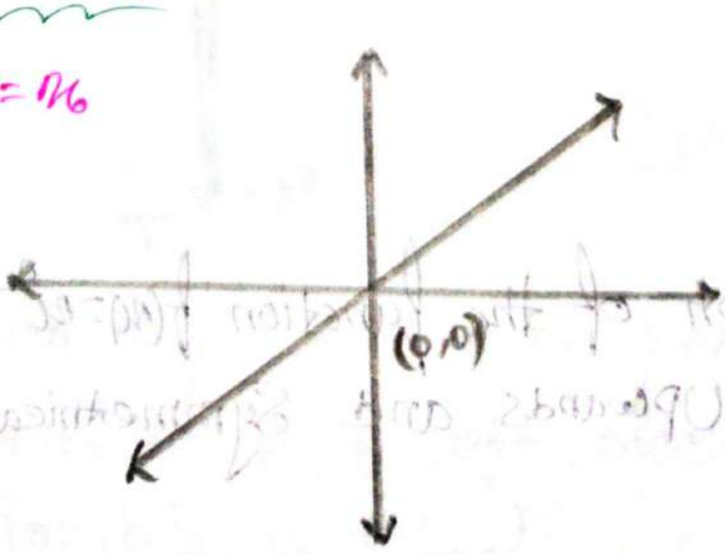
$(1, 4] \rightarrow 1 < x \leq 4$

$[1, 4) \rightarrow 1 \leq x < 4$

Sketch the graph:

Power function:

1: $y = f(x) = x$



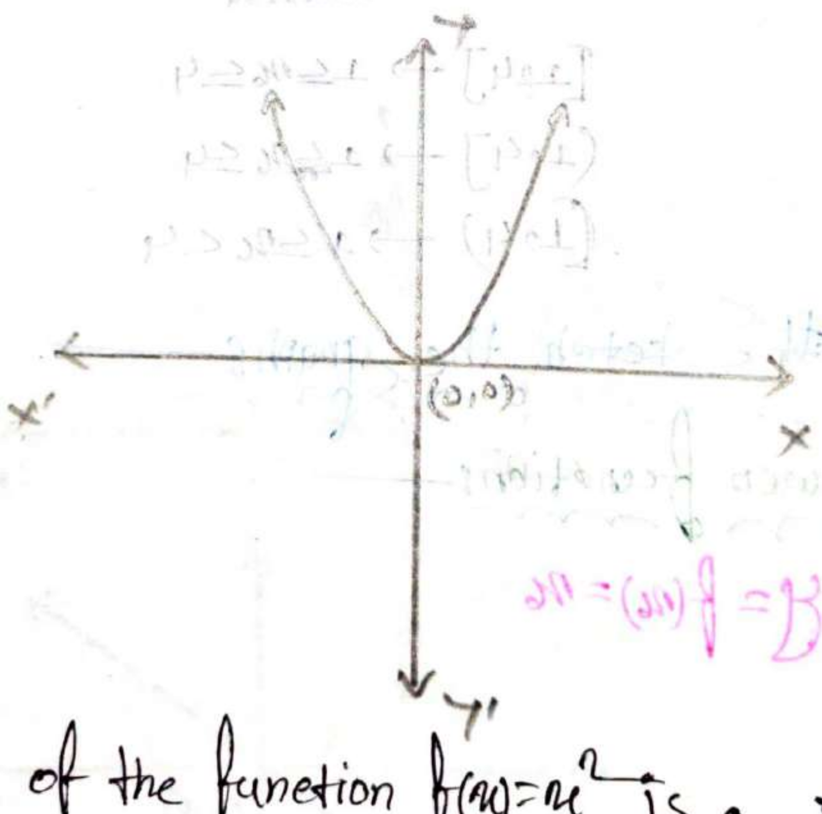
⇒ The graph of the function $f(x) = x$ is a Straight line, that passes through the Origin (0,0) and has a Slope of 1.

if ⇒ $x = 1$ then $f(x) = 1$

$x = -1$ " $f(x) = -1$

the line represents a one to one relationship between x and $f(x)$, meaning that for every value of x there is a Unique value of $f(x)$, and vice versa

② $y = x^2$



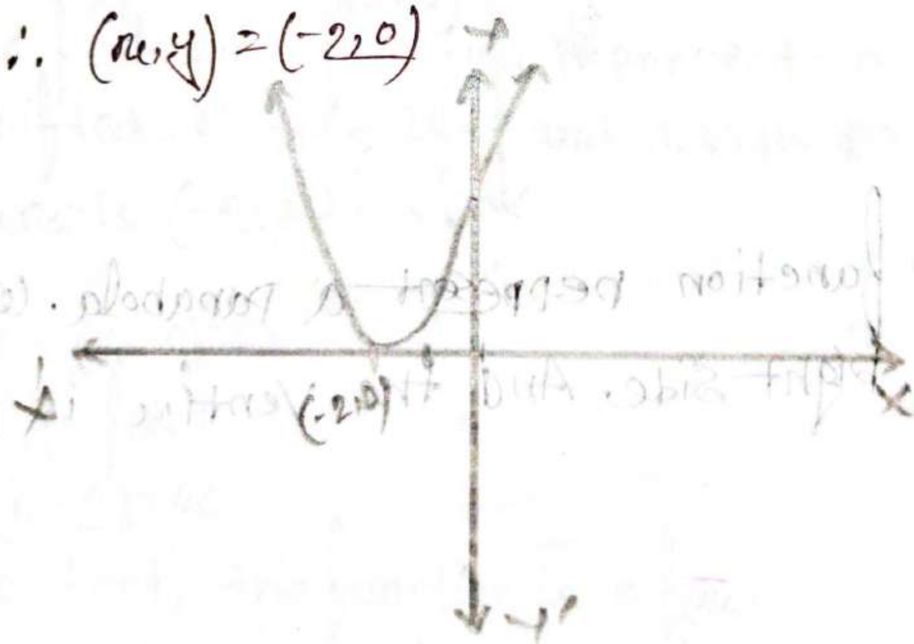
⇒ The graph of the function $f(x) = x^2$ is a Parabola. that opens upwards and Symmetrical about the x-axis.

The Vertex of the Parabola is at the origin $(0,0)$
Example

Q:- What is the name of following functions and sketch it?

** $y=(x+2)^2$

Let, $y=f(x)$
and $x+2=x_0$
So that, function is $f(x_0)=x_0^2$
if $x_0=0$
then $x+2=0$
$x=-2$
and $y=(-2+2)^2$
$=0$
$\therefore (x,y)=(-2,0)$



This given function represent a Parabola which is shifted two units on left side. and the Vertex of the Parabola is $(-2,0)$.

* $y=(x-2)^2$

$\Rightarrow$ Let $y=f(x)$

$(x-2)=x$

then, the function is $f(x)=x^2$

if $x=0$

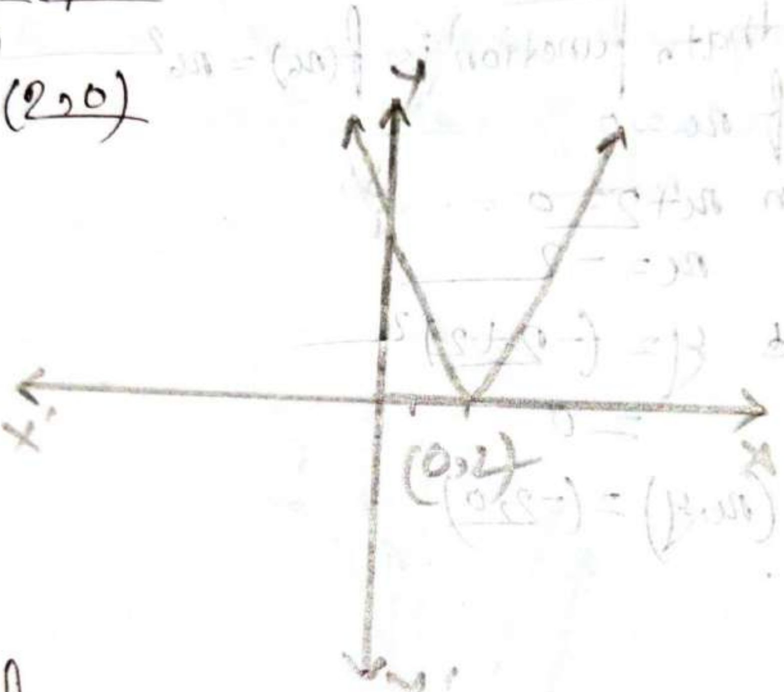
so $x-2=0$

or $x=2$

and $y=(2-2)^2$

$=0$

So $(x,y)=(2,0)$



$\Rightarrow$ this given function ~~represent~~ a parabola. Which is shifted 2 Units on right side. And the vertix is $(2,0)$

* $y=(x+2)^2+1$

$\Rightarrow$ Let, $y=f(x)$

$(x+2)=x$

then the function is $f(x)=x^2+1$

if $x = 0$

So $x+2=0$

or $x=-2$

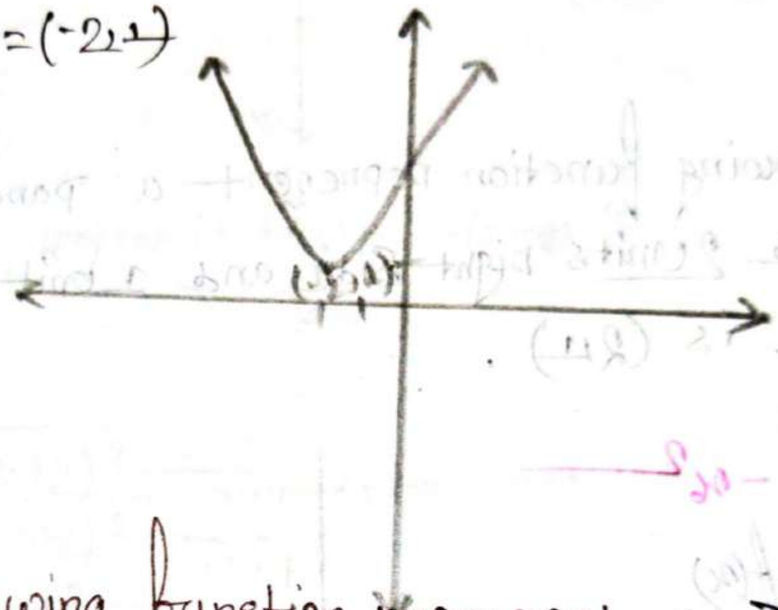
and $y = (x+2)^2+1$

$= (-2+2)^2+1$

$= 0+1$

$= 1$

So, $(x,y) = (-2,1)$



→ The following function represent a parabola. which is shifted 2 units left and 1 units top side. And the vertex is $(-2,1)$.

* $y = (x-2)^2+1$

⇒ Let, $y = f(x)$

$(x-2) = x$

So that, the function is $f(x) = x^2+1$

if $x = 0$

So $x-2=0$

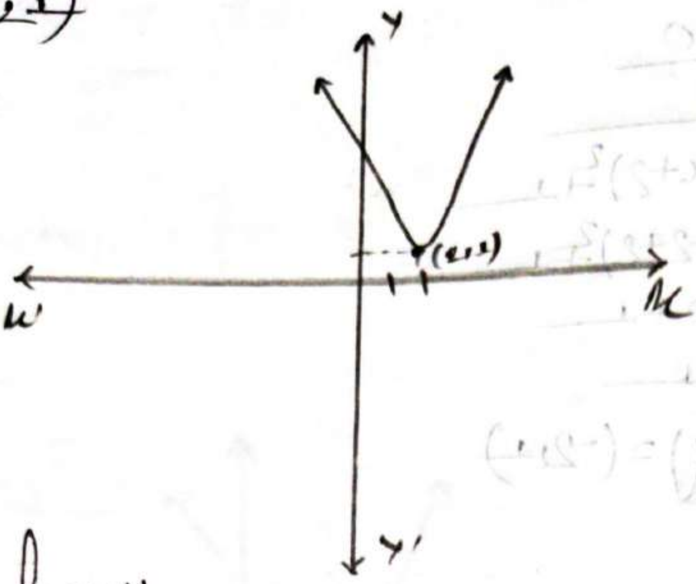
or $x=2$

and $y = (x-2)^2+1$

$= (2-2)^2+1$

$= 0+1 = 1$

So, $(x, y) = (2, 1)$



→ The following function represent a parabola. which is shifted 2 units right side and 1 unit top side. and the vertex is (2, 1).

Q. $y = -x^2$

Let, $y = f(x)$

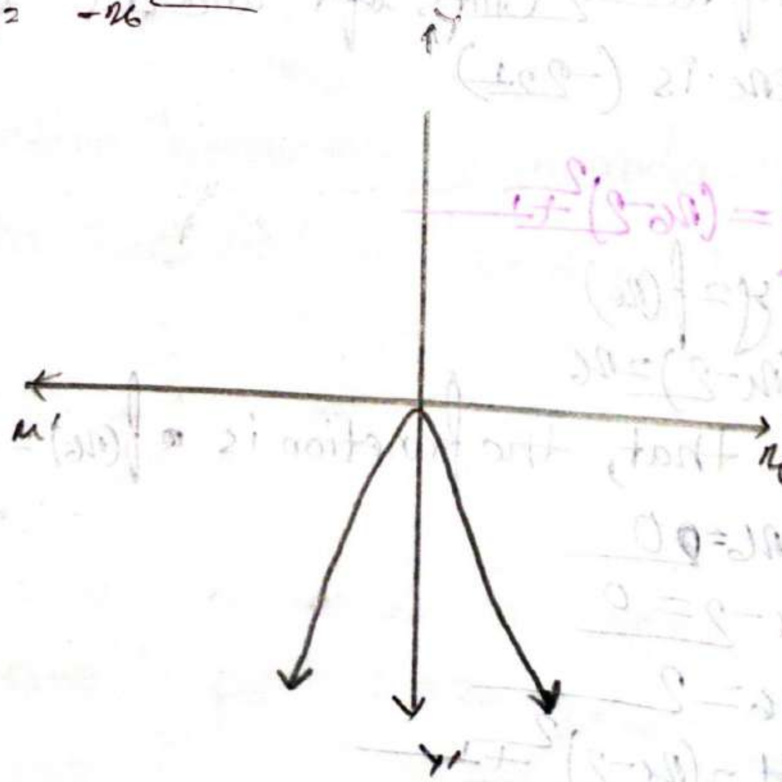
$x = x$

function is $f(x) = -x^2$

if $x = 0$

then $y = 0$

So, $(x, y) = (0, 0)$



⇒ The graph of the function $f(x) = -x^2$ is a Parabola that Opens downwards and is symmmetrical about the x-axis. The Vertex of Parabola is at the origin(0,0).

Q8 What is the name of a following function and Sketch it.

* $y = -(x+2)^2$

Let, $y = f(x)$

$x+2 = x$

So that the function is $f(x) = -(x\;\;)^2$

if $x = 0$

or, $x+2 = 0$

or, $x = -2$

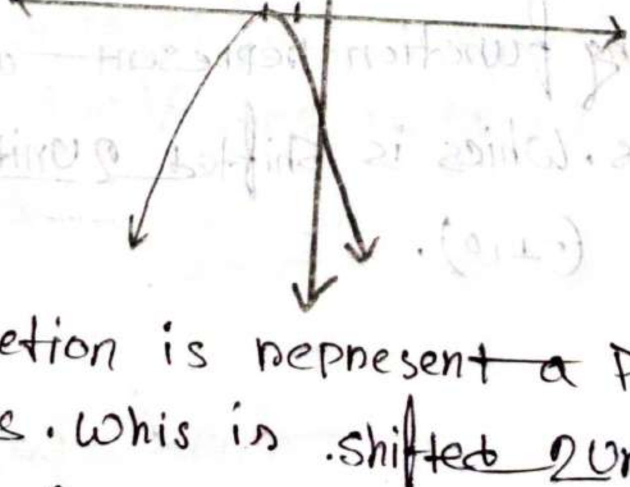
and $y = -(x+2)^2$

$= -(-2+2)^2$

$= -(0)^2$

$= 0$

So $(x,y) = (-2,0)$



⇒ the following function is represent a Parabola. that Opens downwards. Which is shifted 2 Units on left Side and the Vertex is $(-2,0)$.

Q8- * $y = -(x+1)^2$

Let, $y = f(x)$

$x + 1 = x$

So, the function is $f(x) = -x^2$

if $x = 0$

then; $x + 1 = 0$

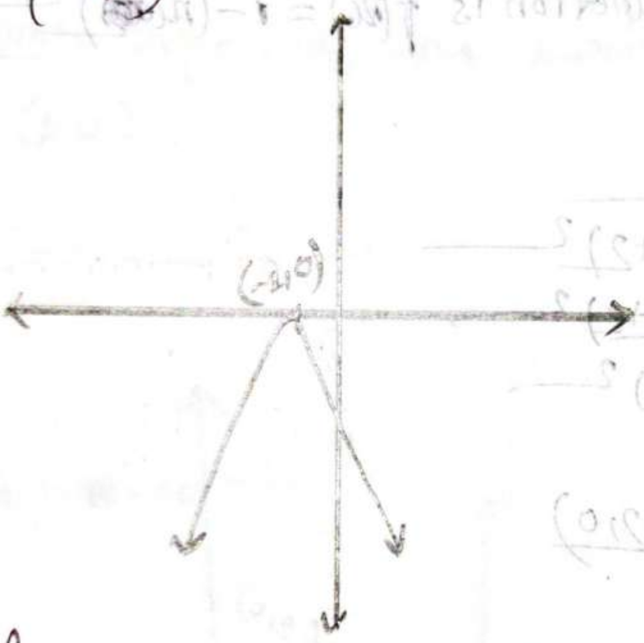
or, $x = -1$

and $y = -(x+1)$

$= -(-1+1)$

$= 0$

So, $(x, y) = (-1, 0)$



→ the following function represent a Parabola. ~~that~~ opens downwards. Whics is shifted 2 units on left, and the Ventex is $(-1, 0)$.

* $y = -(x-1)^2+1$

Let, $y = f(x)$

$x-1 = x$

So the function is $f(x) = -(x-1)^2+1$

if, $x = 0$

then $x-1 = 0$

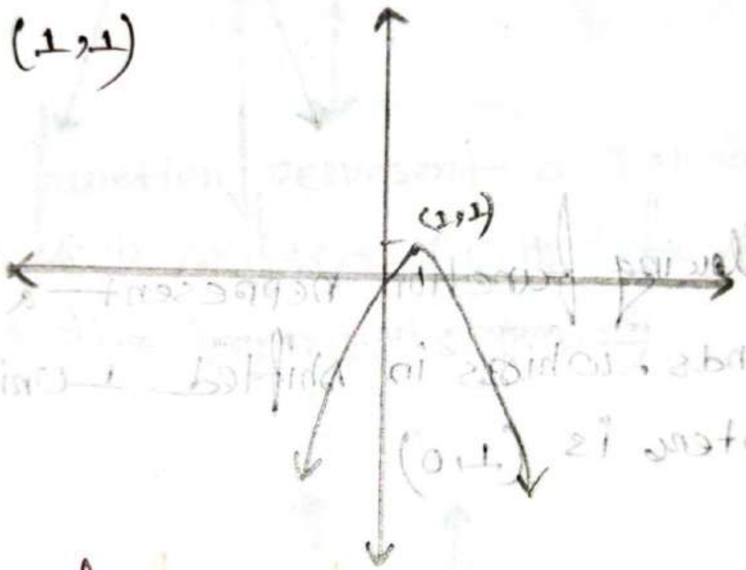
$x = 1$

and $y = -(x-1)^2+1$

$= -(1-1)^2+1$

$= 1$

So, $(x,y) = (1,1)$



→ The following function represent a parabola, that opens downwards. Which is shifted 1 units up and 1 units right side. and the Vertex is (1,1)

* $y = -(x-1)^2$

Let, $y = f(x)$

$x-1 = x$

So, function is $f(x) = -x^2$

if, $x = 0$

$x-1 = 0$

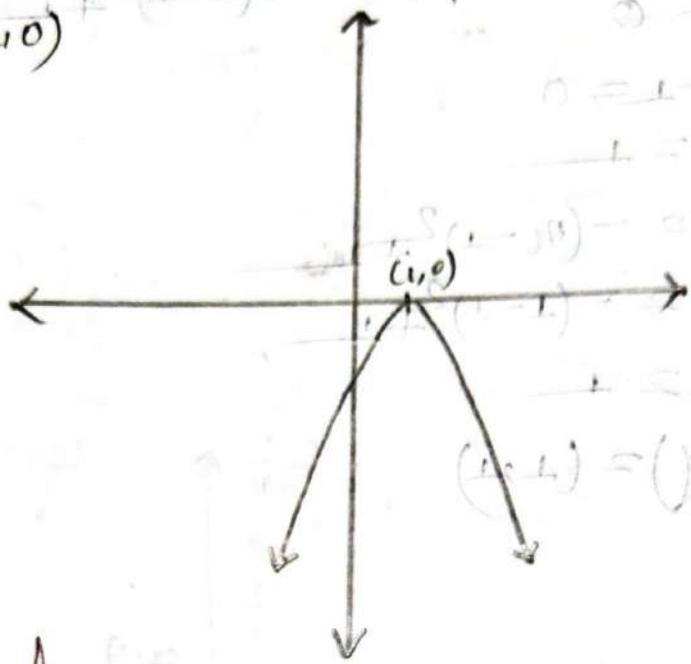
or $x = 1$

and $y = -(x-1)^2$

$= -(1-1)^2$

$= 0$

So, $(x,y) = (1,0)$



⇒ The following function represent a parabola that opens downwards. which is shifted 1 units right side. and the Vertex is $(1,0)$

Q8 Sketch the graph of the functions $f(x) = x^2+6x+10$.

⇒

$$f(x) = x^2+6x+10$$
$$= x^2+2\cdot x\cdot 3+3^2+1$$
$$= (x+3)^2+1$$

Let, $x+3 = x$, $f(x) = y$

So, the function is $f(x) = x^2+1$

if $x+3=0$

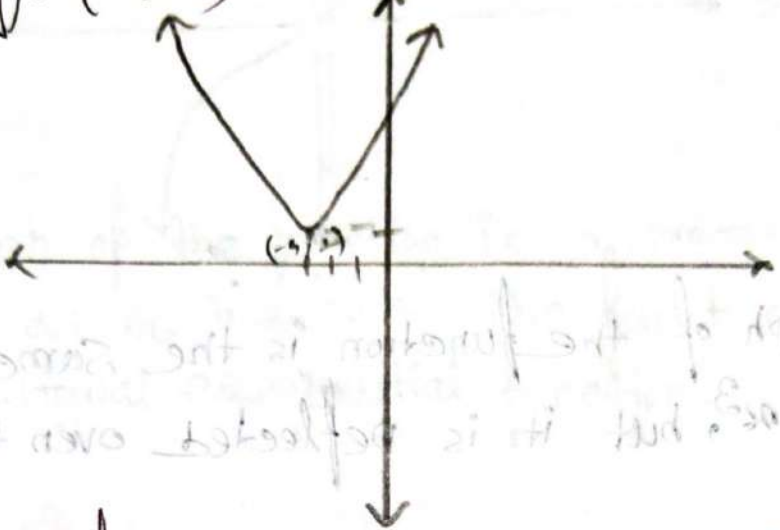
$x = -3$

and $y=(x+3)^2+1$

$=(-3+3)^2+1$

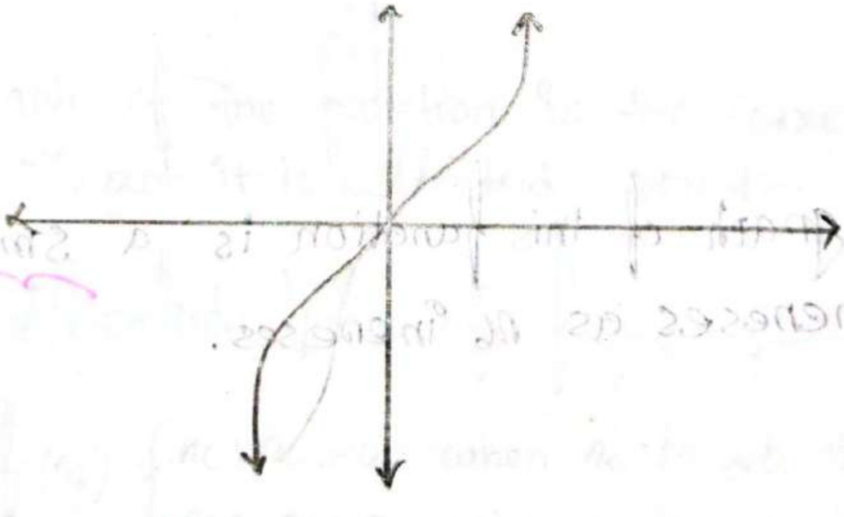
$=1$

So, $(x,y)=(-3,1)$



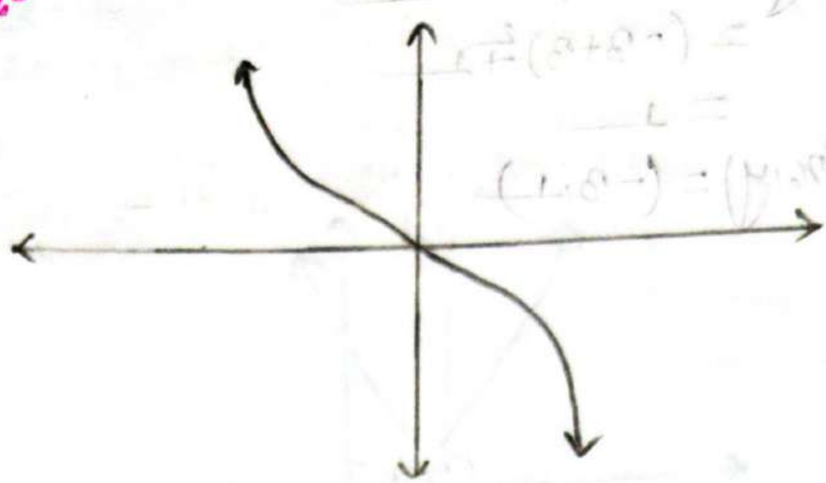
⇒ The following function represent a parabola, that opens upwards. Which is shifted 3 units left and 1 units top side, and the vertex is $(-3,1)$.

(5) $y=x^3$



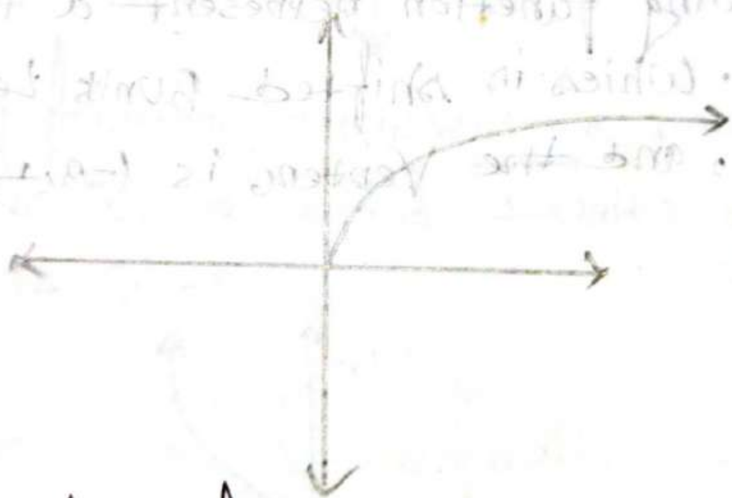
⇒ The following function represent a curve. That increases or decreases depending on the sign of x.

5) $y = -x^3$



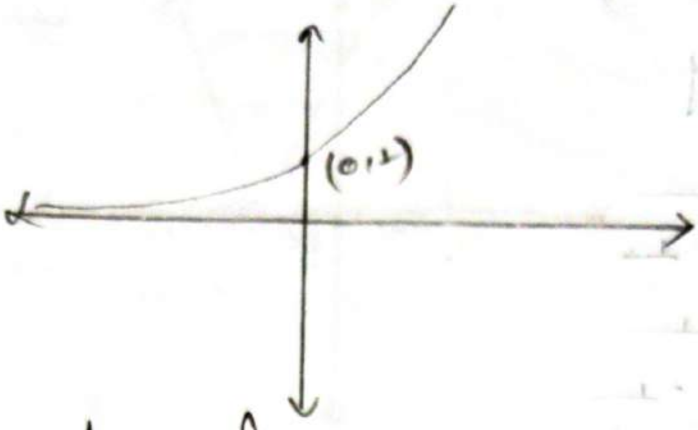
⇒ The graph of the function is the same as the graph of $f(x) = x^3$, but it is reflected over the x-axis.

6) $y = \sqrt{x}$



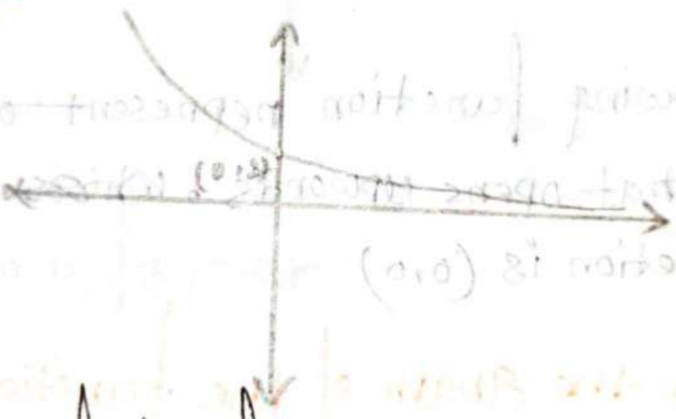
⇒ the graph of this function is a smooth curve. that increses as x increses.

(7) $y = e^{x}$



⇒ The graph of the function is a smooth curve that increases as x increases. The function is also known as the natural exponential function.

(8) $y = -e^{x}$



⇒ The graph of the function is the same as the graph of $f(x) = e^{x}$, but it is reflected over the x-axis.

Absolute Value of function:

$$f(x) \begin{cases} x; & x \geq 0 \quad \text{when } x \text{ is } \geq 0 \text{ then } f(x) = x \\ -x; & x < 0 \quad \text{when } x \text{ is } < 0 \text{ then } f(x) = -x \end{cases}$$

(i) $y = |x|$

Let $f(x) = y$

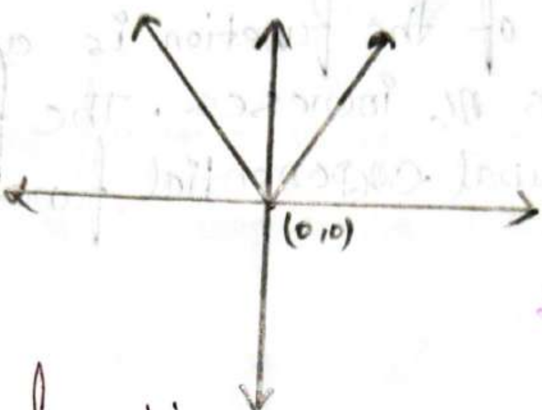
$x = x$

Now, $x = 1$

$f(x) = 1$

or, $x = -1$

$f(x) = -1$



⇒ The following function represent a Absolute value function. that opens upwards, ~~whiss~~ and the vertex of the function is (0,0)

Q:- Sketch the graph of the function

A. $y = |x+2|$

Let, $y = f(x)$

$x+2 = x$

Now, function is $f(x) = |x|$

if $x = 0$

so $x+2 = 0$

or $x = -2$

and $y = 0$

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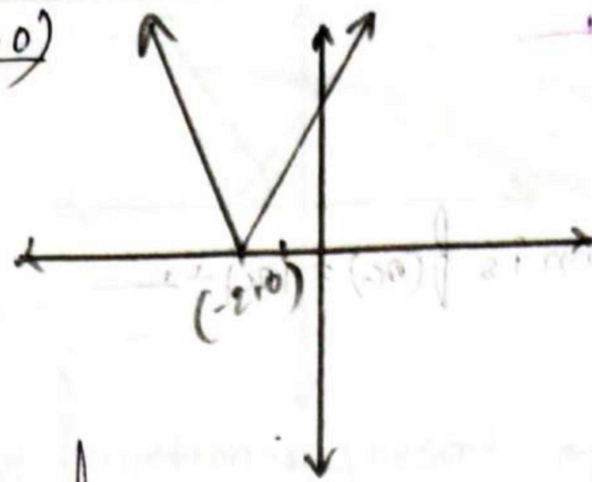
MQ1 - lecture 02

So, $(x,y) = (-2,0)$



⇒ the following function represent a Absolute Value function that opens upwards, which is shifted 2 units on left. and the vertex is $(-2,0)$

Ans

* $y = |x-2|$

Let $y = f(x)$

$x-2 = x$

So the function is $f(x) = |x|$.

if $x = 0$

so $x-2 = 0$

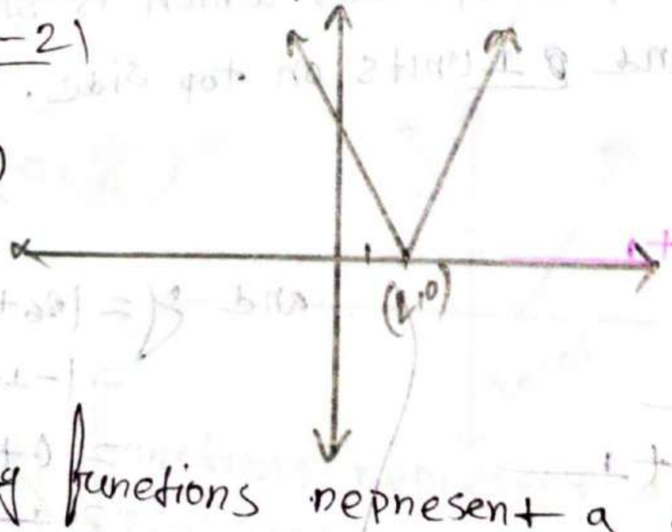
or $x = 2$

and $y = |x-2|$

$= |2-2|$

$= 0$

so, $(x,y) = (2,0)$



⇒ The following functions represent a absolute value function. that opens upwards, which is shifted 2 units on right side and the vertex is $(2,0)$

* $y=|x-2|+1$

Let, $y=f(x)$

$x-2=x$

So the function is $f(x)=|x|+1$

if $x=0$

so, $x-2=0$

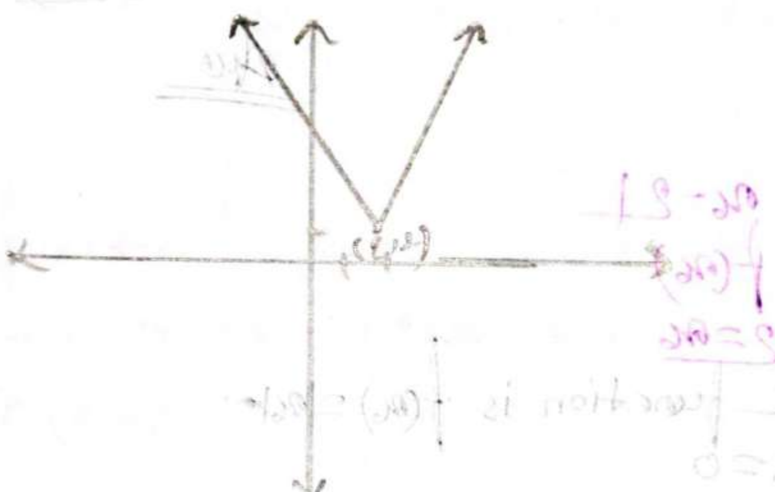
or $x=2$

and $y=|x-2|+1$

$=|2-2|+1$

$=1$

$(x,y)=(2,1)$



⇒ the following function represent a Absolute value function. that opens upwards. which is shifted 2 Units on right side and 1 Units on top side. and the vertex is (2,1).

* $y=|x+1|+1$

Let, $f(x)=y$

$x=x+1$

so, $f(x)=|x|+1$

if $x=0$

$x+1=0$

$x=-1$

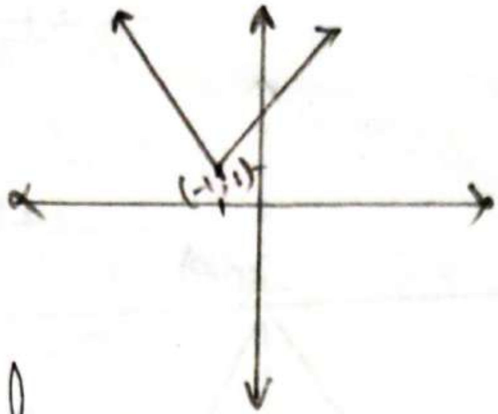
and $y=|x+1|+1$

$=|-1+1|+1$

$=0+1$

$=1$

$\therefore (x, y) = (-1, 1)$



⇒ The following function represent a Absolute value function. that open upwords. which is Shifted 1 unit ~~to~~ left and 1 unit Top side. and the vertenc is $(-1, 1)$

(47) $g(t) = |1 - 3t|$

Let, $g(t) = f(x)$

$1 - 3t = x$

So, $f(x) = |x|$

if $x = 0$

So $1 - 3t = 0$

or $3t = 1$

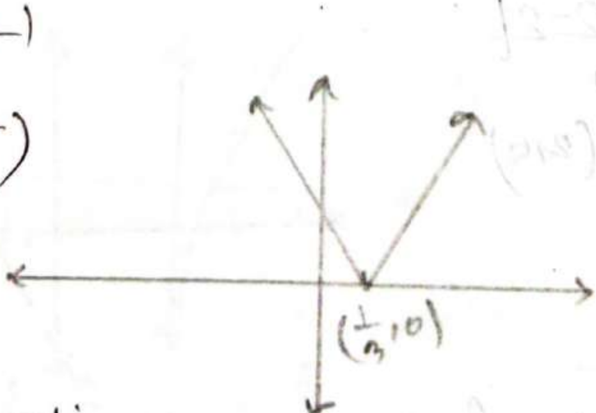
or $t = \frac{1}{3}$

and $g(t) = |1 - 3 \times \frac{1}{3}|$

$= |1 - 1|$

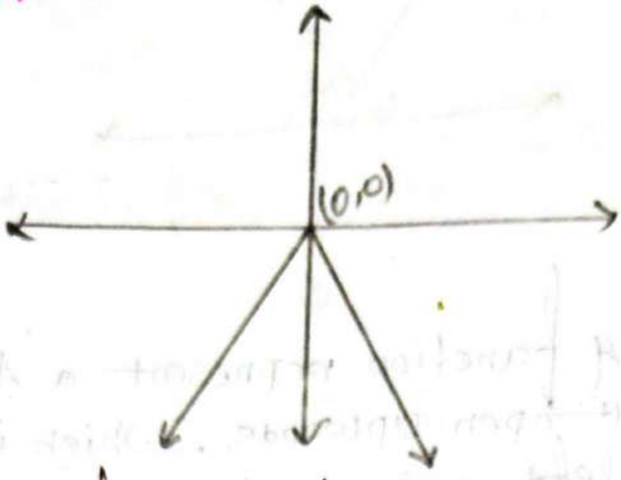
$= 0$

$(g(t), t) = (0, \frac{1}{3})$



⇒ The following functions represent a absolute value function. that opens upwords. which in Shifted $\frac{1}{3}$ units on Right side. the vertex is $(\frac{1}{3}, 0)$

(10) $y = -|x|$



⇒ the following function of the graph is a absolute value function. that opens downwoords. and the ventere is (0,0).

Q: Sketch the graph of the function:

(1) $y = -|x-2|$

Let $y = f(x)$

$x - 2 = x$

So $f(x) = -|x|$

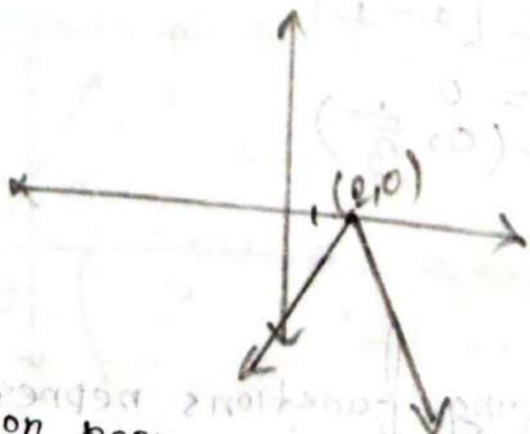
if $x = 0$

so $x - 2 = 0$

or $x = 2$

and $y = -|2-2|$
$= 0$

so $(x, y) = (2, 0)$



⇒ The following function represent a absolute value function that opens downwoonds. which is shifted 2 units on Right Side. and the ventere is (2,0)

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* $y = -|x+2|+1$

Let, $y = f(x)$

$x+2 = x$

So, $f(x) = -|x|+1$

if $x = 0$

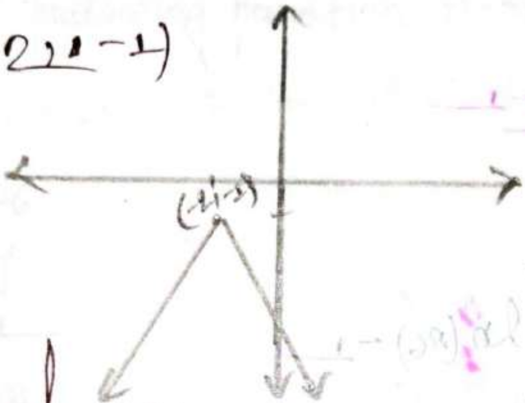
or $x+2 = 0$

$x = -2$

and $y = -|-2+2|+1$

$= -1$

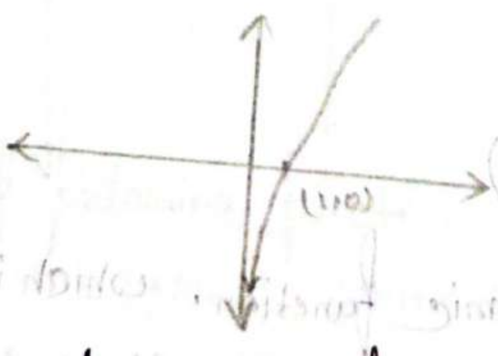
$\therefore (x,y) = (-2, -1)$



⇒ The following function represent a absolute value function. that open's downwords. which is shifted 2 units on left and 1 units down side. and the vertex is (-2, -1).

(11) $y = \ln(x)$

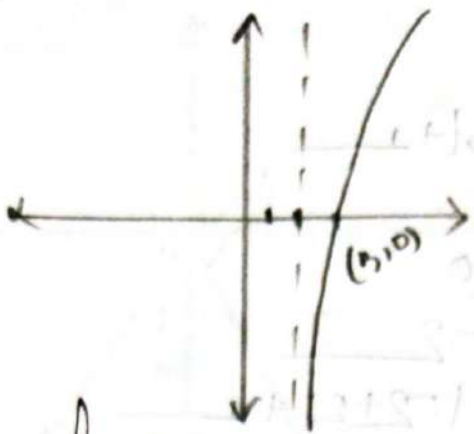
$x = 1$
$y = 0$



⇒ It's a type of a logarithmic function.

* $y = \ln(x-2)$

$x = 3$
$y = 0$



→ It's a logarithmic function. which is shifted 3 units on Right side.

* $y = \ln(x-2) - 1$

Let $f(x) = y$

$(x-2) = x$

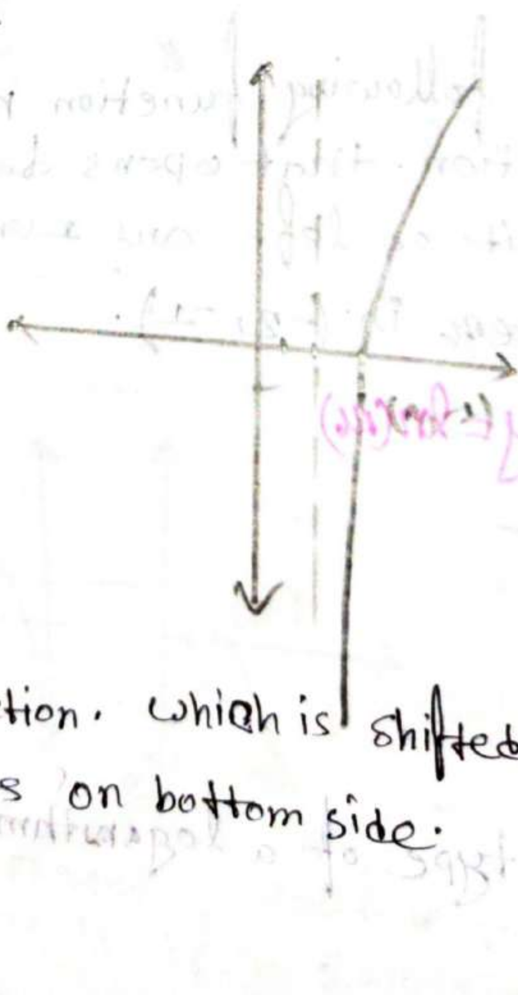
So, $f(x) = \ln(x) - 1$

if $x - 2 = 0$
$x = 2$

so $y = \ln(x-2) - 1$
$= \infty$

and $x = 3$
$y = -1$

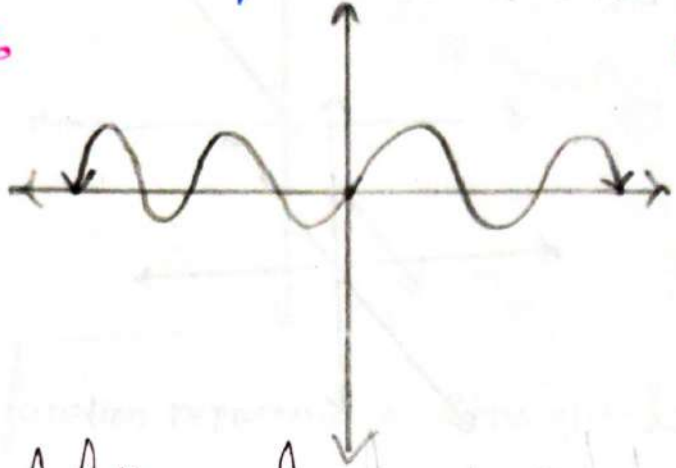
$(x, y) = (3, -1)$
$(x, y) = (2, \infty)$



→ It's a logarithmic function. which is shifted 3 unit on Right side and 1 units on bottom side.

Trigonometric functions:

(11.) $y = \sin x$



⇒ The graph of following function represent a smooth curve.

(*) $y = 1 - \sin x$

Let, $f(x) = y$

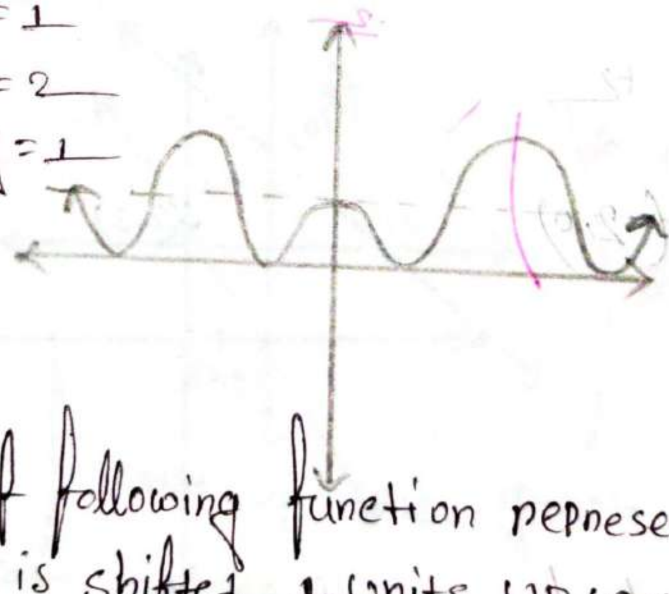
now, $x = 0, y = 1$

$x = \frac{\pi}{2}, y = 0$

$x = \pi, y = 1$

$x = \frac{3\pi}{2}, y = 2$

$x = 2\pi, y = 1$



⇒ The graph of following function represent ~~a smooth~~ curve. which is shifted 1 units up words.

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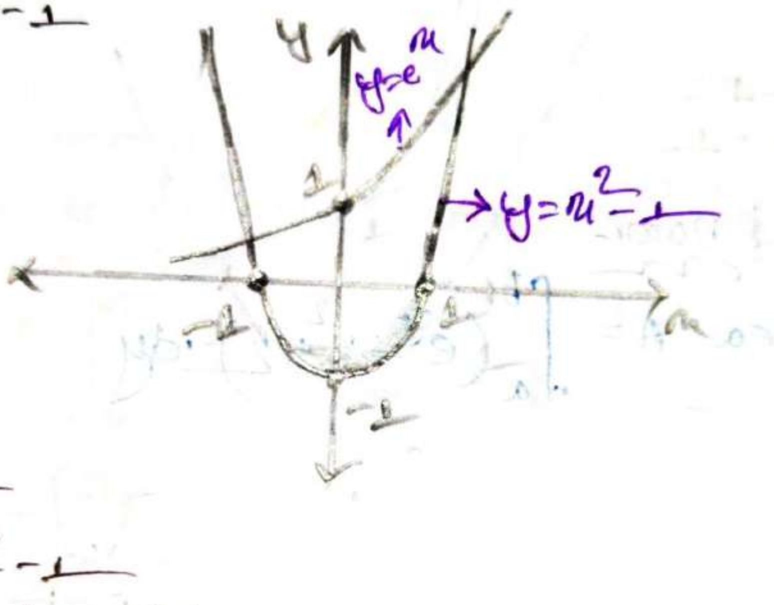
Q:- Find the Area of the region bounded by the Curves, Also sketch the region:—

(5) $y=e^x$, $y=x^2-1$, $x=-1$, $x=1$

⇒ here:—

$b=1$

$a=-1$



⇒ is given:—

$a=-1$

$b=1$

We know:—

$$\text{Area } A=\int_a^b (e^x-x^2+1)\cdot dx$$

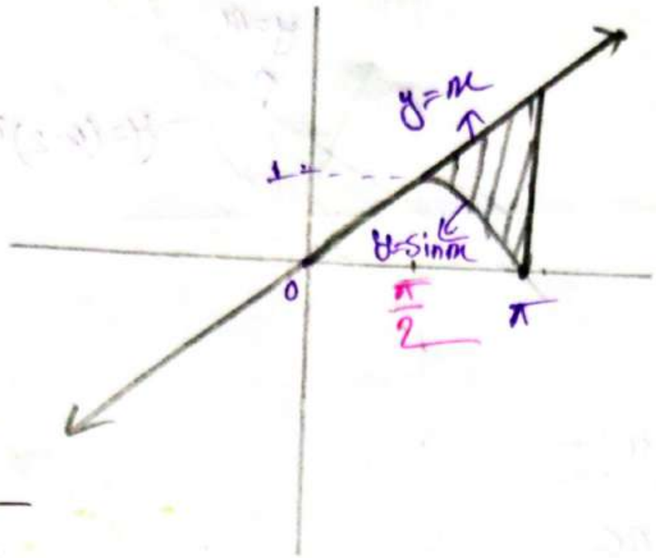
$$\therefore A=\int_{-1}^{1}(e^x-x^2+1)\cdot dx$$

$$=\left[e^x\right]_{-1}^{1}-\frac{1}{3}\left[x^3\right]_{-1}^{1}+\left[x\right]_{-1}^{1}$$

$$=e^1-e^{-1}-\frac{1}{3}(1^3+1^3)+(1+1)$$

$$=3.68$$

(6.) $y = \sin x$, $y = x$, $x = \pi/2$, $x = \pi$



is given:—

$b = \pi$

$a = \frac{\pi}{2}$

We know:—

$$\text{Area } A = \int_a^b (x - \sin x) \cdot dx$$

$$\therefore A = \int_{\pi/2}^{\pi} (x - \sin x) \cdot dx$$

$$= \frac{1}{2}[x^2]_{\pi/2}^{\pi} - [-\cos x]_{\pi/2}^{\pi}$$

$$= \frac{1}{2}\left(\pi^2 - \frac{\pi^2}{4}\right) + \cos\pi - \cos\frac{\pi}{2}$$

$$= \frac{1}{2} \times \left(\frac{3\pi^2}{4}\right) + (-1)$$

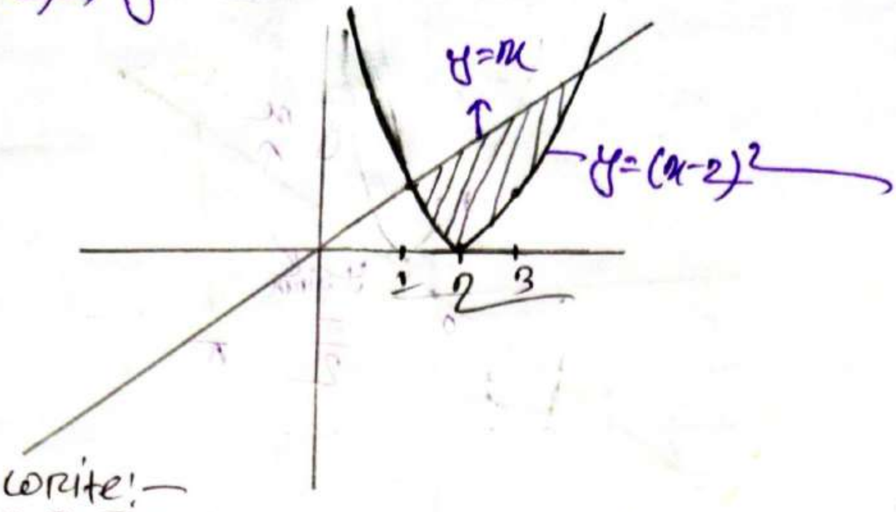
$$= \frac{3\pi^2}{8} - 1$$

$$= \frac{3 \times (3.1416)^2}{8} - 1$$

$$= 2.7011$$

(7.) $y=(x-2)^2,\ y=x$

$\Rightarrow$



$\Rightarrow$ We can write:-

$$(x-2)^2=x$$
$$\Rightarrow x^2-4x+4=x$$
$$\Rightarrow x^2-5x+4=0$$
$$\therefore x=4,\ 1$$
$$\therefore a=1$$
$$b=4$$

We know:-

$$\text{Area } A=\int_a^b (x-(x-2)^2)\cdot dx$$

$$\therefore A=\int_1^4 x\cdot dx-\int_1^4 (x^2-4x+4)\cdot dx$$

$$=\frac{1}{2}[x^2]_1^4-\left[\frac{1}{3}[x^3]_1^4-4\times\frac{1}{2}[x^2]_1^4+4[x]_1^4\right]$$

$$=\frac{1}{2}[4^2-1^2]-\frac{1}{3}(4^3-1^3)+2[4^2-1]-4\times(4-1)$$

$$=\frac{15}{2}-21+30-12$$

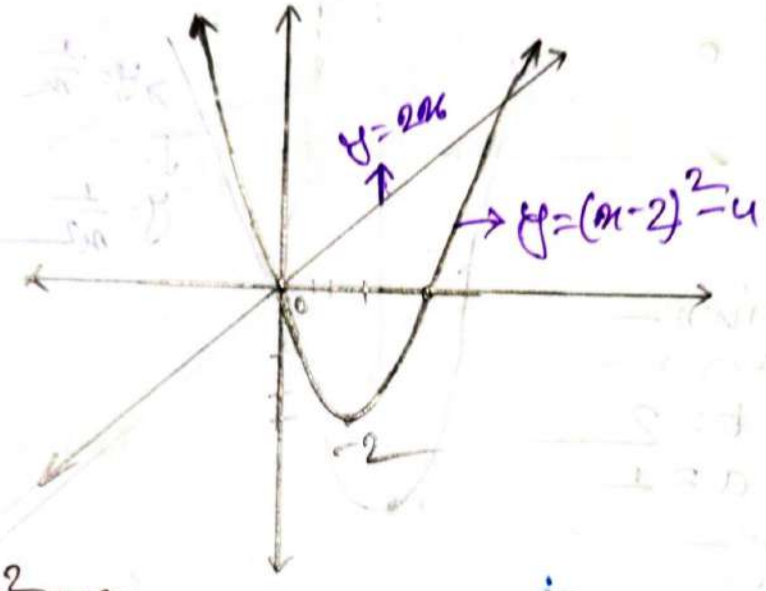
$$=4.5$$

(8.) $y = x^2 - 4x,\ y = 2x$

$\Rightarrow\ y = x^2 - 4x$

$= x^2 - 2 \cdot x \cdot 2 + (2)^2 - 4$

$y = (x-2)^2 - 4$



is given:-

$y = x^2 - 4x$

$y = 2x$

so, $x^2 - 4x = 2x$

$x^2 - 6x = 0$

$\therefore x = 0, 6$

$\therefore b = 6$

$a = 0$

We know:- Area $A = \int_a^b \left(2x - \{(x-2)^2 - 4\}\right) \cdot dx$

$A = \int_0^6 (6x - x^2) \cdot dx$

$= 6 \times \frac{1}{2} [x^2]_0^6 - \frac{1}{3} [x^3]_0^6$

$= 3(6^2 - 0^2) - \frac{1}{3}(6^3 - 0^3)$

$= 36$

(9) $y = \frac{1}{x}$, $y = \frac{1}{x^2}$, $x = 2$

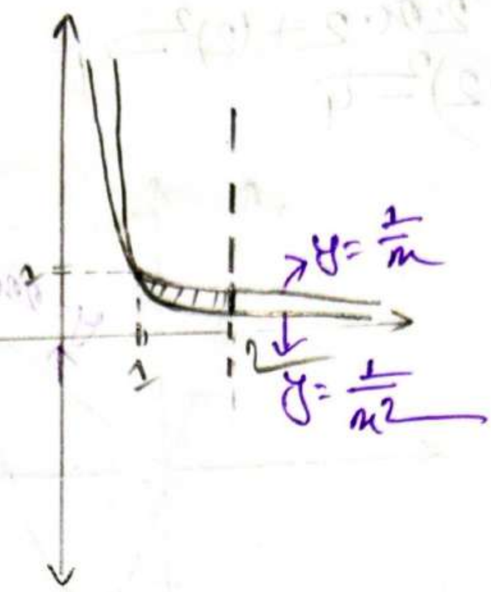
$\Rightarrow \frac{1}{x} = \frac{1}{x^2}$

$\Rightarrow \frac{1}{x^2} - \frac{1}{x} = 0$

$\Rightarrow \frac{x^2}{x^2} - \frac{x^2}{x} = 0$

$\Rightarrow 1 - x = 0$

$\therefore x = 1$



$\therefore$ is given:-

$b = 2$

$a = 1$

We know:-

Area $A = \int_a^b \left(\frac{1}{x} - \frac{1}{x^2}\right) \cdot dx$

$A = \int_1^2 \left(\frac{1}{x} - \frac{1}{x^2}\right) \cdot dx$

$= \left[\ln|x|\right]_1^2 - \left[\frac{x^{-2+1}}{-2+1}\right]_1^2$

$= \ln 2 - \ln 1 + \frac{1}{2} - \frac{1}{1}$

$= 0.193$

(11) $x = 1 - y^2$, $x = y^2 - 1$

$\Rightarrow x = -y^2 + 1$

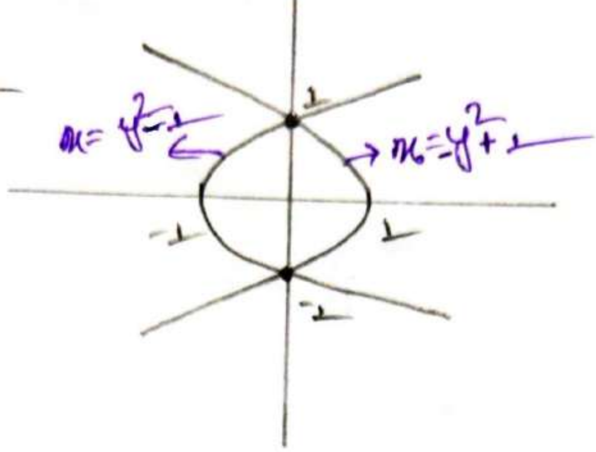
Since $x = y^2 - 1$

So, $-y^2 + 1 = y^2 - 1$

$2y^2 - 2 = 0$

$\therefore y = \sqrt{\frac{2}{2}}$

$= \pm 1$



$\therefore a = -1$

$b = 1$

when $y = 1$, $x = 0$ $= (0, 1)$

when $y = -1$, $x = 0$ $= (0, -1)$

We know:

Area $A = \int_a^b [(-y^2+1) - (+y^2-1)] \cdot dx$

$A = \int_{-1}^{1} (-y^2+1 - y^2+1) \cdot dx$

$= -2 \times \frac{1}{3}[y^3]_{-1}^{1} + 2[x]_{-1}^{1}$

$= \frac{-2}{3}(1+1) + 2 \times (1+1)$

$= +\frac{8}{3}$

(13) $y = 12 - x^2,\ y = x^2 - 6$

$\Rightarrow y = -x^2 + 12,\ y = x^2 - 6$

Find the intersectionspoint:-

$-x^2 + 12 = x^2 - 6$

$2x^2 = 18$

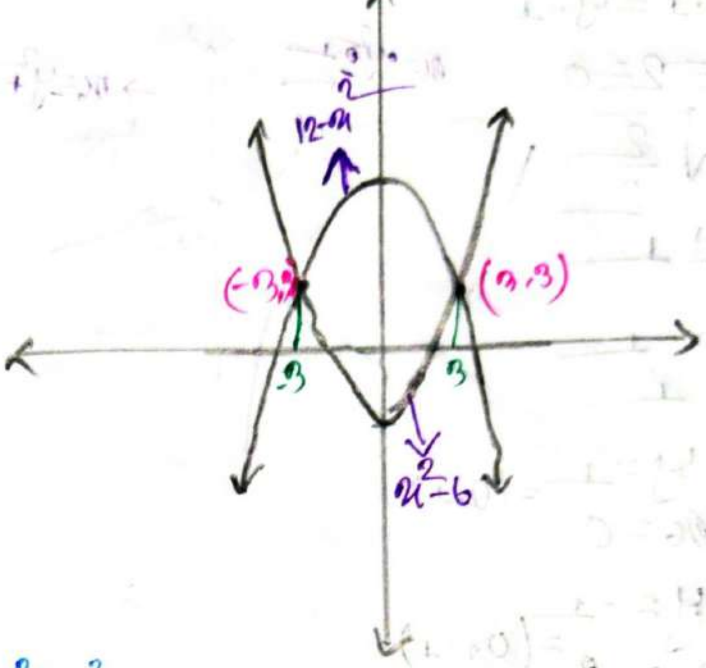
$x^2 = 9$

$x = \pm 3$

So, limit is:-

$b = 3$

$a = -3$



We know:-

Area $A = \int_a^b (12 - x^2 - x^2 + 6) \cdot dx$

$= \int_{-3}^{3} (12 - 2x^2 + 6) \cdot dx$

$= 2\int_0^3 (18 - 2x^2) \cdot dx$

Let,
$f(x) = 18 - 2x^2$
$f(-x) = 18 - 2 \times (-x)^2$
$= 18 - 2x^2$
$= f(x)$
So, its a even function

$= 2 \times \left\{ 18[x]_0^3 - 2 \times \left[\frac{1}{2+1} \times x^3\right]_0^3 \right\}$

$= 2 \times \left\{ 18 \times 3 - \frac{2}{3} \times 27 \right\}$

$= 2 \times (54 - 18)$

$= 72$

(14) $y = x^2,\ y = 4x - x^2$

$\Rightarrow y = x^2,\ y = -(x^2 - 4x)$

$= -\left(x^2 - 2\cdot x\cdot 2 + (2)^2 - 4\right)$

$= -\{(x-2)^2 - 4\}$

$= -(x-2)^2 + 4$

Find the intersection point:—

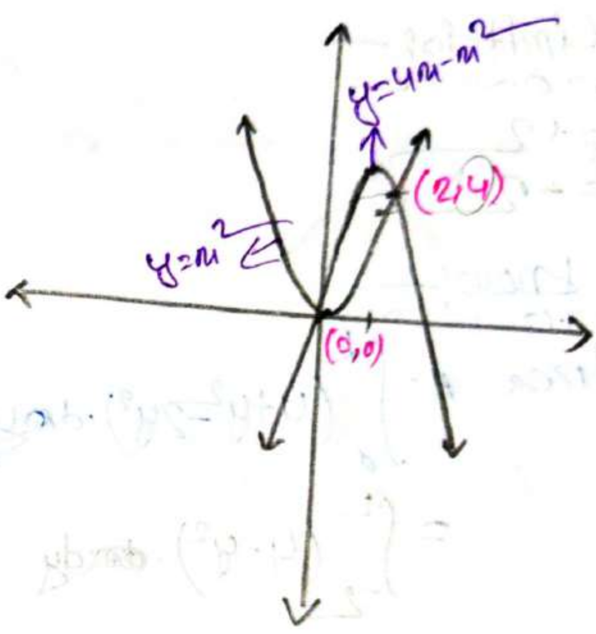
$x^2 = 4x - x^2$

$2x^2 - 4x = 0$

$x^2 - 2x = 0$

$x(x-2) = 0$

$x = 0, 2$



So, Limit is:—

$b = 2$

$a = 0$

We know:—

$$\text{Area } A = \int_a^b (4x - x^2 - x^2)\cdot dx$$

$$= \int_0^2 (4x - 2x^2)\cdot dx$$

$$= 4\times\frac{1}{2}[x^2]_0^2 - 2\times\frac{1}{2+1}[x^3]_0^2$$

$$= 2\times(4-0) - \frac{2}{3}\times[8-0]$$

$$= 8 - \frac{2}{3}\times 8$$

$$= \frac{8}{3}$$

(17) $x = 2y^2$, $x = 4 + y^2$

Find the intersections point:—

$2y^2 = 4 + y^2$

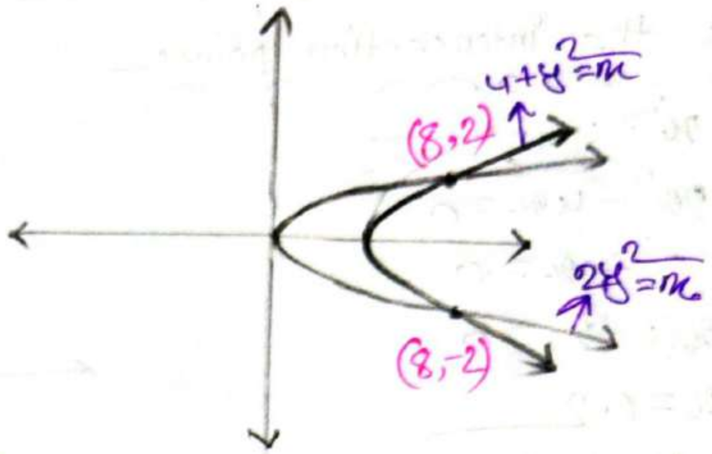
$\Rightarrow y^2 = 4$

$\Rightarrow y = \pm 2$

So, Limit is:—

$b = +2$

$a = -2$



We know:—

Area $A = \int_a^b (4 + y^2 - 2y^2) \cdot dy$

$= \int_{-2}^{2} (4 - y^2) \, dy$

$= 2 \times \int_0^2 (4 - y^2) \, dy$

$= 2 \times \left\{ 4[y]_0^2 - \frac{1}{2+1} \times [y^3]_0^2 \right\}$

$= 2 \times \left\{ 8 - \frac{1}{3} \times 2^3 \right\}$

$= 2 \times \left(8 - \frac{8}{3} \right)$

$= -\frac{8}{3} \times 2 + 16$

$= \frac{32}{3}$

$= 10.667$

Let:—

$f(y) = 4 - y^2$

$f(-y) = 4 - (-y)^2$

$= 4 - y^2$

$= f(y)$

So, its a even function

(18) $y = \sqrt{x-1}$, $x - y = 1$

$\Rightarrow y = \sqrt{x-1}$, $y = x-1$

Find the intersection point:—

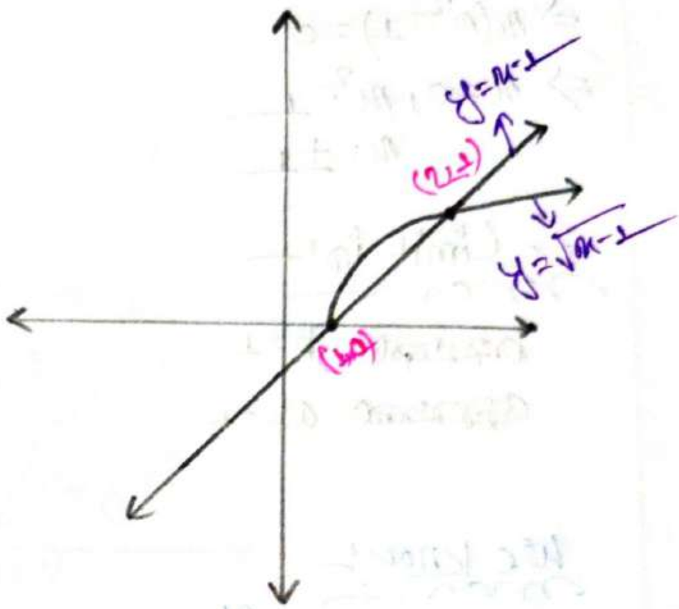
$\sqrt{x-1} = x-1$

$\Rightarrow (x-1)^2 = x-1$

$\Rightarrow x^2 - 2x + 1 - x + 1 = 0$

$\Rightarrow x^2 - 3x + 2 = 0$

$\therefore x = 1, 2$



So, the limit is:—

$b = 2$

$a = 1$

We know:—

Area $A = \int_a^b (\sqrt{x-1} - x + 1) \cdot dx$

$= \int_1^2 \{(x-1)^{\frac{1}{2}} - x + 1\} \cdot dx$

$= \frac{2}{3}\left[(x-1)^{\frac{3}{2}}\right]_1^2 - \frac{1}{2}[x^2]_1^2 + [x]_1^2$

$= \frac{2}{3}\left[(2-1)^{\frac{3}{2}} - (1-1)^{\frac{3}{2}}\right] - \frac{1}{2}(2^2 - 1) + 2 - 1$

$= \frac{2}{3} \times 1 - \frac{3}{2} + 1$

$= \frac{1}{6}$

(22) $y = x^3, y = x$

Find the intersection points—

$x^3 = x$

$\Rightarrow x(x^2 - 1) = 0$

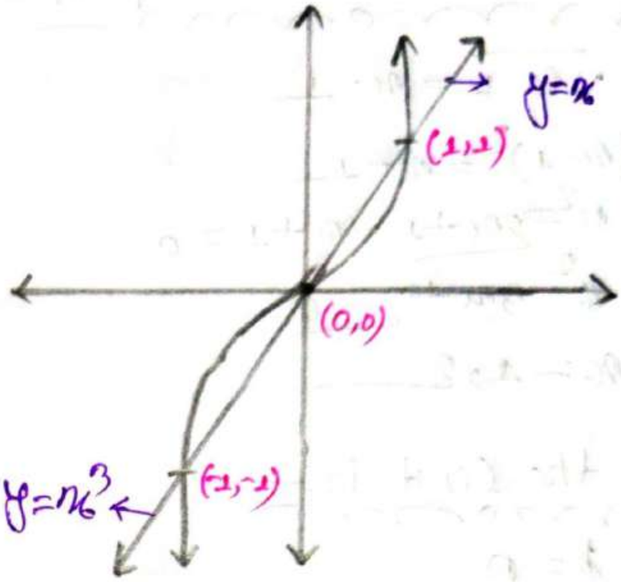
$\Rightarrow x = 0, x^2 = 1$

$x = \pm 1$

So, Limit is:—

$b = 1$

$a = -1$



We know:—

Area $A = \int_a^b (x - x^3) \cdot dx$

$= \int_{-1}^{1} (x - x^3) \cdot dx$

$= \frac{1}{2}[x^2]_{-1}^{1} - \frac{1}{4}[x^4]_{-1}^{1}$

$= \frac{1}{2}(1+1) - \frac{1}{4}(1+1)$

$= 1 - \frac{1}{2}$

$= \frac{1}{2}$

Let,

$f(x) = x - x^3$

$f(-x) = -x + x^3$

$= -(x - x^3)$

$= -f(x)$

So, it's a odd function

(22) $y = x^3$, $y = x$

Find the intersection point:—

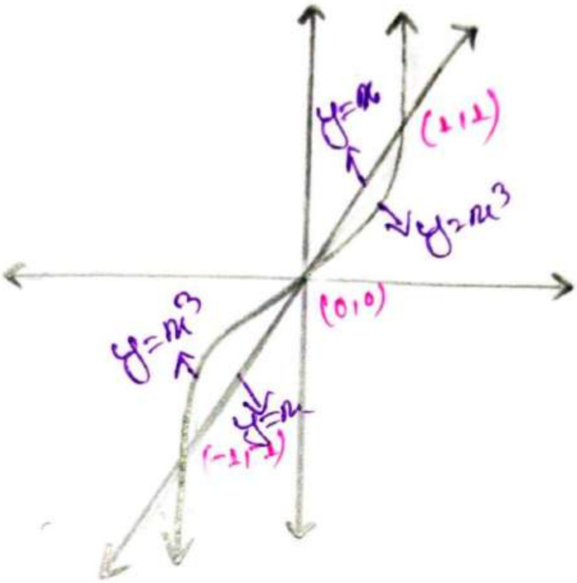
$x^3 = x$

$\Rightarrow x(x^2 - 1) = 0$

$\therefore x = 0,\ x^2 - 1 = 0$

$x = \pm 1$

So, the curve intersect at $(1,1)$ $(0,0)$ $(-1,-1)$



We know:—

$$\text{Area } A = \int_a^b (x^3 - x)\cdot dx + \int_b^c (x - x^3)\cdot dx$$

$$= \int_{-1}^{0} (x^3 - x)\cdot dx + \int_0^1 (x - x^3)\cdot dx$$

$$= \frac{1}{4}[x^4]_{-1}^{0} - \frac{1}{2}[x^2]_{-1}^{0} + \frac{1}{2}[x^2]_0^1 - \frac{1}{3+1}\times[x^4]_0^1$$

$$= -\frac{1}{4}\times 1 + \frac{1}{2}\times 1 + \frac{1}{2}\times 1 - \frac{1}{4}\times 1$$

$$= \frac{1}{2}$$

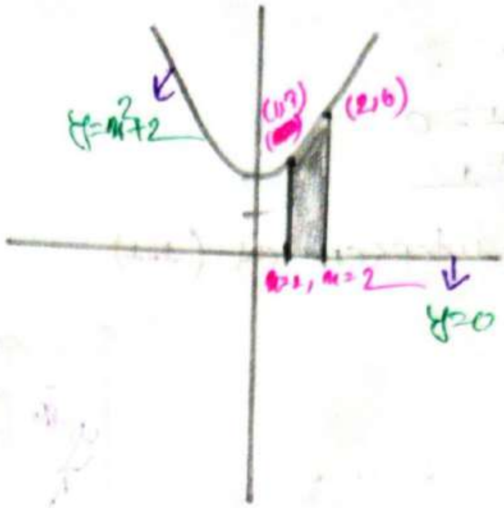
Sketch the Region bounded by the Curved $y = x^2+2$ and the x-asis and the Lines $x=1$, $x=2$

$\Rightarrow$ $y = x^2+2$

$y = 0$

$x^2+2=0$

$x = -2, 0$



So, Limit is:-

$a=1$

$b=2$

We know:-

$$\text{Area } A = \int_a^b (x^2+2)\cdot dx$$

$$= \int_1^2 (x^2+2)\cdot dx$$

$$= \left[\tfrac{1}{3}x^3 + 2x\right]_1^2$$

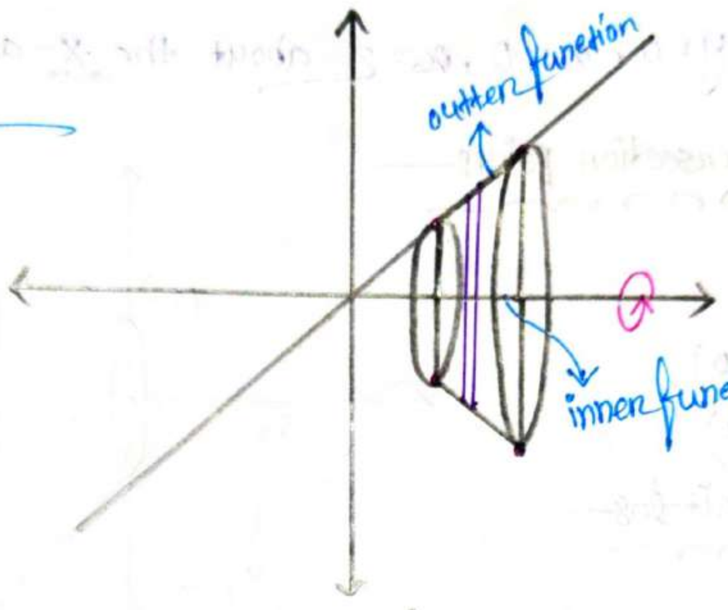
$$= \tfrac{1}{3}(8-1) + 2\times(2-1)$$

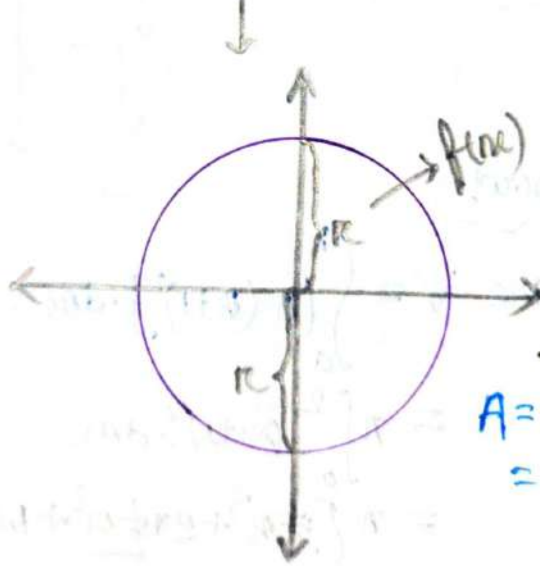
$$= \frac{7}{3} + 2$$

$$= \frac{13}{3}$$

Volume of Solids of Revolution:—

$f(x) = x$
$x = 1,\ x = 2$





$$A = \pi r^2$$
$$= \pi \times [f(x)]^2$$

Volume:—

$$\therefore V = A \times \Delta x$$
$$= \int_a^b \pi\left[(\text{outter function})^2 - (\text{inner function})^2\right] \cdot dx$$
$$= \int_a^b \pi [f(x)]^2 \cdot dx$$

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⇒ Find the volume of the solid obtained by rotating the region by the following curves. Also sketch the region.

(1) $y = x+1$, $y = 0$, $x = 0$, $x = 2$ about the x-axis:—

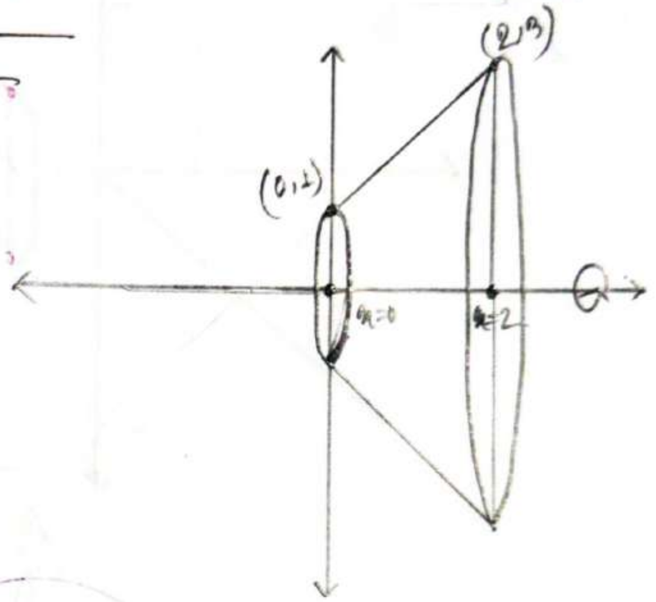
Find the Intersection points:—

$x + 1 = 0$

$x = -1$

$(x, y) = (-1, 0)$

$(x, y) = (0, 1)$



So, the limiting:—

$b = 2$

$a = 0$

So, we know—

$$\text{Volume } V = \int_a^b [\pi (x+1)^2] \cdot dx$$

$$= \pi \int_0^2 (x+1)^2 \cdot dx$$

$$= \pi \left[\frac{1}{3}x^3 + 2 \times \frac{1}{2}x^2 + x\right]_0^2$$

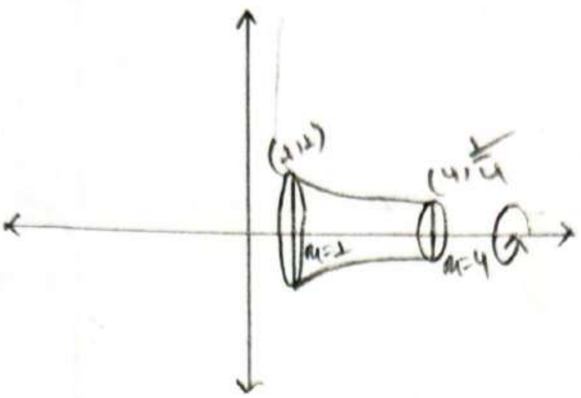
$$= \pi \left[\frac{8}{3} + 4 + 2\right]$$

$$= \frac{26\pi}{3}$$

② $y=\frac{1}{x}$ · $y=0$, $x=1$, $x=4$ · about the x axis :-

Find the intersection point:—

$\frac{1}{x}=0$



$(x,y)=(1,1)$

$(x,y)=(4,\frac{1}{4})$

Limit is:—

$b=4$

$a=1$

We know:—

$$\text{Volume } V=\int_a^b \left[\pi\cdot\left(\frac{1}{x}\right)^2-0^2\right]\cdot dx$$

$$V=\int_1^4\left[\pi\times\left(\frac{1}{x^2}\right)\right]\cdot dx$$

$$=\pi\left[-\frac{1}{x}\right]_1^4$$

$$=-\pi\left[\frac{1}{4}-\frac{1}{1}\right]$$

$$=\frac{3\pi}{4}$$

(3) $y=\sqrt{x-1}$, $y=0$, $x=5$ about the x axis:—

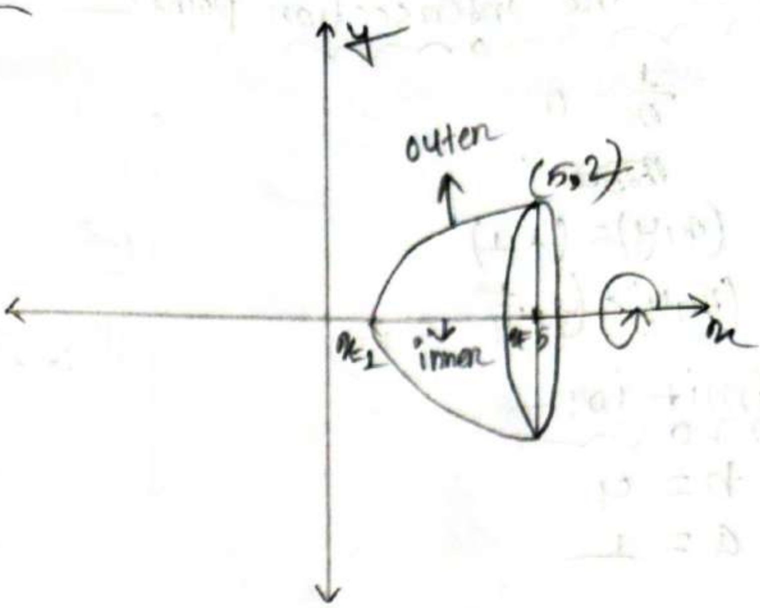
Find the intersection point:—

$\sqrt{x-1} = 0$

$x = 1$

$(x,y) = (1,0)$

$(x,y) = (5,2)$



So limit is:—

$a=1$

$b=5$

We know:—

$$\text{Volume } V = \pi\int_a^b \left[\sqrt{x-1}\right]^2 \cdot dx$$

$$= \pi\int_1^5 (x-1)\cdot dx$$

$$= \pi\left[\tfrac{1}{2}x^2 - x\right]_1^5$$

$$= \pi\left[\tfrac{1}{2}(24) - 4\right]$$

$$= 8\pi$$

(4) $y = e^x$, $y = 0$, $x = -1$, $x = 1$ about the x axis:—

Find the intersection points:—

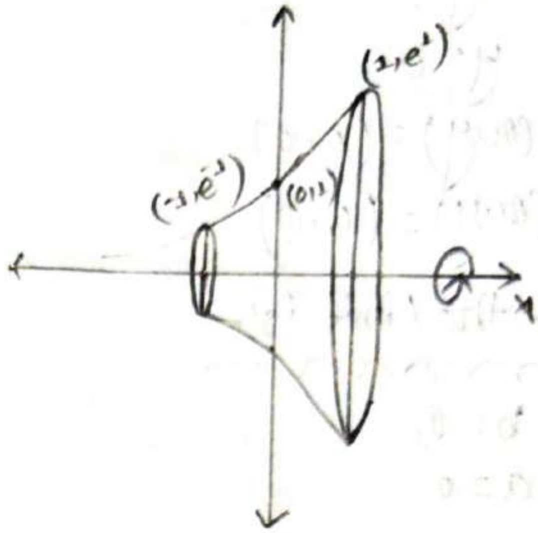
$$e^x = 0$$

$$y = e^x$$

$$(x, y) = (0, 1)$$

$$(x, y) = (-1, e^{-1})$$

$$(x, y) = (1, e^1)$$



Limit is:—

$a = -1$

$b = 1$

We know:—

$$\text{Volume } V = \pi \int_a^b (e^x)^2 \cdot dx$$

$$V = \pi \int_{-1}^{1} e^{2x} \cdot dx$$

$$= \pi \left[\frac{1}{2} e^{2x} \right]_{-1}^{1}$$

$$= \pi \times \frac{1}{2} \left[e^2 - e^{-2} \right]$$

$$= \frac{\pi}{2} \left[e^2 - e^{-2} \right]$$

(5) $x = 2\sqrt{y}$, $x = 0$, $y = 9$ about the y axis:—

Find intersection point:—

$2\sqrt{y} = 0$

$y = 0$

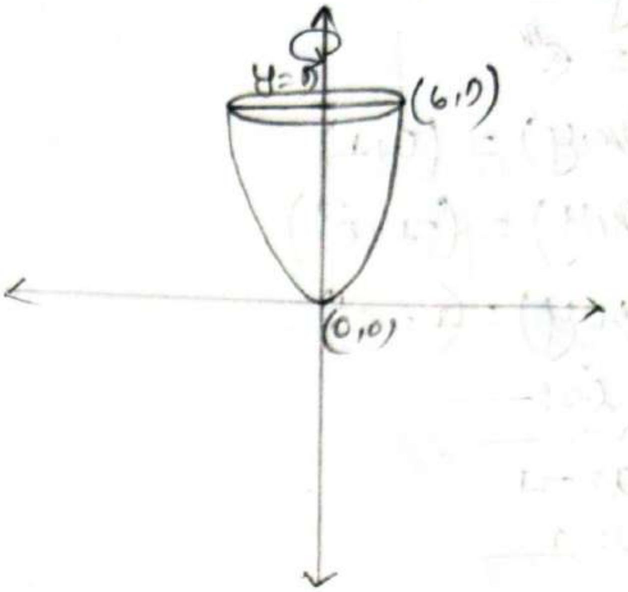
$\therefore (x, y) = (0, 0)$

$(x, y) = (6, 9)$

So, the limit is:—

$b = 9$

$a = 0$



We know:—

$$\text{Volume } V = \int_a^b \left[\pi (2\sqrt{y})^2\right] \cdot dy$$

$$= \int_0^9 [\pi \times 4y] \cdot dy$$

$$= \pi \left[2y^2\right]_0^9$$

$$= \pi \times 162$$

$$= 162\pi$$

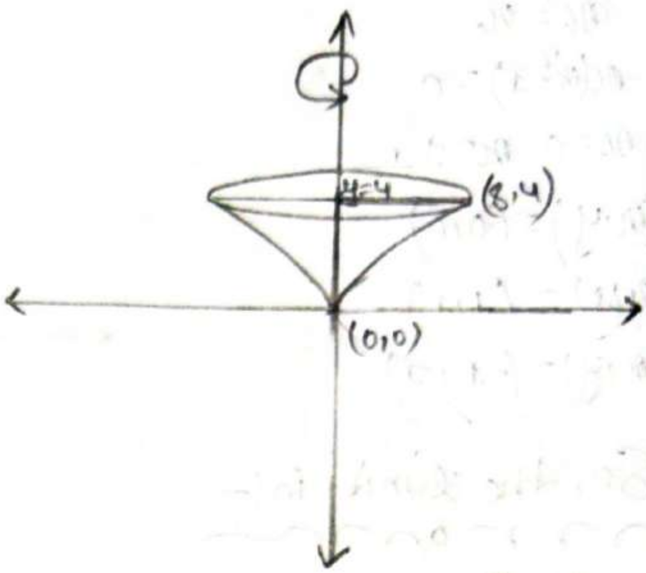
(6) $2x = y^2;\ x = 0,\ y = 4$ about the y axis:—

Find the intersection point:—

$x = \frac{1}{2}y^2$

$(x, y) = (8, 4)$

$(x, y) = (0, 0)$



So Limit is:—

$b = 4$

$a = 0$

We know:—

$$\text{Volume } V = \int_a^b \pi \left(\tfrac{1}{2}y^2\right)^2 \cdot dy$$

$$= \int_0^4 \pi \left(\tfrac{1}{4} \cdot y^4\right) \cdot dy$$

$$= \pi \left[\tfrac{1}{4} \times \tfrac{1}{5} \times y^5\right]_0^4$$

$$= \pi \left(\tfrac{1}{20} \times 4^5\right)$$

$$= \frac{256\pi}{5}$$

⑦ $y = x^3,\ y = x,\ x \geq 0$ about the x axis:—

Find the intersection points:—

$x^3 = x$

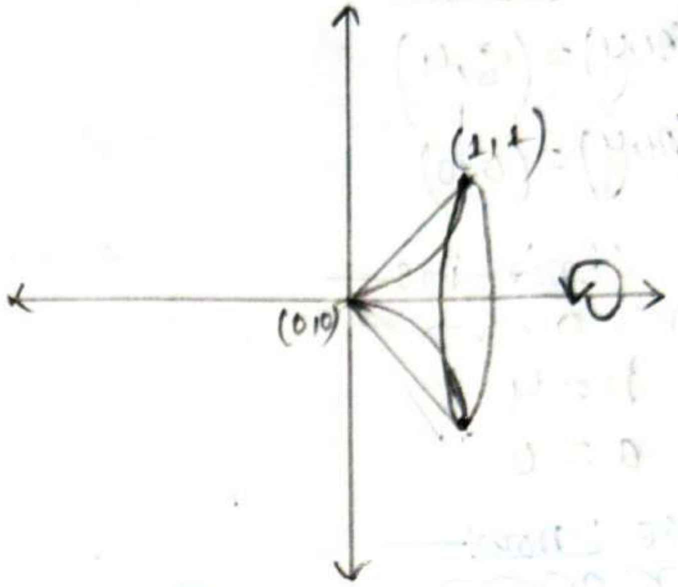
$x(x^2 - 1) = 0$

$x = 0,\ x = \pm 1$

$(x, y) = (0, 0)$

$(x, y) = (1, 1)$

$(x, y) = (-1, -1)$



So, the limit is:—

$a = 0$

$b = 1$

We know:—

$$\text{Volume } V = \int_a^b \pi \{(x)^2 - (x^3)^2\} \cdot dx$$

$$= \pi \int_0^1 (x^2 - x^6) \cdot dx$$

$$= \pi \times \left[\frac{1}{3} x^3 - \frac{1}{7} x^7\right]_0^1$$

$$= \pi \times \left(\frac{1}{3} - \frac{1}{7}\right)$$

$$= \frac{4\pi}{21}$$

(8) $y=6-x^2$; $y=2$ about x asis:—

Find the intersection point:—

$y=-x^2+6$

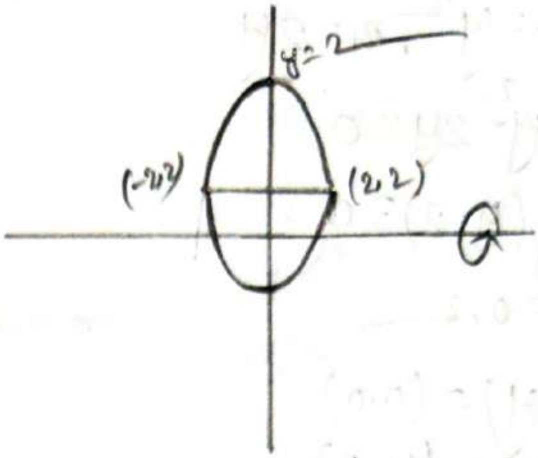
$y=2$

$x^2=4$

$x=\pm 2$

$(x,y)=(2,2)$

$(x,y)=(-2,2)$



So, limit is:—

$b=2$

$a=-2$

We know:—

Volume $V=\pi\int_a^b [(6-x^2)^2-2^2]\,dx$

$V=\pi\int_{-2}^{2}[36-12x^2+x^4-4]\cdot dx$

$=2\pi\left[32x-12\times\frac{1}{3}x^3+\frac{1}{5}\times x^5\right]_0^2$

$=2\pi\left(32\times 2-12\times\frac{1}{3}\times 2^3+\frac{1}{5}\times 2^5\right)$

$=2\pi\left(64-32+\frac{32}{5}\right)$

$=\frac{384}{5}\pi$

(5) $y^2 = x,\ x = 2y$. about the y axis:—

Find the intersection point:—

$x = y^2,\ x = 2y$

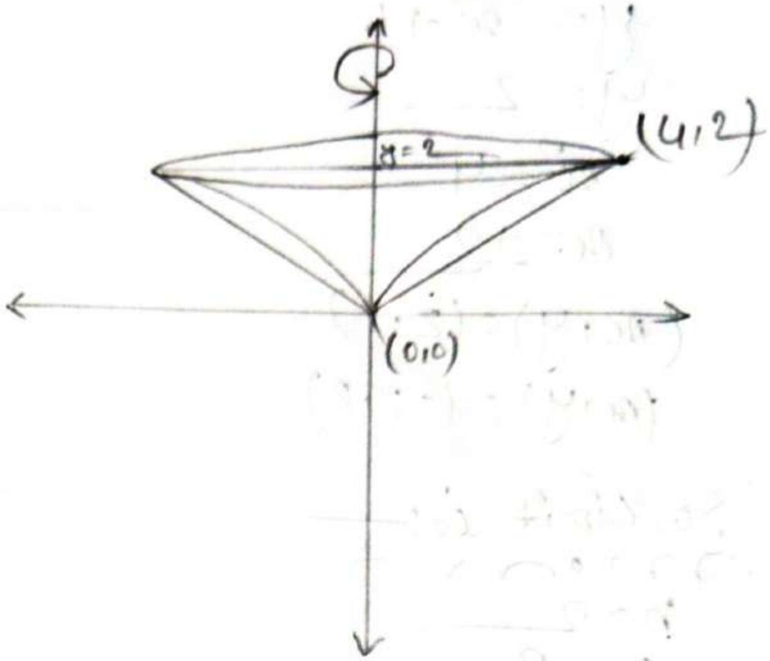
$y^2 - 2y = 0$

$y(y-2) = 0$

$y = 0, 2$

$(x, y) = (0, 0)$

$(x, y) = (4, 2)$



So, Limit is:—

$a = 0$

$b = 2$

We know:—

Volume $V = \pi \int_a^b [(2y)^2 - (y^2)^2] \cdot dy$

$= \pi \int_0^2 [4y^2 - y^4] \cdot dy$

$= \pi \left[4 \times \frac{1}{3} y^3 - \frac{1}{5} y^5 \right]_0^2$

$= \pi \left(4/3 \times 2^3 - \frac{1}{5} \times (2^5) \right)$

$= \pi \left(\frac{32}{3} - \frac{32}{5} \right)$

$= \frac{64}{15} \pi$

(10) $x = 2 - y^2$, $x = y^4$ about the y axis:—

Find the intersection points:—

$2 - y^2 = y^4$

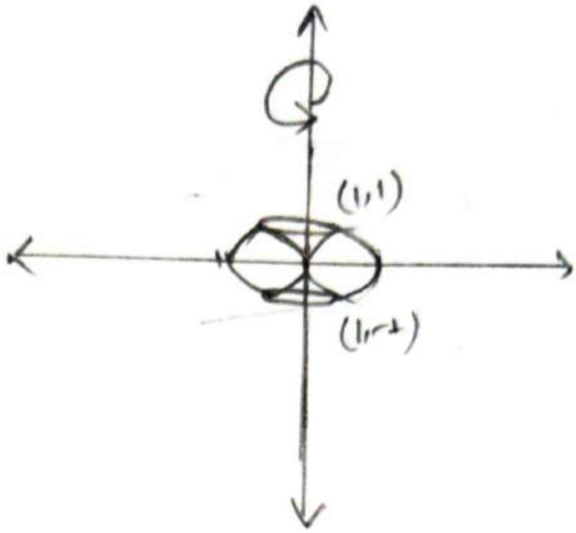
$y^4 + y^2 = 2$

$\therefore y = \pm 1$

$(x, y) = (1, 1)$

$(x, y) = (0, 0)$

$(x, y) = (1, -1)$



So, the limit is:—

$b = 1$

$a = -1$

We know:—

Volume $V = \pi \int_a^b [(2-y^2)^2 - (y^4)^2] \cdot dy$

$= \pi \int_{-1}^{1} [4 - 4y^2 + y^4 - y^8] \cdot dy$

$= 2\pi \times \int_0^1 [4 - 4y^2 + y^4 - y^8] \cdot dy$

$= 2\pi \left[4y - \frac{4}{3}y^3 + \frac{1}{5}y^5 - \frac{1}{9}y^9\right]_0^1$

$= 2\pi \left[4 - \frac{4}{3} + \frac{1}{5} - \frac{1}{9}\right]$

$= \frac{248\pi}{45}$

Gamma And Beta Function.

Improper Integrals:—

(1.) $\int_0^\infty e^{-x} \cdot dx$

$= \lim_{b\to\infty} \int_0^b e^{-x} \cdot dx = \lim_{b\to\infty} [-e^{-x}]_0^b$

$= \lim_{b\to\infty} [-e^{-b} + 1]$

Ex:—

(2) $\int_{-\infty}^{1}$

(3) $\int_{-1}^{\infty}$

(4.) $\int_0^1 \frac{1}{x} \cdot dx$

Gamma:—

⇒ The gamma function is denoted $\Gamma(n)$ and it is defined by

$$\boxed{\Gamma n = \int_0^\infty x^{n-1} \cdot e^{-x} \cdot dx} \quad ; n > 0$$

where, for convergence of the integral $n > 0$

Some useful formulas—

(1) $\Gamma(n+1) = n\Gamma n$ or, $\Gamma n = (n-1)\Gamma n-1$; when n is fraction

(2) $\Gamma n+1 = n!$ or $\Gamma n = (n-1)!$; When n is an Integer

(3) $\Gamma n = \frac{\Gamma(n+1)}{n}$; when n is negative.

(4) $\Gamma \frac{1}{2} = \sqrt{\pi}$; (5) $\Gamma n = \int_0^\infty x^{n-1} \cdot e^{-x} \cdot dx$

Evaluate the following:-

① $\Gamma\frac{5}{2} = \frac{5}{2}-1\,\Gamma\left(\frac{5}{2}-1\right)$

$= \frac{3}{2}\,\Gamma\frac{3}{2}$

$= \frac{3}{2}\times\left(\frac{3}{2}-1\right)\Gamma\left(\frac{3}{2}-1\right)$

$= \frac{3}{2}\times\frac{1}{2}\,\Gamma\frac{1}{2}$

$= \frac{3}{4}\times\sqrt{\pi}$

$= \frac{3\sqrt{\pi}}{4}$

② $\Gamma 7$

$\Rightarrow \Gamma 7 = (7-1)!$

$= 6!$

$= 720$

③ $\Gamma -\frac{5}{2}$

$\Rightarrow \Gamma -\frac{5}{2} = \frac{\Gamma\left(-\frac{5}{2}+1\right)}{-\frac{5}{2}}$

$= \frac{\Gamma\left(-\frac{3}{2}+1\right)}{-\frac{5}{2}\times\left(-\frac{3}{2}\right)}$

$= \frac{\Gamma -\frac{1}{2}}{\frac{15}{4}}$

$= \frac{\Gamma\left(-\frac{1}{2}+1\right)}{\frac{15}{4}\times -\frac{1}{2}}$

$= \frac{\Gamma\frac{1}{2}}{-\frac{15}{8}}$

$= -\frac{8\sqrt{\pi}}{15}$

④ (a) $\Gamma\left(-\frac{3}{2}\right)$

$$\Rightarrow \Gamma\left(-\frac{3}{2}\right) = \frac{\Gamma\left(-\frac{3}{2}+1\right)}{-\frac{3}{2}}$$

$$= \frac{\Gamma\left(-\frac{1}{2}+1\right)}{-\frac{3}{2}\times-\frac{1}{2}}$$

$$= \frac{\Gamma\left(\frac{1}{2}\right)}{\frac{3}{4}}$$

$$= 4\sqrt{\pi}/3$$

(b) $\frac{\Gamma\left(\frac{5}{2}\right)}{\Gamma\left(\frac{3}{2}\right)}$

$$= \frac{\Gamma\left(\frac{5}{2}\right)}{\Gamma\left(\frac{3}{2}\right)} = \frac{\frac{5}{2}-1\,\Gamma\left(\frac{5}{2}-1\right)}{\frac{3}{2}-1\,\Gamma\left(\frac{3}{2}-1\right)}$$

$$= \frac{3/2\,\Gamma(3/2)}{\frac{1}{2}\times\Gamma\left(\frac{1}{2}\right)}$$

$$= \frac{\frac{3}{2}\times\frac{3}{2}-1\,\Gamma\left(\frac{3}{2}-1\right)}{\frac{1}{2}\times\sqrt{\pi}}$$

$$= \frac{3/2\times\frac{1}{2}\times\sqrt{\pi}}{\frac{1}{2}\times\sqrt{\pi}}$$

$$= \frac{3}{2}$$

(b) $\frac{\Gamma\left(\frac{5}{2}\right)}{\Gamma\left(\frac{3}{2}\right)}$

$$= \frac{\frac{3}{2}\times\frac{1}{2}\times\Gamma\left(\frac{1}{2}\right)}{\frac{1}{2}\times\Gamma\left(\frac{1}{2}\right)}$$

$$= \frac{\frac{3}{4}\times\sqrt{\pi}}{\frac{1}{2}\times\sqrt{\pi}}$$

$$= \frac{3}{2}$$

(3) $\dfrac{6\Gamma\frac{8}{3}}{5\Gamma\frac{2}{3}}$

$$\Rightarrow \frac{6\Gamma\frac{8}{3}}{5\Gamma\frac{2}{3}} = \frac{6\times\left(\frac{8}{3}-1\right)\cdot\Gamma\frac{8}{3}-1}{5\times\left(\frac{2}{3}-1\right)\Gamma\frac{2}{3}-1}$$

$$\Rightarrow \frac{6\times\frac{5}{3}\times\left(\frac{5}{3}-1\right)\Gamma\frac{5}{3}-1}{5\times-\frac{1}{3}\times\dfrac{\Gamma-\frac{1}{3}+1}{-\frac{1}{3}}}$$

$$\Rightarrow \frac{6\times\frac{5}{3}\times\frac{2}{3}\times\left(\frac{2}{3}-1\right)\times\Gamma\frac{2}{3}-1\times-\frac{1}{3}}{5\times-\frac{1}{3}\Gamma\frac{2}{3}}$$

$$\Rightarrow \frac{\frac{20}{3}\times-\frac{1}{3}\dfrac{\Gamma-\frac{1}{3}+1}{-\frac{1}{3}}\times-\frac{1}{3}}{5\times-\frac{1}{3}\Gamma\frac{2}{3}}$$

$$\Rightarrow \frac{\frac{20}{3}\times-\frac{1}{3}\times\Gamma\frac{2}{3}\times 1}{5\times-\frac{1}{3}\Gamma\frac{2}{3}}$$

$$\Rightarrow \frac{20/3}{5}$$

$$\Rightarrow \frac{4}{3}$$

ⓒ $\dfrac{6\Gamma\frac{8}{3}}{5\Gamma\frac{2}{3}}$

$= \dfrac{6\times\frac{5}{3}\times\frac{2}{3}\times-\frac{1}{3}\Gamma\frac{-1}{3}}{5\times-\frac{1}{3}\times\Gamma-\frac{1}{3}}$

$= \dfrac{6\times\frac{5}{3}\times\frac{2}{3}}{5}$

$= \dfrac{4}{3}$

(B.) Evaluate the followings:—

ⓐ $\int_0^\infty x^4 \cdot e^{-x} \cdot dx$

$= \int_0^\infty x^{5-1} \cdot e^{-x} \cdot dx$

$= \Gamma 5$

$= 4!$

$= 24$

we know:—

$\Gamma n = \int_0^\infty x^{n-1} \cdot e^{-x} \cdot dx$

ⓑ $\int_0^\infty \sqrt{x} \cdot e^{-x^3} \cdot dx$

$= \int_0^\infty x^{\frac{1}{2}} \cdot e^{-x^3} dx$

$= \int_0^\infty (\sqrt[3]{z})^{\frac{1}{2}} \cdot e^{-z} \cdot \dfrac{1}{3\times(\sqrt[3]{z})^2} \cdot dz$

$= \frac{1}{3}\int_0^\infty (\sqrt[3]{z})^{\frac{1}{2}-2} \cdot e^{-z} \cdot dz$

let:—

$z = x^3 \Rightarrow x = \sqrt[3]{z}$

$\dfrac{dz}{dx} = 3x^2$

$\dfrac{dz}{3x^2} = dx$

x	0	∞
z	0	∞

$$\Rightarrow \frac{1}{3}\int_0^\infty \left(z^{\frac{1}{3}}\right)^{-\frac{3}{2}} . e^{-z} . dz$$

$$\Rightarrow \frac{1}{3}\int_0^\infty (z)^{-\frac{1}{2}} . e^{-z} . dz$$

$$\Rightarrow \frac{1}{3}\int_0^\infty (z)^{\frac{1}{2}-1} . e^{-z} . dz$$

$$\Rightarrow \frac{1}{3} \times \Gamma\frac{1}{2}$$

$$\Rightarrow \frac{\sqrt{\pi}}{3}$$

Ans:-

(c) $\int_0^\infty x^{\frac{3}{2}} e^{-x} . dx$

$$= \int_0^\infty x^{\frac{5}{2}-1} . e^{-x} . dx$$

$$= \Gamma\frac{5}{2}$$

$$= \frac{3}{2} \times \Gamma\frac{1}{2} \times \frac{1}{2}$$

$$= \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}$$

$$= \frac{3\sqrt{\pi}}{4}$$

Ans:-

(d) $\int_0^\infty x^6 \cdot e^{-2x} \cdot dx$

$= \int_0^\infty \left(\frac{1}{2}z\right)^6 \cdot e^{-z} \cdot \times \frac{1}{2}\, dz$

$= \int_0^\infty \frac{1}{2^6} \times \frac{1}{2} \times (z)^{7-1} \cdot e^{-z} \cdot dz$

$= \frac{1}{2^7} \Gamma 7$

$= \frac{1}{2^7} \times 6!$

$= \frac{45}{8}$

Let:-

$z = 2x \Rightarrow x = \frac{z}{2}$

$\frac{dz}{dx} = 2$

$\frac{1}{2} \times dz = dx$

x	0	∞
z	0	∞

(e) $\int_0^\infty \sqrt{y} \cdot e^{-y^2} \cdot dy$

$= \int_0^\infty \left(z^{\frac{1}{2}}\right)^{\frac{1}{2}} \cdot e^{-z} \cdot \frac{1}{2\left(z^{\frac{1}{2}}\right)} \cdot dz$

$= \frac{1}{2} \int_0^\infty z^{\frac{1}{4} - \frac{1}{2}} \cdot e^{-z} \cdot dz$

$= \frac{1}{2} \int_0^\infty z^{-\frac{1}{4}} \cdot e^{-z} \cdot dz$

$= \frac{1}{2} \int_0^\infty z^{\frac{3}{4} - 1} \cdot e^{-z} \cdot dz$

$= \frac{1}{2} \times \Gamma \frac{3}{4}$

Let:-

$z = y^2 \Rightarrow y = \sqrt{z}$

$\frac{dz}{dy} = 2y$

$\therefore dy = dz \times \frac{1}{2y}$

y	0	∞
z	0	∞

Ans:-

Beta function:—

⇒ The Beta function is denoted by $\beta(m,n)$ and is defined by $\beta(m,n) = \int_0^1 x^{m-1} \cdot (1-x)^{n-1} \cdot dx \quad m, n > 0$

Relationship between Gamma and Beta function.

1. $\beta(m,n) = \dfrac{\Gamma m \times \Gamma n}{\Gamma m+n}$

2. $\int_0^{\frac{\pi}{2}} \sin^p x \cdot \cos^q x \cdot dx = \frac{1}{2} \beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$

Evaluate the following:—

(4.)

(d) $\beta\left(\frac{3}{2}, 2\right)$

$= \dfrac{\Gamma\frac{3}{2} \times \Gamma 2}{\Gamma\frac{3}{2}+2}$

$= \dfrac{\frac{1}{2} \times \sqrt{\pi} \times 1!}{\frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}}$

$= \dfrac{8}{15} \times \dfrac{1}{2}$

$= \dfrac{4}{15}$

⑤ (i) $\int_0^1 x^4 (1-x)^3 \cdot dx$

$= \int_0^1 x^{5-1} \cdot (1-x)^{4-1} \cdot dx$

$= \beta(5,4)$

$= \dfrac{\Gamma 5 \times \Gamma 4}{\Gamma 5+4}$

$= \dfrac{4! \times 3!}{8!}$

$= \dfrac{1}{280}$

(ii) $\int_0^2 \dfrac{x^2}{\sqrt{2-x}} \cdot dx$

$= \int_0^2 x^2 \times (2-x)^{-\frac{1}{2}} \cdot dx$

$= \int_0^2 x^2 \times \left\{2\left(1-\frac{x}{2}\right)\right\}^{-\frac{1}{2}} \cdot dx$

$= \dfrac{1}{\sqrt{2}} \int_0^2 x^2 \left(1-\frac{x}{2}\right)^{-\frac{1}{2}} \cdot dx$

$= \dfrac{1}{\sqrt{2}} \int_0^1 4z^2 \times (1-z)^{-\frac{1}{2}} \cdot 2 \cdot dz$

$= \dfrac{2}{\sqrt{2}} \times 4 \int_0^1 z^{3-1} \cdot (1-z)^{\frac{1}{2}-1} \cdot dz$

$= \dfrac{8}{\sqrt{2}} \times \beta\left(3, \frac{1}{2}\right)$

$= \dfrac{8}{\sqrt{2}} \times \dfrac{\Gamma 3 \times \Gamma \frac{1}{2}}{\Gamma 3+\frac{1}{2}}$

$= \dfrac{8}{\sqrt{2}} \times \dfrac{2! \times \sqrt{\pi}}{\frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}} = \dfrac{64\sqrt{2}}{15}$

Let:-

$z = \dfrac{x}{2} \Rightarrow x = 2z$

$\dfrac{dz}{dx} = \dfrac{1}{2}$

$dx = 2 \cdot dz$

x	0	2
z	0	1

(h) $\int_0^{\pi/2} \cos^6\theta \cdot d\theta.$

$= \int_0^{\pi/2} \sin^0\theta \cdot \cos^6\theta \cdot d\theta$

$= \frac{1}{2} B\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$

$= \frac{1}{2} B\left(\frac{0+1}{2}, \frac{6+1}{2}\right)$

$= \frac{1}{2} \times \frac{\Gamma\frac{1}{2} \times \Gamma\frac{7}{2}}{\Gamma\frac{7}{2}+\frac{1}{2}}$

$= \frac{1}{2} \times \frac{\sqrt{\pi} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}}{3!}$

$= \frac{5\pi}{32}$

(i) $\int_0^{\pi/2} \sin^8\theta \cdot d\theta$

$= \int_0^{\pi/2} \sin^8\theta \cdot \cos^0\theta \cdot d\theta$

$= \frac{1}{2} B\left(\frac{9}{2}, \frac{1}{2}\right)$

$= \frac{1}{2} \times \frac{\Gamma\frac{9}{2} \times \Gamma\frac{1}{2}}{\Gamma\frac{9}{2}+\frac{1}{2}}$

$= \frac{1}{2} \times \frac{\frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi} \times \sqrt{\pi}}{4!}$

$= \frac{35\pi}{256}$

(j) $\int_0^1 x^4 (\sqrt{1-x^2}) \cdot dx$

$= \int_0^1 x^4 \cdot (1-x^2)^{\frac{1}{2}} \cdot dx$

$= \int_0^1 \left\{(z)^{\frac{1}{2}}\right\}^3 \cdot (1-z)^{\frac{1}{2}} \cdot \frac{1}{2} dz$

$= \frac{1}{2} \int_0^1 z^{\frac{5}{2}-1} \cdot (1-z)^{\frac{3}{2}-1} \cdot dz$

$= \frac{1}{2} \times \beta(5/2, 3/2)$

$= \frac{1}{2} \times \frac{\Gamma 5/2 \cdot \Gamma 3/2}{\Gamma 5/2 + 3/2}$

$= \frac{1}{2} \times \frac{\frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi} \times \frac{1}{2} \times \sqrt{\pi}}{3!}$

$= \frac{\pi}{32}$

Let:-

$z = x^2 \Rightarrow x = z^{\frac{1}{2}}$

$\frac{dz}{dx} = 2x$

$\frac{1}{2} \cdot dz = x \cdot dx$

x	0	1
z	0	1

(k) $\int_0^{\frac{\pi}{2}} \sin^7\theta \cdot \cos^5\theta \cdot d\theta$

$= \frac{1}{2} \beta\left(\frac{8}{2}, \frac{6}{2}\right)$

$= \frac{1}{2} \times \frac{\Gamma 4 \times \Gamma 3}{\Gamma 7}$

$= \frac{1}{2} \times \frac{3! \times 2!}{6!}$

$= \frac{1}{120}$

(c) $\int_{-\pi/2}^{\pi/2} \sin^6\phi \cdot \cos^4\phi \cdot d\phi$

$= \cancel{\phi}\, 2\int_{0}^{\pi/2} \sin^6\phi \cdot \cos^4\phi \cdot d\phi$

$= 2 \times \frac{1}{2} \beta\left(\frac{7}{2}, \frac{5}{2}\right)$

$= \dfrac{\Gamma\frac{7}{2} \times \Gamma\frac{5}{2}}{\Gamma\left(\frac{7}{2}+\frac{5}{2}\right)}$

$= \dfrac{\frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi} \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}}{5!}$

$= \dfrac{3\pi}{256}$

Ans:-

Let:-

$f(x) = \sin^6 x \cdot \cos^4 x$

$f(-x) = \sin(-x)^6 \cdot \cos(-x)^4$

$= \sin^6 x \cdot \cos^4 x$

$= f(x)$

So, its a even function.

Techniques of Integration :-

Integrations Techniques:—

1. Integration by parts
2. Using Trigonometric Substitution
3. Using Trigonometric identities.
4. Partial Fraction

1. Integration By Parts:—

Formula:—

1. $\int uv \cdot dx = u\int v \cdot dx - \int \left\{ \frac{d}{dx}(u) \cdot \int v \cdot dx \right\} \cdot dx$

Evaluate the following integral:—

1. $\int xe^{2x} \cdot dx$

$= x\int e^{2x} \cdot dx - \int \left\{ \frac{d}{dx} \cdot x \cdot \int e^{2x} \cdot dx \right\} \cdot dx$

$= x \times \frac{1}{2}e^{2x} - \int 1 \cdot \frac{1}{2}e^{2x} \cdot dx$

$= \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + c$

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(3) $\int x \cdot \cos 5x \cdot dx$

$= x\int \cos 5x - \int\left\{\frac{d}{dx}x \cdot \int \cos 5x \cdot dx\right\} \cdot dx$

$= x \times \frac{1}{5}\sin 5x - \int 1 \times \frac{1}{5}\sin 5x \cdot dx$

$= \frac{1}{5}x \sin 5x - \frac{1}{5} \times -\frac{1}{5}\cos 5x \cdot dx$

$= \frac{1}{5}x\sin 5x + \frac{1}{25}\cos 5x + C$

* $\int x^2 \cdot \cos x \cdot dx$

$= x^2\int \cos x \cdot dx - \int\left\{\frac{d}{dx}x^2 \cdot \int \cos x \cdot dx\right\} dx$

$= x^2 \sin x - \int 2x \cdot \sin x \cdot dx$

$= x^2 \sin x - 2\int x \cdot \sin x \cdot dx$

$= x^2 \sin x - 2\left[x\int \sin x - \int\left\{\frac{d}{dx} \cdot x \cdot \int \sin x \cdot dx\right\} \cdot dx\right]$

$= x^2 \sin x - 2\left[-x\cos x - \int -\cos x \cdot dx\right]$

$= x^2 \cdot \sin x - 2\left[-x\cos x + \sin x\right] + C$

$= x^2 \cdot \sin x + 2x\cos x - 2\sin x + C$

Ans:

A short cut way:—

* $\int x^2 \cos x \cdot dx$

	$u = \frac{d}{dx}$	$v = \int$
+	x^2	$\cos x$
−	$2x$	$\sin x$
+	2	$-\cos x$
−	0	$-\sin x$

$= x^2 \sin x + 2x\cos x - 2\sin x + c$

Ans:—

* We can use shortcuts if the following types exists.

1. $\int x^n \cdot \cos ax \cdot dx$
2. $\int x^n \cdot \sin bx \cdot dx$
3. $\int x^n \cdot e^{ax} \cdot dx$

* $\int x^3 \cdot \cos 2x \cdot dx$

	$u = \frac{d}{dx}$	$v = \int$
+	x^3	$\cos 2x$
−	$3x^2$	$\frac{1}{2}\sin 2x$
+	$6x$	$-\frac{1}{4}\cos 2x$
−	6	$-\frac{1}{8}\sin 2x$
+	0	$\frac{1}{16}\cos 2x$

$$= x^3 \times \frac{1}{2}\sin 2x + 3x^2 \times \frac{1}{4}\cos 2x - 6x \times \frac{1}{8}\sin 2x + \frac{6}{16}\cos 2x + c$$

$$= \frac{1}{2}x^3 \sin 2x + \frac{3}{4}x^2 \cos 2x - \frac{3}{4}\sin 2x + \frac{3}{8}\cos 2x + c$$

* $\int x^3 \cdot e^{5x} \cdot dx$

	$u = \frac{d}{dx}$	$v = \int$
+	x^3	e^{5x}
−	$3x^2$	$\frac{1}{5}e^{5x}$
+	$6x$	$\frac{1}{25}e^{5x}$
−	6	$\frac{1}{125}e^{5x}$
+	0	$\frac{1}{625}e^{5x}$

$$= \frac{1}{5}x^3 e^{5x} - \frac{3}{25}x^2 e^{5x} + \frac{6}{125}x e^{5x} + \frac{6}{625}e^{5x} + c$$

(5) $\int te^{-3t} \cdot dt$

$= t\int e^{-3t} \cdot dt - \int \{ \frac{d}{dt} t \cdot \int e^{-3t} \cdot dt \} \cdot dt$

$= t \times -\frac{1}{3} e^{-3t} - \int -\frac{1}{3} e^{-3t} \cdot dt$

$= -\frac{t}{3} e^{-3t} - \frac{1}{9} e^{-3t} + c$

(7) $\int (x^2 + 2x) \cdot \cos x \cdot dx$

$= \int (x^2 \cos x + 2x \cdot \cos x) \, dx$

$= \int x^2 \cos x \cdot dx + 2\int x \cdot \cos x \cdot dx$

	$u = \frac{d}{dx}$	$v = \int$
+	x^2	$\cos x$
−	$2x$	$\sin x$
+	2	$-\cos x$
−	0	$-\sin x$

	$u = \frac{d}{dx}$	$v = \int$
+	x	$\cos x$
−	1	$\sin x$
+	0	$-\cos x$

$= x^2 \sin x + 2x \cos x - 2\sin x + 2[x \sin x + \cos x] + c$

$= x^2 \sin x + 2x \cos x - 2\sin x + 2x\sin x + 2\cos x + c$

(17) $\int e^{2\theta}\cdot \sin 3\theta \cdot d\theta$

$= \frac{e^{2\theta}(2\sin 3\theta - 3\cos 3\theta)}{2^2+3^2} + c$

$= \frac{e^{2\theta}(2\sin 3\theta - 3\cos 3\theta)}{13} + c$

Formula:—

1. $\int e^{ax} \sin bx \cdot dx = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2} + c$

2. $\int e^{ax} \cos bx \cdot dx = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2} + c$

(18) $\int e^{2\theta}\cdot \cos 5\theta \cdot d\theta$

$= \frac{e^{2\theta}(2\cos 5\theta - 5\sin 5\theta)}{2^2+5^2} + c$

$= \frac{e^{2\theta}(2\cos 5\theta - 5\sin 5\theta)}{29} + c$

(23) $\int_0^{\frac{1}{2}} x\cos \pi x \cdot dx$

	$u = \frac{d}{dx}$	$v = \int$
+	x	$\cos \pi x$
−	1	$\frac{1}{\pi}\sin \pi x$
+	0	$-\frac{1}{\pi^2}\cos \pi x$
−		

$= \left[\frac{1}{\pi} x \sin \pi x + \frac{1}{\pi^2}\cos \pi x\right]_0^{\frac{1}{2}}$

$= \frac{1}{\pi}\times\frac{1}{2}\sin\frac{\pi}{2} + \frac{1}{\pi^2}\cos \pi\times\frac{1}{2} - \frac{1}{\pi^2}\cos \pi\times 0$

$= \frac{1}{2\pi}\times 1 + 0 - \frac{1}{\pi^2}$

$= \frac{1}{2\pi} - \frac{1}{\pi^2}$ Ans:—

Integration by using Trigonometric Identities:

Formula:

1. $\sin^2 x + \cos^2 x = 1$
2. $\sec^2 x - \tan^2 x = 1$
3. $\operatorname{cosec}^2 x - \cot^2 x = 1$
4. $\sin 2A = 2\sin A \cdot \cos A$
5. $\cos 2A = \cos^2 A - \sin^2 A$
6. $\quad = 1 - 2\sin^2 A$
7. $\quad = 2\cos^2 A - 1$
8. $2\sin A \cdot \cos B = \sin(A+B) + \sin(A-B)$
9. $2\cos A \cdot \cos B = \cos(A+B) + \cos(A-B)$
10. $2\sin A \cdot \sin B = \cos(A-B) - \cos(A+B)$
11. $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

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(7) $\int_0^{\frac{\pi}{2}} \cos^2\theta \cdot d\theta$

$= \int_0^{\frac{\pi}{2}} (1-\sin^2\theta)\, d\theta$

$= \int_0^{\frac{\pi}{2}} 1 \cdot d\theta - \int_0^{\frac{\pi}{2}} \sin^2\theta \cdot d\theta$

$= [\theta]_0^{\frac{\pi}{2}} - \int_0^{\pi/2} \frac{1}{2}(1-\cos 2\theta) \cdot d\theta$

$= \frac{\pi}{2} - \frac{1}{2}\int_0^{\pi/2} 1 \cdot d\theta + \frac{1}{2}\int_0^{\pi/2} \cos 2\theta \cdot d\theta$

$= \frac{\pi}{2} - \frac{1}{2}[\theta]_0^{\frac{\pi}{2}} + \frac{1}{2} \times \frac{1}{2}[\sin 2\theta]_0^{\frac{\pi}{2}}$

$= \frac{\pi}{2} - \frac{\pi}{4} + \frac{1}{4}\sin\theta$

$= \frac{\pi}{4}$

(11) $\int_0^{\frac{\pi}{2}} \sin^2 x \cdot \cos^2 x \cdot dx$

$= \int_0^{\pi/2} (\sin x \cdot \cos x)^2 \cdot dx$

$= \int_0^{\pi/2} \frac{1}{4}(2\sin x \cdot \cos x)^2 \cdot dx$

$= \frac{1}{4}\int_0^{\pi/2} (\sin 2x)^2 \cdot dx$

$$= \frac{1}{4}\int_0^{\pi/2} \sin^2 2x \cdot dx$$

$$= \frac{1}{4}\int_0^{\pi/2} \frac{1}{2}(1 - \cos 4x) \cdot dx$$

$$= \frac{1}{8}\int_0^{\pi/2} 1 \cdot dx - \frac{1}{8}\int_0^{\pi/2} \cos 4x \cdot dx$$

$$= \frac{1}{8}[x]_0^{\frac{\pi}{2}} - \frac{1}{8} \times \frac{1}{4}[\sin 4x]_0^{\pi/2}$$

$$= \frac{\pi}{16} - \frac{1}{32} \times 0$$

$$= \frac{\pi}{16}$$

(23) $\int \tan^2 x \cdot dx$

$$= \int (\sec^2 x - 1) \cdot dx$$

$$= \int \sec^2 x \cdot dx - \int 1 \cdot dx$$

$$= \tan x - x + c$$

(35) $\int_{+\frac{\pi}{6}}^{\frac{\pi}{2}} \cot^2 x \cdot dx$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (\operatorname{cosec}^2 x - 1) \cdot dx$$

$$= -[\cot x]_{\pi/6}^{\pi/2} - [x]_{\pi/6}^{\pi/2}$$

$$= -\left[\cot\frac{\pi}{2} - \cot\frac{\pi}{6}\right] - \left(\frac{\pi}{2} - \frac{\pi}{6}\right)$$

$$= 0 + \sqrt{3} - \frac{\pi}{3}$$

$$= \sqrt{3} - \frac{\pi}{3} \quad \text{Ans.}$$

(39) $\int \operatorname{cosec}x \cdot dx$

$= \int \operatorname{cosec}x \cdot \frac{\operatorname{cosec}x + \cot x}{\operatorname{cosec}x + \cot x} \cdot dx$

$= \int \frac{\operatorname{cosec}^2 x + \operatorname{cosec}x \cdot \cot x}{\operatorname{cosec}x + \cot x} \cdot dx$

$= \int - \frac{1}{z} \cdot dz$

$= - \ln|\operatorname{cosec}x + \cot x| + c$

Let

$z = \operatorname{cosec}x + \cot x$

$\frac{dz}{dx} = -(\cot x \cdot \operatorname{cosec}x + \operatorname{cosec}^2 x)$

$dz = -(\cot x \cdot \operatorname{cosec}x + \operatorname{cosec}^2 x)\, dx$

(41) $\int \sin 8x \cdot \cos 5x \cdot dx$

$= \int \frac{1}{2}\left[\sin(8x+5x) + \sin(8x-5x)\right] \cdot dx$

$= \frac{1}{2}\int (\sin 13x + \sin 3x) \cdot dx$

$= \frac{1}{2} \times \frac{-1}{13} \cos 13x + \frac{1}{2} \times -\frac{1}{3} \cos 3x + c$

$= \frac{-1}{26} \cos 13x - \frac{1}{6} \cos 3x + c$

(43) $\int_0^{\pi/2} \cos 5t \cdot \cos 10t \cdot dt$

$= \int_0^{\pi/2} \frac{1}{2}\left[\cos(10t+5t) + \cos(10t - 5t)\right] \cdot dt$

$= \frac{1}{2} \int_0^{\pi/2} (\cos 15t + \cos 5t) \cdot dt$

$= \frac{1}{2} \times \frac{1}{15} \left[\sin 15t\right]_0^{\pi/2} + \frac{1}{2} \times \frac{1}{5} \left[\sin 5t\right]_0^{\pi/2}$

$= \frac{1}{30}\left[\sin 15 \times \frac{\pi}{2}\right] + \frac{1}{10}\left[\sin 5 \times \frac{\pi}{2}\right]$

$= -\frac{1}{30} + \frac{1}{10}$

$= \frac{1}{15}$

(45) $\int_0^{\pi/6} \sqrt{1+\cos 2x}\cdot dx$

$= \int_0^{\pi/6} \sqrt{1+(1-2\sin^2 x)}\cdot dx$

$= \int_0^{\pi/6} \sqrt{2-2\sin^2 x}\cdot dx$

$= \int_0^{\pi/6} \sqrt{2(1-\sin^2 x)}\cdot dx$

$= \int_0^{\pi/6} \sqrt{2}\times\sqrt{(1-\sin^2 x)}\cdot dx$

$= \sqrt{2}\int_0^{\pi/6} \sqrt{\cos^2 x}\cdot dx$

$= \sqrt{2}\int_0^{\pi/6} \cos x\cdot dx$

$= \sqrt{2}\left[\sin x\right]_0^{\pi/6}$

$= \sqrt{2}\left[\sin\frac{\pi}{6}\right]$

$= \frac{\sqrt{2}}{2}$

(46) $\int_0^{\pi/4} \sqrt{1-\cos 4\theta}\cdot d\theta$

$= \int_0^{\pi/4} \sqrt{1-\cos 2.2\theta}\cdot d\theta$

$= \int_0^{\pi/4} \sqrt{1-(2\cos^2 2\theta-1)}\cdot d\theta$

$= \int_0^{\pi/4} \sqrt{1-2\cos^2 2\theta+1}\cdot d\theta$

$= \int_0^{\pi/4} \sqrt{2(1-\cos^2 2\theta)}\cdot d\theta$

$= \int_0^{\pi/4} \sqrt{2}\sqrt{\sin^2 2\theta}\cdot d\theta$

$= \sqrt{2}\int_0^{\pi/4} \sin 2\theta\cdot d\theta$

$= \sqrt{2}\times\frac{-1}{2}\left[\cos 2\theta\right]_0^{\pi/4}$

$= \frac{-\sqrt{2}}{2}\left[\cos\frac{\theta}{2}-\cos 0\right]$

$= -\frac{\sqrt{2}}{2}\times -1$

$= \frac{1}{\sqrt{2}}$

অ Integration of Rational function using Trigonometric Substitution:—

Formula:—

1. $\sqrt{a^2-x^2} \Rightarrow x = a\sin\theta$

2. $\sqrt{a^2+x^2} \Rightarrow x = a\tan\theta$

3. $\sqrt{x^2-a^2} \Rightarrow x = a\sec\theta$

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(1) $\int \frac{dx}{x^2\sqrt{4-x^2}}$

$= \int \frac{dx}{x^2\sqrt{2^2-x^2}}$

$= \int \frac{2\cos\theta \cdot d\theta}{4\sin^2\theta \times \sqrt{4-4\sin^2\theta}}$

$= \frac{1}{2}\int \frac{\cos\theta \cdot d\theta}{\sin^2\theta \times \sqrt{4(1-\sin^2\theta)}}$

$= \frac{1}{2}\int \frac{\cos\theta \cdot d\theta}{2\sin^2\theta \times \sqrt{\cos^2\theta}}$

$= \frac{1}{2}\int \frac{\cos\theta \cdot d\theta}{2\sin^2\theta \cdot \cos\theta}$

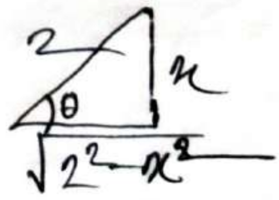
Let,

$x = 2\sin\theta \Rightarrow \theta = \sin^{-1}(x)$

$\frac{dx}{d\theta} = 2\cos\theta$

$dx = 2\cos\theta \cdot d\theta$

$\sin\theta = \frac{x}{2}$



$\tan\theta = \frac{x}{\sqrt{4-x^2}}$

$\cot\theta = \frac{\sqrt{4-x^2}}{x}$

$$= \frac{1}{2}\int \frac{1}{2} \times \frac{1}{\sin^2\theta} \cdot d\theta$$

$$= \frac{1}{4}\int \operatorname{cosec}^2\theta \cdot d\theta$$

$$= \frac{-1}{4} \times \cot\theta + c$$

$$= \frac{-1}{4} \times \frac{\sqrt{4-x^2}}{x} + c$$

(2) $\int \frac{x^3}{\sqrt{x^2+4}} \cdot dx$

$$= \int \frac{x^3}{\sqrt{2^2+x^2}} \cdot dx$$

$$= \int \frac{8\tan^3\theta}{\sqrt{4+4\tan^2\theta}} \cdot 2\sec^2\theta \cdot d\theta$$

$$= \int \frac{8\tan^3\theta}{\sqrt{4(1+\tan^2\theta)}} \cdot 2\sec^2\theta \cdot d\theta$$

$$= \int \frac{8\tan^3\theta}{2\sqrt{\sec^2\theta}} \cdot 2\sec^2\theta \, d\theta$$

$$= \int 8\tan^3\theta \cdot \sec\theta \cdot d\theta$$

$$= 8\int \left[\tan^2\theta \cdot (\tan\theta \cdot \sec\theta)\right] d\theta$$

$$= 8\int \left[(\sec^2\theta - 1) \cdot (\tan\theta \cdot \sec\theta)\right] \cdot d\theta$$

$$= 8\int \left[z^2 - 1\right] \cdot dz$$

Let:

$x = 2\tan\theta$.

$\frac{dx}{d\theta} = 2\sec^2\theta$.

$dx = 2\sec^2\theta \cdot d\theta$

$\tan\theta = \frac{x}{2}$

$\sec\theta = \frac{\sqrt{4+x^2}}{x}$

Let

$z = \sec\theta$

$\frac{dz}{d\theta} = \tan\theta \cdot \sec\theta$

$dz = \tan\theta \cdot \sec\theta \cdot d\theta$

P.T.O

$$= 8\times\left[\frac{1}{3}z^3 - z\right]$$

$$= 8\times\left[\frac{1}{3}\sec^3\theta - \sec\theta\right]$$

$$= 8\times\left[\frac{1}{3}\times\left(\frac{\sqrt{4+x^2}}{x}\right)^3 - \frac{\sqrt{4+x^2}}{x}\right]$$

(3) $\int \frac{\sqrt{x^2-4}}{x}\,dx$

$$= \int \frac{\sqrt{x^2-2^2}}{x}\,dx$$

Let

$x = 2\sec\theta$

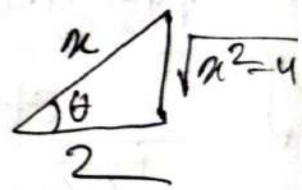
$\frac{dx}{d\theta} = 2\sec\theta\cdot\tan\theta$

$dx = 2\sec\theta\cdot\tan\theta$

$$= \int \frac{\sqrt{4\sec^2\theta-4}}{2\sec\theta}\cdot 2\sec\theta\cdot\tan\theta\cdot d\theta$$

$$= \int \frac{\sqrt{4(\sec^2\theta-1)}}{2\sec\theta}\cdot 2\sec\theta\cdot\tan\theta\cdot d\theta$$

$\sec\theta = \frac{x}{2}$



$\theta = \sec^{-1}\left(\frac{x}{2}\right)$

$\tan\theta = \frac{\sqrt{x^2-4}}{2}$

$$= \int \frac{2\sqrt{\tan^2\theta}}{2\sec\theta}\times 2\sec\theta\cdot\tan\theta\cdot d\theta$$

$$= 2\int \tan^2\theta\cdot d\theta$$

$$= 2\int(\sec^2\theta - 1)\,d\theta$$

$$= 2\int \sec^2\theta\cdot d\theta - 2\int 1\cdot d\theta$$

$$= 2\tan\theta - 2\theta + c$$

$$= 2\times\frac{\sqrt{x^2-4}}{2} - 2\times\sec^{-1}\left(\frac{x}{2}\right) + c$$

$$= \sqrt{x^2-4} - 2\sec^{-1}\left(\frac{x}{2}\right) + c$$

(4) $\int \frac{x^2}{\sqrt{9-x^2}} \cdot dx$

$= \int \frac{x^2}{\sqrt{3^2-x^2}} \cdot dx$

$= \int \frac{9\sin^2\theta}{\sqrt{9(1-\sin^2\theta)}} \cdot 3\cos\theta \cdot d\theta$

$= \int \frac{9\sin^2\theta}{3\cos\theta} \cdot 3\cos\theta \cdot d\theta$

$= \int 9\sin^2\theta \cdot d\theta$

$= 9\int \frac{1}{2}(1-\cos 2\theta) \cdot d\theta$

$= \frac{9}{2}\int (1-\cos 2\theta) \cdot d\theta$

$= \frac{9}{2}\times\theta - \frac{9}{2}\times\frac{1}{2}\sin 2\theta + c$

$= \frac{9}{2}\times\theta - \frac{9}{4}\, 2\times\sin\theta\cdot\cos\theta + c$

$= \frac{9}{2}\times\theta - \frac{9}{2}\sin\theta\cdot\cos\theta + c$

$= \frac{9}{2}\sin^{-1}\left(\frac{x}{3}\right) - \frac{9}{2}\times\frac{x}{3}\times\frac{\sqrt{9-x^2}}{3} + c$

$= \frac{9}{2}\sin^{-1}\left(\frac{x}{3}\right) - \frac{9x}{6}\times\frac{\sqrt{9-x^2}}{3} + c$

Ans.

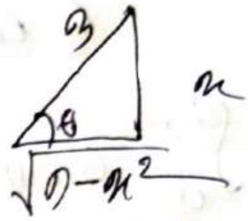
Let

$x = 3\sin\theta$

$\frac{dx}{d\theta} = 3\cos\theta$

$dx = 3\cos\theta \cdot d\theta$

$\sin\theta = \frac{x}{3}$



$\cos 2\theta = 1 - 2\sin^2\theta$

$2\sin^2\theta = 1 - \cos 2\theta$

$\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$

$\theta = \sin^{-1}\left(\frac{x}{3}\right)$

$\cos\theta = \frac{\sqrt{9-x^2}}{3}$

(6) $\displaystyle\int_0^3 \frac{x}{\sqrt{36-x^2}}\cdot dx$

$$= \int_0^{\frac{\pi}{6}} \frac{6\sin\theta}{\sqrt{36(1-\sin^2\theta)}}\cdot 6\cos\theta\cdot d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{6\sin\theta}{6\cos\theta}\cdot 6\cos\theta\cdot d\theta$$

$$= \int_0^{\pi/6} 6\sin\theta\cdot d\theta$$

$$= 6\left[-\cos\theta\right]_0^{\pi/6}$$

$$= -6\left[\cos\frac{\pi}{6}-\cos 0\right]$$

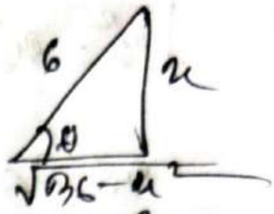
$$= -6\times\left[\frac{-2+\sqrt{3}}{2}\right]$$

$$= 6-3\sqrt{3}$$

Let

$x = 6\sin\theta$

$dx = 6\cos\theta\cdot d\theta$



$\sin\theta = \frac{x}{6}$

$\theta = \sin^{-1}\left(\frac{x}{6}\right)$

x	0	3
θ	0	$\frac{\pi}{6}$

(10) $\displaystyle\int_0^{2/3} \sqrt{4-9x^2}\cdot dx$

$$= \int_0^{2/3} \sqrt{2^2-(3x)^2}\cdot dx$$

$$= \int_0^{\pi/2} \sqrt{4(1-\sin^2\theta)}\cdot\frac{2}{3}\cos\theta\cdot d\theta$$

$$= \int_0^{\pi/2} 2\cos\theta\cdot\frac{2}{3}\cos\theta\cdot d\theta$$

$$= \frac{4}{3}\int_0^{\pi/2}\cos^2\theta\cdot d\theta$$

Let

$3x = 2\sin\theta$

$x = \frac{2}{3}\sin\theta$

$dx = \frac{2}{3}\cos\theta\cdot d\theta$

x	0	$\frac{2}{3}$
θ	0	$\frac{\pi}{2}$

$\theta = \sin^{-1}\left(\frac{3x}{2}\right)$

$$= \frac{4}{3}\int_0^{\pi/2} \frac{1}{2}(\cos 2\theta + 1)\cdot d\theta$$

$$= \frac{2}{3}\int_0^{\pi/2} (\cos 2\theta + 1)\cdot d\theta$$

$$= \frac{2}{3}\times\frac{1}{2}\Big[\sin 2\theta\Big]_0^{\frac{\pi}{2}} + \frac{2}{3}\Big[\theta\Big]_0^{\frac{\pi}{2}}$$

$$= \frac{2}{6}\sin\frac{2\pi}{2} + \frac{2}{3}\times\frac{\pi}{2}$$

$$= 0 + \frac{\pi}{3}$$

$$= \frac{\pi}{3}$$

$\cos 2\theta = 2\cos^2\theta - 1$

$\cos^2\theta = \frac{1}{2}(\cos 2\theta + 1)$

(11) $\int_0^{1/2} x\sqrt{1-4x^2}\cdot dx$

Let

$2x = \sin\theta$

$x = \frac{1}{2}\sin\theta$

$dx = \frac{1}{2}\cos\theta \times d\theta$

$\theta = \sin^{-1}(2x)$

x	0	$\frac{1}{2}$
θ	0	$\frac{\pi}{2}$

$$= \int_0^{\pi/2} \frac{1}{2}\sin\theta\sqrt{1-4\times\frac{1}{4}\sin^2\theta}\times\frac{1}{2}\cos\theta\cdot d\theta$$

$$= \int_0^{\pi/2} \frac{1}{2}\sin\theta\cdot\cos\theta\cdot\frac{1}{2}\cos\theta\cdot d\theta$$

$$= \frac{1}{4}\int_0^{\pi/2} 2\sin\theta\cdot\cos\theta\cdot\frac{1}{2}\cos\theta\cdot d\theta$$

$$= \frac{1}{8}\int_0^{\pi/2} (\sin 2\theta \times \cos\theta)\, d\theta$$

Let

$z = \cos\theta$

$\frac{dz}{d\theta} = -\sin\theta$

$-\frac{dz}{1} = \sin\theta\cdot d\theta$

$$= \frac{1}{8}\int_0^{\pi/2} 2\sin\theta\cdot\cos\theta\cdot\cos\theta\cdot d\theta$$

$$= \frac{1}{4}\int_0^{\pi/2} \sin\theta\cdot\cos^2\theta\cdot d\theta$$

$$= -\frac{1}{4}\int_0^{\pi/2} z^2\cdot dz$$

$$= -\frac{1}{4}\times\frac{1}{3}\Big[z^3\Big]_0^{\pi/2}$$

$$= \Big[\cos^3\theta\Big]_0^{\pi/2} \times -\frac{1}{12}$$

$$= \frac{1}{12}$$

(12) $\displaystyle\int_0^2 \frac{dt}{\sqrt{4+t^2}}$

$$= \int_0^{\pi/4} \frac{2\sec^2\theta \cdot d\theta}{\sqrt{4(1+\tan^2\theta)}}$$

$$= \int_0^{\pi/4} \frac{2\sec^2\theta \cdot d\theta}{2\sec\theta}$$

T.P

$$= \int_0^{\pi/4} \sec\theta \cdot d\theta$$

Let

$t = 2\tan\theta$

$dt = 2\sec^2\theta \cdot d\theta$

$\theta = \tan^{-1}\left(\frac{t}{2}\right)$

t	0	2
θ	0	$\frac{\pi}{4}$

Let

$u = \sec\theta + \tan\theta$

$\frac{du}{d\theta} = \sec\theta \cdot \tan\theta + \sec^2\theta$

$du = (\sec\theta \cdot \tan\theta + \sec^2\theta)\, d\theta$

$$= \int_0^{\pi/4} \frac{\sec\theta \cdot (\sec\theta + \tan\theta)}{(\sec\theta + \tan\theta)} \cdot d\theta$$

$$= \int_0^{\pi/4} \frac{\sec^2\theta + \sec\theta\ \tan\theta}{\sec\theta + \tan\theta} \cdot d\theta$$

$$= \int_0^{\pi/4} \frac{d\theta}{u}$$

$$= \left[\ln u\right]_0^{\pi/4}$$

$$= \left[\ln\left|\sec\theta + \tan\theta\right|\right]_0^{\frac{\pi}{4}}$$

$$= \ln\left|\sec\frac{\pi}{4} + \tan\frac{\pi}{4}\right| - \ln|\sec 0 + \tan 0|$$

$$= \ln|\sqrt{2}+1| - \ln|1+0|$$

$$= \ln|\sqrt{2}+1|$$

(13) $\int \frac{\sqrt{x^2-9}}{x^3} \cdot dx$

$= \int \frac{\sqrt{9\sec^2\theta - 9}}{27 \sec^3\theta} \cdot 3\sec\theta \cdot \tan\theta \cdot d\theta$

$= \int \frac{3\sqrt{(\sec^2\theta - 1)}}{27\sec^3\theta} \cdot 3\sec\theta \cdot \tan\theta \cdot d\theta$

$= \int \frac{3\tan\theta}{27\sec^3\theta} \cdot 3\sec\theta \tan\theta \cdot d\theta$

$= \int \frac{1}{9} \times 3 \frac{\tan^2\theta}{\sec^2\theta} \cdot d\theta$

$= \frac{1}{3} \int \frac{\sin^2\theta}{\cos^2\theta} \times \frac{\cos^2\theta}{1} \times d\theta$

$= \frac{1}{3} \int \sin^2\theta \cdot d\theta$

$= \frac{1}{3} \times \int \frac{1}{2}(1 - \cos 2\theta) \cdot d\theta$

$= \frac{1}{6}\theta - \frac{1}{6} \times \frac{1}{2} \sin 2\theta + c$

$= \frac{1}{6}\theta - \frac{1}{12} \times 2\sin\theta \cdot \cos\theta + c$

$= \frac{1}{6}\sec^{-1}\left(\frac{x}{3}\right) - \frac{1}{6} \times \frac{\sqrt{x^2-9}}{x} \times \frac{3}{x} + c$

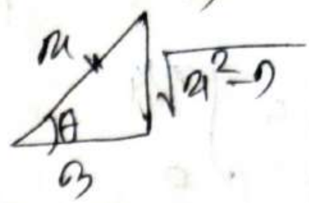
Ans

Let

$x = 3\sec\theta$

$dx = 3\sec\theta \cdot \tan\theta \cdot d\theta$

$\sec\theta = \frac{x}{3}$



$\theta = \sec^{-1}\left(\frac{x}{3}\right)$

$\sin\theta = \frac{\sqrt{x^2-9}}{x}$

$\cos\theta = \frac{3}{x}$

(14) $\int_0^1 \frac{dx}{(x^2+1)^2}$

Let

$x = \tan\theta$

$dx = \sec^2\theta \cdot d\theta$

$\theta = \tan^{-1}(x)$

x	0	1
θ	0	$\pi/4$

$= \int_0^{\pi/4} \frac{\sec^2\theta \cdot d\theta}{(\tan^2\theta+1)^2}$

$= \int_0^{\pi/4} \frac{\sec^2\theta \cdot d\theta}{(\sec^2\theta)^2}$

$= \int_0^{\pi/4} \frac{1}{\sec^2\theta} \cdot d\theta$

$= \int_0^{\pi/4} \cos^2\theta \cdot d\theta$

$= \int_0^{\pi/4} \frac{1}{2}(\cos 2\theta + 1) \cdot d\theta$

$= \frac{1}{2} \times \frac{1}{2}[\sin 2\theta]_0^{\frac{\pi}{4}} + \frac{1}{2}(\theta]_0^{\frac{\pi}{4}}$

$= \frac{1}{4}\sin\frac{\pi}{2} + \frac{1}{8}\pi$

$= \frac{1}{4} + \frac{1}{8}\pi$

(21) $\int_0^{0.6} \frac{x^2}{\sqrt{9-25x^2}}\cdot dx$

Let

$5x = 3\sin\theta$

$x = \frac{3}{5}\sin\theta$

$dx = \frac{3}{5}\cos\theta\cdot d\theta$

x	0	0.6
θ	0	$\pi/2$

$\theta = \sin^{-1}\left(\frac{5x}{3}\right)$

$(5x)^2 = (3\sin\theta)^2$

$25x^2 = 9\sin^2\theta$

$= \int_0^{\pi/2} \frac{\left(\frac{3}{5}\right)^2 \sin^2\theta}{\sqrt{9 - 9\sin^2\theta}}\cdot\frac{3}{5}\cos\theta\cdot d\theta$

$= \int_0^{\pi/2} \frac{\frac{9}{25}\sin^2\theta}{\sqrt{9(1-\sin^2\theta)}}\times\frac{3}{5}\cos\theta\cdot d\theta$

$= \int_0^{\pi/2} \frac{9\sin^2\theta}{25\times 3\times\cos\theta}\times\frac{3}{5}\cos\theta\cdot d\theta$

$= \frac{9}{125}\int_0^{\pi/2}\sin^2\theta\cdot d\theta$

$= \frac{9}{125}\times\frac{1}{2}\int_0^{\pi/2}(1-\cos 2\theta)\cdot d\theta$

$= \frac{9}{250}\left[\theta - \left(\frac{1}{2}\sin 2\theta\right)\right]_0^{\pi/2}$

$= \frac{9}{250}\times\frac{\pi}{2} - \frac{9}{250}\times\frac{1}{2}\times\sin 2\times\frac{\pi}{2}$

$= \frac{9\pi}{500}$

Ans:

(23) $\displaystyle\int \frac{dx}{\sqrt{x^2+2x+5}}$

$$= \int \frac{dx}{\sqrt{(x^2+2x.1+1^2)+4}}$$

$$= \int \frac{dx}{\sqrt{(x+1)^2+4}}$$

$$= \int \frac{dx}{\sqrt{2^2+(1+x)^2}}$$

$$= \int \frac{2\sec^2\theta \cdot d\theta}{\sqrt{4(1+\tan^2\theta)}}$$

$$= \int \frac{2\sec^2\theta \cdot d\theta}{2\sec\theta}$$

$$= \int \sec\theta \cdot d\theta$$

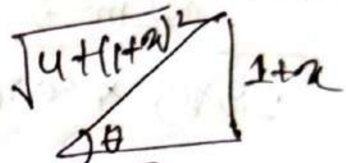
Let:

$1+x = 2\tan\theta$

$\therefore x = 2\tan\theta - 1$

$dx = (2\sec^2\theta - 0) \cdot d\theta$

$\tan\theta = \dfrac{1+x}{2}$



$\sin\theta = \dfrac{1+x}{\sqrt{4+(1+x)^2}}$

$\tan\theta = \dfrac{1+x}{2}$

$$= \int \sec\theta \cdot d\theta$$

$$= \ln|\sec\theta + \tan\theta| + c = \ln\left|\frac{\sqrt{x^2+2x+5}}{2} + \frac{x+1}{2}\right|$$

Integration of Rational function by using partial fractions:—

Formula:—

(1) $\int \frac{5x+1}{(2x+1)(x-1)} \cdot dx$

Let, $\frac{5x+1}{(2x+1)(x-1)} = \frac{A}{2x+1} + \frac{B}{x-1}$

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(1) $\int \frac{5x+1}{(2x+1)(x-1)} \cdot dx$

Let, $\frac{5x+1}{(2x+1)(x-1)} = \frac{A}{2x+1} + \frac{B}{x-1}$ ——— (I)

$\Rightarrow \frac{5x+1}{(2x+1)(x-1)} = \frac{A(x-1)+B(2x+1)}{(2x+1)(x-1)}$

$\Rightarrow 5x+1 = A(x-1) + B(2x+1)$ ——— (II)

if

$2x+1 = 0$

$x = -\frac{1}{2}$

Now, put the value in equation (II) $\Rightarrow$

$-\frac{5}{2}+1 = A\left(-\frac{5}{2}-1\right)+0$

$$A = \frac{-\frac{3}{2}}{-\frac{3}{2}}$$

$$= 1$$

When:- $x - 1 = 0$

$x = 1$

Put the value of $x = 1$ in equation (ii) $\Rightarrow$

$5 + 1 = B(2 \times 1 + 1)$

$\therefore B = \frac{6}{3} = 2$

now Substitute the value of A and B in equation (1)

$$\Rightarrow \int \frac{5x+1}{(2x+1)(x-1)} \cdot dx = \int \frac{1}{2x+1} \cdot dx + \int \frac{2}{x-1} \cdot dx$$

$$= \int \frac{1}{2x+1} dx + \int \frac{2}{x-1} \cdot dx$$

$$= \frac{1}{2} \ln|2x+1| + 2\ln|x-1| + C$$

Ans.

Short cut:-

(9) $\int \frac{5x+1}{(2x+1)(x-1)} \cdot dx$

Let, $\frac{5x+1}{(2x+1)(x-1)} = \frac{A}{(2x+1)} + \frac{B}{x-1}$ ——— (1)

Using cover up rule:-

$x = -\frac{1}{2}$; $A = \frac{5x+1}{x-1} = 1$

$x = 1$; $B = \frac{5x+1}{2x+1} = 2$

Substitute the value A and B in eq (1) =>

$= \int \frac{5x+1 \cdot dx}{(2x+1)(x-1)} = \int \frac{1 \cdot dx}{2x+1} + \frac{2 \cdot dx}{x-1}$

$= \frac{1}{2} \ln|2x+1| + 2\ln|x-1| + c$

(10) $\int \frac{y}{(y+4)(2y-1)} \cdot dy$

Let, $\frac{y}{(y+4)(2y-1)} = \frac{A}{y+4} + \frac{B}{2y-1}$ ——— (1)

using cover up rule:-

$y = -4$; $A = \frac{y}{2y-1} = \frac{4}{9}$

$y = \frac{1}{2}$; $B = \frac{y}{y+4} = \frac{1}{9}$

Substitute the value of A and B in the eq (1)

$$\int \frac{y \cdot dy}{(y+4)(2y-1)} = \int \frac{4/9}{y+4} \cdot dy + \int \frac{\frac{1}{9}}{2y-1} \cdot dy$$

$$= \frac{4}{9} \ln|y+4| + \frac{1}{9} \times \frac{1}{2} \ln|2y-1| + c$$

$$= \frac{4}{9} \ln|y+4| + \frac{1}{18} \ln|2y-1| + c$$

(11) $\int_0^1 \frac{2}{2x^2+3x+1} \cdot dx$

$$= \int_0^1 \frac{2}{(2x^2+2x+x+1)} \cdot dx$$

$$= \int_0^1 \frac{2}{2x(x+1)+1(x+1)} \cdot dx$$

$$= \int_0^1 \frac{2}{(2x+1)(x+1)} \cdot dx$$

Let,

$$\frac{2}{(2x+1)(x+1)} = \frac{A}{2x+1} + \frac{B}{x+1} \quad \text{— (1)}$$

Using cover up rule:—

$x = -\frac{1}{2}$ $\quad A = \frac{2}{x+1} = 4$

$x = -1$ $\quad B = \frac{2}{2x+1} = -2$

Substitute the value of A and B in the eq (1) =>

$$\Rightarrow \int_0^1 \frac{2 \cdot dx}{(2x+1)(x+1)} = \int_0^1 \frac{4 \cdot dx}{2x+1} + \int_0^1 \frac{-2 \cdot dx}{x+1}$$

$$= \left[\frac{4}{2} \ln|2x+1| - 2\ln|x+1|\right]_0^1$$

$$= \frac{4}{2}\{\ln 3 - \ln 1\} - 2\{\ln 2 - \ln 1\}$$

$$= \frac{4}{2} \ln 3 - 2\ln 2$$

$$= 2\ln 3 - 2\ln 2$$

Ans:—

(12) $\int_0^1 \frac{x-4}{x^2-5x+6} \cdot dx$

$$= \int_0^1 \frac{x-4}{x^2-3x-2x+6} \cdot dx$$

$$= \int_0^1 \frac{x-4}{x(x-3)-2(x-3)} \cdot dx$$

$$= \int_0^1 \frac{x-4}{(x-3)(x-2)} \cdot dx$$

Let,

$$\frac{x-4}{(x-3)(x-2)} = \frac{A}{x-3} + \frac{B}{x-2} \quad \text{—— (1)}$$

Using coverup rule:—

$$x=3; \; A = \frac{x-4}{x-2} = -1$$

$$x=2; \; B = \frac{x-4}{x-3} = 2$$

Substitute the Value of A and B in equation (1)

$$\Rightarrow \int_0^1 \frac{(x-4)\cdot dx}{(x-3)(x-2)} = \int_0^1 \frac{-1}{x-3}\cdot dx + \int_0^1 \frac{2}{x-2}\cdot dx$$

$$= \left[-\ln|x-3| + 2\ln|x-2|\right]_0^1$$

$$= \left[2\ln|x-2|\right]_0^1 - \left[\ln|x-3|\right]_0^1$$

$$= 2\left[\ln|1-2| - \ln|0-2|\right] - \left\{\ln|1-3| - \ln|0-3|\right\}$$

$$= 2\left[(\ln 1 - \ln 2)\right] - \left[\ln 2 + \ln 3\right]$$

$$= -2\ln 2 - \ln 2 + \ln 3$$

$$= \boxed{\ln 3 - 3\ln 2}$$

(20) $$\int_2^3 \frac{x(3-5x)}{(3x-1)(x-1)^2}\cdot dx$$

Let:

$$\frac{x(3-5x)}{(3x-1)(x-1)^2} = \frac{A}{3x-1} + \frac{B}{(x-1)} + \frac{C}{(x-1)^2}$$

multiply $(3x-1)(x-1)^2$ from both side:-

$$x(3-5x) = A(x-1)^2 + B(x-1)(3x-1) + C(3x-1) \quad \text{—— (1)}$$

if:- $3x-1=0$

$x = \frac{1}{3}$ put in eq (1) $\Rightarrow$

$$\frac{1}{3}\left(3-5\times\frac{1}{3}\right)=A\left(\frac{1}{3}-1\right)^2+B\left(\frac{1}{3}-1\right)\left(3\times\frac{1}{3}-1\right)$$

$$\frac{4}{9}=\frac{4}{9}A+B\times 0$$

$$A=1$$

When, $x-1=0$

$x=1$ put in the eq ① $\Rightarrow$

$$\Rightarrow 1(3-5)=A(1-1)^2+B(1-1)(3-1)+C(3-1)$$

$$\Rightarrow -2=A(0)+B(0)\times(2)+C\times 2$$

$$\therefore C=-1$$

Now:-

$$3x-5x^2=Ax^2-2xA+A+3x^2B-4xB+B+3xC-C$$

Equating the co-efficint of $x^2, x, x^0,$:-

$$-5x^2=Ax^2+3x^2B$$

$$=x^2(A+3B)$$

$$-5=(1+3B)$$

$$B=\frac{-5-1}{3}$$

$$=-2$$

Substitute the value of A and B in eq ① $\Rightarrow$

$$\int_2^3\frac{x(3-5x)}{(3x-1)(x-1)^2}\cdot dx=\int_2^3\frac{1}{3x-1}\cdot dx-\int_2^3\frac{2}{(x-1)}\cdot dx-\int_2^3\frac{1}{(x-1)^2}$$

$$=\left[\ln|3x-1|\times\frac{1}{3}-2\ln|x-1|+\frac{1}{x-1}\right]_2^3$$

$= \frac{1}{3}[\ln 8 - \ln 5] - 2[\ln 2 - \ln 1] + \left(\frac{1}{2} - \frac{1}{1}\right)$

$= \frac{1}{3}\ln 8 - \frac{1}{3}\ln 5 - 2\ln 2 + \frac{1}{2} - 1$

Ans:—

(23) $\int \frac{10}{(x-1)(x^2+9)} \cdot dx$

Let:—

$\frac{10}{(x-1)(x^2+9)} = \frac{A}{(x-1)} + \frac{Bx+C}{x^2+9}$

multiply $(x-1)(x^2+9)$ from both side:—

$10 = A(x^2+9) + (Bx+c)(x-1)$ ——— (1)

If $\Rightarrow$ $x-1=0$

$x=1$ put in eq (1) $\Rightarrow$

$\Rightarrow 10 = A(1+9) + (B+c)(1-1)$

$10 = 10A + 0$

$\therefore A = 1$

Now:—

$10 = Ax^2 + 9A + Bx^2 - Bx + cx - c$

equting the coefficients of $x^2, x^1, x^0 \Rightarrow$

$0 = Ax^2 + Bx^2$

$= x^2(A+B)$

$= B+1$

$\therefore B = -1$

$0 = -Bx + cx$

$= x(-B+c)$

$= -B+c$

$\therefore C = B$

$= -1$

Substitute the Value of A, B, C in eq (1) ⇒

$$\int \frac{10}{(x-1)(x^2+9)}\cdot dx = \int \frac{A}{x-1}\cdot dx + \int \frac{Bx+C}{x^2+9}\cdot dx$$

$$= \int \frac{1}{x-1}\cdot dx + \int \frac{-x-1}{x^2+9}\cdot dx$$

$$= \int \frac{1}{x-1}\cdot dx - \left[\int \frac{x+1}{x^2+9}\cdot dx\right]$$

$$= \int \frac{1}{x-1}\cdot dx - \left[\int \frac{x\cdot dx}{x^2+9} + \int \frac{1}{x^2+3^2}\cdot dx\right]$$

$$= \ln|x-1| - \frac{1}{2}\ln|x^2+9| - \frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right) + c$$

(24) $\int \frac{x^2-x+6}{x^3+3x}\cdot dx$

$$= \int \frac{x^2-x+6}{x(x^2+3)}\cdot x$$

Let, $\frac{x^2-x+6}{x(x^2+3)} = \frac{A}{x} + \frac{Bx+C}{x^2+3}$ —— (1)

multiply $x(x^2+3)$ from both sides

$$\Rightarrow x^2-x+6 = A(x^2+3) + (Bx+C)x$$

$$= Ax^2+3A+Bx^2+Cx$$

$x^2 = Ax^2+Bx^2$

$= x^2(A+B)$

$B = 1-A$

$= 1-2 = -1$

$3A = 6$

$A = 2$

$Cx = -x$

$C = -1$

Substitute the value of A, B, C in eq (I) ⇒

$$\int \frac{x^2-x+6}{x^3+3x} = \int \frac{2}{x} + \int \frac{-x-1}{x^2+3}$$

$$= 2\ln|x| - \int \frac{x}{x^2+3} - \int \frac{1}{x^2+(\sqrt{3})^2}$$

$$= 2\ln|x| - \frac{1}{2}\ln|x^2+3| - \frac{1}{\sqrt{3}} \times \tan^{-1}\left(\frac{x}{\sqrt{3}}\right) + C$$

Ans:-

(25)

$$\int \frac{4x}{x^3+x^2+x+1} \cdot dx$$

$$= \int \frac{4x}{x^2(x+1)+1(x+1)} \cdot dx$$

$$= \int \frac{4x}{(x^2+1)(x+1)} \cdot dx$$

Let $\frac{4x}{(x^2+1)(x+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$ ——— (I)

Multiplying $(x^2+1)(x+1)$ from both side:-

$$4x = A(x^2+1) + (Bx+C)(x+1) \quad \text{——— (II)}$$

$$= Ax^2 + A + Bx^2 + Bx + Cx + C$$

(iii) $x+1=0$

$x=-1$ put in eq (ii) $\Rightarrow$

$-4 = A(1+1)+0$

$\therefore A = -2$

Now:- $0 = Ax^2 + Bx^2$

$= x^2(A+B)$

$B = -A$

$= -(-2) = 2$

$\therefore \quad 4x = (Bx + cx)$

$= x(B+c)$

$4 = B + c$

$\therefore c = 2$

Substitute the value of A and B and c in eq (i):

$$\Rightarrow \int \frac{4x \cdot dx}{(x^2+1)(x+1)} = \int \frac{-2 \cdot dx}{x+1} + \int \frac{2x+2 \cdot dx}{x^2+1}$$

$$= -2\ln|x+1| + 2 \times \frac{1}{2}\ln|x^2+1| + 2\tan^{-1}(x) + c$$

$$= -2\ln|x+1| + \ln|x^2+1| + 2\tan^{-1}(x) + c$$

Ans:-

End